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A Study on

A Context-Aware Selective RAG Framework for Emotion Reaction Analysis in Comments on YouTube Videos Related to Adolescent Topics

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CERTIFICATE OF ORIGINAL AUTHORSHIP

I, *Farahnaz Vahdati* declare that this thesis, submitted in partial fulfilment of the requirements for the award of Doctor of Philosophy, in the *School of Computer Science, Faculty of Engineering and Information Technology* at the University of Technology Sydney, Australia, is wholly my own work unless otherwise referenced or acknowledged. In addition, I certify that all information sources and literature used are indicated in the thesis. This document has not been submitted for qualifications at any other academic institution. This research is supported by the Australian Government Research Training Program.

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ABSTRACT

Analysing emotional reactions in short-form social video comments is challenging when text is context-dependent, inherently ambiguous, or uses informal language. Retrieval-Augmented Generation (RAG) systems often retrieve for every input, increasing cost and the risk that irrelevant context perturbs Large Language Model (LLM) outputs.

This thesis presents **CAS-RAG** (Context-Aware Selective RAG), also implemented as RAG-GATE: a two-layer framework that decides *when* retrieval is warranted for context-aware emotional reaction analysis on comments on videos related to adolescent topics. Layer 1 performs semantic three-way routing (explicit, ambiguous, context-dependent) using SentenceTransformer embeddings and seven complementary similarity metrics. Layer 2 commits the retrieval policy: for context-dependent comments, spaCy-based checks distinguish RAG-resolvable, RAG-unnecessary, and RAG-infeasible cases (including *video mismatch*, where a comment appears context-dependent but is not grounded in the available scenario); explicit and ambiguous categories map directly to RAG-unnecessary and RAG-infeasible outcomes.

Six annotated datasets (approximately 100 comments each; 600 comments in total) were collected from different video-topic categories. **Routing evaluation** covers all six datasets (CAS-RAG Layer 1 and cost audit). **Emotion-classification evaluation** covers two datasets—Tutoring/Diner and Bullying/Bus Stop (N=100 each)—where Algorithms 1–4 and CAS-RAG are compared under a shared contract. The four remaining annotated topic datasets support Layer 1 topic-diversity checks only.[panel] After systematic prompt refinement informed by borderline-case analysis, Tutoring routing accuracy is unchanged (CAS-

RAG: 72.0%) while emotion reaction classification improves across all pipelines (CAS-RAG: 77.6% on 98 evaluable gold labels, up from 62.9% on 97 before refinement). On Tutoring, CAS-RAG retains the highest routing accuracy; Algorithm 1 achieves the highest post-refinement emotion accuracy (79.6%). On Bullying, CAS-RAG again leads three-way routing (72.0%; 28/100 mistakes) and leads evaluable emotion accuracy (75.0% on 88 gold labels; Table 6.6). Binary retrieval-decision accuracy (context-dependent vs. not) is 76% for CAS-RAG and 75% for Algorithm 4 (Table 6.7).[panel] The remaining annotated topic datasets are used as a complementary topic-diversity check of CAS-RAG Layer 1 behaviour (pooled Layer 1 accuracy 69.5%, macro-F1 69.8%, weighted-F1 69.1%; under the stated cost model, 44.6% cost reduction and 66.8% avoided retrieval, with a wrong-retrieval audit of 51/199 RAG-triggered cases). They were *not* subjected to the same full five-pipeline emotion comparison as Tutoring and Bullying; four datasets serve Layer 1 topic-diversity checks only. A large background corpus (152,594 comments via the YouTube Data API) supported iterative development. An earlier 26-comment single-scenario pilot (~77% accuracy, 48.7% cost saving) illustrates earlier-stage design choices.

The thesis also provides comparator methodologies, production-oriented LLM pipelines, the complete CAS-RAG implementation, and the accompanying `CAS_RAG_Project` software platform, together with reproducible code and evaluation scripts. An offline RQ1 multi-LLM consensus audit (Section 6.7) complements the primary CAS-RAG evaluation.[panel] Findings are grounded in Tutoring and Bullying in-depth emotion evaluations (Tables 6.3 and 6.6) and complementary Layer 1 topic-diversity checks across six annotated video-topic datasets. The thesis does not claim clinical, diagnostic, or adolescent-population conclusions.

CHAPTER 1

INTRODUCTION

This chapter establishes the research context, motivation, problem statement, research questions, objectives, and methodology for emotion reaction detection in comments on videos related to adolescent topics using contemporary AI technologies. The chapter provides the foundation for understanding the research scope, significance, and approach, setting the stage for the comprehensive literature review and subsequent empirical chapters.

1.1 Background and Context

YouTube hosts large volumes of public comments under videos on many topics, including education, bullying prevention, safety, substance-use prevention, and related themes that are often discussed in relation to adolescents. These comments can contain emotional reactions that are informal, ambiguous, or dependent on video context. Accurately classifying such reactions is therefore a practical natural language processing problem: short comments may refer to events, characters, or dialogue that appear only in the video, and always-on retrieval can add cost without improving labels.

This thesis does *not* study teenager psychology, adolescent wellbeing, adolescent com-

munication styles, or age-specific behavioural patterns. Adolescent-related themes appear only because the *selected videos* concern those topics. The unit of analysis is the comment text under those videos; commenter age is not known and is not modelled. The contribution is a context-aware selective RAG framework (CAS-RAG) for emotion reaction analysis and retrieval gating.

Informal language, context-dependent references, and ambiguity make automated analysis difficult. For example, a comment such as “that hit different” or “I’m dead” may be figurative and hard to interpret without conversational or video grounding. Traditional emotional reaction analysis methods, designed for more formal text, often struggle with informal, context-dependent comments of this kind Mohammad and Turney, 2013; Pang, Lee, et al., 2008. Surveys of platform popularity among teenagers Anderson, Jiang, et al., 2018; Vogels et al., 2022 motivate why videos on adolescent-related themes are a relevant *content* setting for collecting comments; they do not establish that comment authors are adolescents, and this thesis does not claim that they are.

Recent advances in Large Language Models (LLMs) such as GPT-4, Gemini, and LLaMA offer strong few-shot and zero-shot classification capabilities Achiam et al., 2023; Brown et al., 2020; Team et al., 2023. Retrieval-Augmented Generation (RAG) combines external retrieval with generation Lewis et al., 2020. Applying RAG to emotion reaction classification—especially through intelligent gating that decides *when* retrieval is warranted—remains comparatively underexplored and is the focus of this research.

1.1.1 Motivation and Research Positioning

The motivation is methodological: uniform retrieval for every comment is costly and can inject irrelevant context, while some comments are self-contained and others are inherently ambiguous. This thesis sits primarily in natural language processing and affective computing. Developmental psychology and adolescent mental-health literatures may appear as background for *video topic selection* only; they are not used to claim findings about adolescents as a population. Social validation in this thesis means agreement with human

gold labels under a written non-clinical schema, not validation of emotional meaning "as experienced by adolescents."

1.1.2 Scope statement

Videos were selected because their topics relate to adolescents. The analysed units are public YouTube comments under those videos. Commenter age is not collected. This thesis does not claim findings about teenager psychology, adolescent wellbeing, or adolescent-specific communication styles. The contribution is CAS-RAG: selective retrieval gating, routing decisions, and emotion reaction classification with efficiency and accuracy trade-offs.

1.2 Problem Statement

Despite the promise of contemporary AI technologies, three critical gaps limit the effective application of emotional reaction analysis systems to comments associated with videos related to adolescent topics. Addressing these gaps matters because always-on retrieval is costly and can inject irrelevant context, while unsupported confident labels undermine trust in automated pipelines. This thesis evaluates selective retrieval gating and emotion reaction classification against human gold labels; it does *not* claim mental-health screening, adolescent wellbeing monitoring, or teenager-appropriate psychological assessment.

1.2.1 Gap 1: Multi-LLM Reliability and Hallucination Detection

First, while LLMs show impressive capabilities for emotion classification, there is insufficient research on their reliability—specifically regarding hallucination detection—and comparative evaluation across multiple models (GPT-4, Gemini, LLaMA) for YouTube comments under the topic-selected video settings used in this thesis. Hallucination in this context refers to confident but incorrect classifications: for instance, a model might confidently label a sarcastic comment as expressing genuine joy, or misread informal wording and assign an emotion that contradicts a reasonable reading of the text. Current research evalu-

ates models in isolation or on general benchmarks, leaving unanswered questions about how different models perform on this comment setting and how ensemble approaches might improve reliability and reduce hallucination Huang et al., 2025; Zhao et al., 2023. The consequence of this gap is that practitioners lack evidence-based guidance on which models to deploy, and systems may produce systematically unreliable classifications without any mechanism to detect or filter them.

1.2.2 Gap 2: RAG for Emotion Classification with Specialised Gating

Second, RAG architectures have demonstrated remarkable success in knowledge-intensive tasks such as question answering and document summarisation, yet their application to emotion classification remains largely underexplored. Current RAG research focuses almost exclusively on generative tasks, with few specialised architectures designed for emotion or sentiment classification. Critically, existing gating mechanisms—which determine when retrieval should occur—are designed for factual tasks and do not consider emotion-specific characteristics such as ambiguity, context dependency, or the potential value of retrieving scenario/script context Asai et al., 2024; Gao et al., 2023. For example, a comment like “That kid is so brave” may benefit from retrieving the video context to understand which “kid” is referenced and what actions elicited the reaction; however, a comment like “MY HEART MELTED” may be classifiable without any retrieval. Without emotion-specific gating, systems either over-retrieve (wasting computational resources and latency) or under-retrieve (missing accuracy gains where context would help). The consequence is that emotional reaction analysis systems cannot leverage the proven benefits of retrieval-augmented approaches while remaining efficient at scale.

1.2.3 Gap 3: Selective RAG Evaluation for Emotion Reaction Classification under Topic-Selected Videos

Third, emotional reaction analysis systems on social media rarely evaluate selective retrieval when comments may depend on video scenario or script context. Most systems ei-

ther retrieve uniformly or omit retrieval gating, with limited transparent cost and retrieval-audit metrics relative to comparator pipelines. This gap is methodological. It is *not* a claim that adolescent emotional expression patterns must be modelled as a population effect, and this thesis does not validate classifications against adolescents' self-reported emotional experience.

1.2.4 Interconnected Nature of the Gaps

These gaps are not independent but interact in ways that compound their impact. The absence of reliable hallucination detection (Gap 1) becomes more problematic when retrieval is applied indiscriminately (Gap 2). The lack of selective RAG for emotion classification means systems cannot decide when scenario grounding helps versus when it adds noise. Human gold-label validation under a written schema is required so that reported metrics reflect agreement with those labels, not claimed psychological validity for adolescent populations. The challenge is primarily technical and methodological: multi-model reliability, intelligent retrieval gating, and transparent evaluation.

1.3 Research Questions and Objectives

This research addresses three fundamental research questions that emerge from the identified gaps:

Research Question 1 (RQ1): How can multi-LLM provider comparison and offline consensus analysis improve reliability signalling and support detection of unsupported emotion classifications for comments associated with videos related to adolescent topics, and what simple consensus mechanisms (e.g., majority vote and agreement scoring) are useful under the shared experimental setting?

Research Question 2 (RQ2): How can RAG architectures with intelligent gating mechanisms enhance emotion reaction analysis accuracy while reducing retrieval overhead, and what gating strategies are optimal for emotion classification tasks in this social media com-

ment setting?

Research Question 3 (RQ3): How does a context-aware selective RAG gate (CAS-RAG) affect retrieval decisions and emotion reaction classification accuracy and efficiency for comments associated with videos related to adolescent topics, relative to comparator pipelines?

1.3.1 Research Objectives

To address these research questions, this thesis pursues the following objectives:

Objective 1: To implement and evaluate a **lightweight offline multi-LLM consensus audit** for emotion reaction detection on the Tutoring and Bullying evaluation datasets, using the existing `CAS_RAG_Project` OpenAI and Gemini backends (with optional Ollama/Llama) under the same pipeline prompts. This objective addresses RQ1 by comparing independent provider labels, computing majority-vote consensus and agreement scores, and reporting disagreement-based reliability signals and gold-linked proxies for unsupported classifications. It does *not* claim a full in-pipeline ensemble gate or weighted/confidence-weighted consensus.

Objective 2: To design and implement a RAG-based framework with intelligent gating mechanisms specifically optimised for emotion reaction detection, where retrieval is selectively activated based on content ambiguity, model confidence, and emotion-specific characteristics, with applications to comments associated with videos related to adolescent topics. This objective addresses RQ2 by exploring how RAG principles can be adapted for classification tasks and developing gating mechanisms that consider emotion-specific factors.

Objective 3: To implement and evaluate CAS-RAG (selective retrieval gating with three-way routing) for emotion reaction analysis on comments associated with videos related to adolescent topics, and to compare it with Algorithms 1–4 under a shared experimental contract on the Tutoring and Bullying evaluation datasets (emotion classification) and six annotated datasets (routing)[panel]. This objective addresses RQ3 by measuring routing ac-

curacy, emotion reaction labels, and efficiency trade-offs (cost and avoided retrieval), not by modelling adolescent populations or communication styles.

These objectives are pursued through integrated components that can operate independently: (1) **Model 1 (RQ1):** an offline Multi-LLM Consensus Audit with disagreement-based reliability signalling (Section 3.4; Section 6.7); (2) **Model 2 (RQ2/RQ3):** Gated-RAG for Emotion Detection (GREED), operationalised as CAS-RAG / RAG-GATE (Chapters 5–6); and (3) **Model 3:** scenario-conditioned emotion reaction classification using prompts, prototypes, and human gold labels under a non-clinical schema.

1.4 Research Methodology

This research employs a mixed-methods approach that integrates computational methods, experimental evaluation, and human gold-label validation. The methodology addresses research questions in computer science and NLP; psychology and adolescent-wellbeing literatures are used only as background for *video topic selection*, not as objects of study.

1.4.1 Research Design

The research follows a design science research (DSR) approach, which is well-suited for developing novel technical artefacts (systems, models, methods) to address real-world problems Hevner et al., 2004; Peffers et al., 2007. The DSR approach involves iterative cycles of design, development, and evaluation, allowing for refinement based on empirical findings.

The research design incorporates three main phases:

Phase 1: Foundation and Design involves literature review, gap identification, and initial design of the proposed components (multi-LLM ensemble concepts, selective RAG gating, and scenario-conditioned emotion reaction classification).

Phase 2: Development and Implementation focuses on implementing comparator Algorithms 1–4 and CAS-RAG (RAG-GATE), including gating mechanisms, scenario-conditioned prototypes, and production LLM API integration.

Phase 3: Evaluation and Validation employs computational experiments and comparative analysis against human gold labels for routing and emotion reaction categories under a written non-clinical schema.

1.4.2 Data Collection and Preparation

1.4.2.1 Dataset Development

Publicly available YouTube comments are collected following platform terms of service and institutional ethics guidelines. Comments are collected from selected videos whose *topics* relate to adolescents (e.g., education, bullying prevention, safety, substance-use prevention). Commenter age is not inferred from linguistic markers and is not used as a variable.

Sampling aims for diversity of comment types (clear vs. ambiguous; self-contained vs. context-dependent) that are critical for evaluating retrieval gating. Dataset names such as TeenSafety or TeenPregnancy refer to *video topic themes*, not verified commenter demographics.

Evaluation depth (scope). Six topic datasets were collected and manually annotated to support topic diversity. The **primary in-depth evaluation**—full Algorithms 1–4 versus CAS-RAG comparison, emotion scoring, and qualitative error analysis—is conducted on the Tutoring/Diner dataset (N=100). The remaining annotated datasets are used for complementary CAS-RAG Layer 1 / cost topic-diversity checks and are not claimed to have received the same depth of comparative or qualitative analysis.

1.4.2.2 Annotation and Ground Truth

Emotion and routing annotation follows a predefined non-clinical label schema: emotion reaction categories used in the experimental chapters, and routing labels (explicit / ambiguous / context-dependent). Labels are assigned by trained annotators / supervisors under written guidelines. Inter-annotator checks are used where available. Ambiguous cases

may be marked for special attention. This thesis does not claim adolescent self-validation of emotional meaning and does not employ adolescent psychology diagnosis.

1.4.2.3 Ethical Considerations

Ethical approval is obtained from relevant institutional review boards before data collection where required. All data collection follows principles of: (1) privacy protection through anonymisation and aggregation, (2) informed consent where applicable (recognising challenges with public social media data), (3) data minimisation, and (4) respect for platform terms. Comments are analysed as public text artefacts; no personally identifiable information is retained beyond what is necessary for research purposes. The work does not diagnose individuals or measure wellbeing outcomes.

1.4.3 Experimental Methodology

1.4.3.1 Model Development and Implementation

Pipelines are implemented using contemporary machine learning frameworks and LLM APIs. The primary emotion-classification experiments use a production OpenAI backend (default `gpt-4o-mini`) with optional Gemini and Ollama providers. RQ1 is operationalised as an **offline multi-LLM consensus audit** on dual-provider comparison exports: independent labels, majority-vote consensus, agreement scoring, and disagreement-based reliability signalling (Section 6.7; `scripts/run_rq1_consensus_audit.py`). This is not an in-pipeline ensemble gate.Where multi-LLM ensemble ideas are discussed, they integrate GPT-4 (OpenAI), Gemini (Google), and LLaMA (open-source / local), with consensus mechanisms that aggregate classifications and detect potential hallucinations.[panel]

The gated-RAG system (CAS-RAG) implements retrieval using dense retrieval over a scenario/script corpus, with gating functions that consider ambiguity, confidence, and context dependence. Retrieval is used when beneficial for grounding, while avoiding unnecessary overhead.

Scenario-conditioned classification uses video scenario text, prompts, and scenario-conditioned prototypes—not adolescent-age linguistic dictionaries or teenager communication models.

1.4.3.2 Evaluation Metrics

Evaluation employs multiple metrics to assess different aspects of performance:

Classification Accuracy: Standard metrics including precision, recall, F1-score, and accuracy for routing and emotion categories, as well as macro-averaged and weighted scores where reported.

Reliability Metrics: Measures of consistency including inter-annotator agreement where available, and agreement between model outputs and gold labels.

Hallucination Detection: For RQ1, disagreement between provider labels combined with gold-label mismatch is reported as a *proxy* for unsupported (schema-incorrect) classifications under the offline consensus audit (Section 6.7). This is not clinical or factual-QA hallucination detection.Where multi-LLM consensus is evaluated, metrics for identifying and filtering unsupported classifications.[panel]

Efficiency Metrics: For CAS-RAG, metrics include retrieval frequency (proportion of cases requiring retrieval), relative cost under a stated cost model, and trade-offs between accuracy and efficiency.

Validity Metrics: Agreement with human gold labels under the stated schema. This is not clinical or adolescent-participant validation of emotional experience.

1.4.3.3 Comparative Analysis

Comparative evaluation includes: **Baseline Comparisons** (Algorithms 1–4 and individual LLM paths), **Ablation-style component contrasts** (routing gates, RAG on/off), and **Cross-pipeline Comparison** (different gating policies under a shared experimental contract).

1.4.4 Human Label Validation

Recognising that computational metrics alone do not establish label quality, this research uses human gold labels under written guidelines as the reference for reported accuracy.

1.4.4.1 Expert / Supervisor Labelling

Trained annotators or supervisors independently code comments using the same schema as the evaluation tables. Agreement between automated classifications and gold labels provides the primary validity evidence for this thesis. Disagreements are analysed as model error patterns (e.g., false context-dependency; praise mislabelled as Empathy), not as adolescent psychological findings.

1.4.4.2 Participant Validation

Adolescent participant feedback on classifications is *not* claimed as a completed evaluation component of this thesis. Where future work mentions such studies, they remain optional and ethically constrained; they are not required for the CAS-RAG contribution.

1.4.4.3 Mixed-Methods Analysis

Where qualitative review is reported (e.g., row-level failure analysis), it identifies repeated classifier error habits. Triangulation is limited to metrics plus error analysis of labelled comments.

1.4.5 Limitations and Ethical Considerations

This research acknowledges several limitations that affect the interpretation and generalisability of findings.

First, emotion classification involves inherent subjectivity—what one person perceives as “sad” another might label as “disappointed”—creating challenges for ground truth. Annotation reflects human judgment under a schema, which may vary across coders. The

research addresses this through transparent reporting of annotation challenges and disagreements where available.

Second, social media comments are public expressions that may be performative. Findings pertain to *expressed* emotional reactions in comment text, not to private emotional experience or to adolescent mental states.

Third, ethical constraints govern social media research. Access is governed by platform terms, privacy regulations, and institutional ethics requirements. This thesis does not treat comments as confirmed youth-authored clinical data.

1.5 Significance and Contributions

This research makes several contributions to NLP and affective computing. Existing work has explored LLM-based emotion classification, RAG for knowledge-intensive tasks, and social media sentiment analysis—but often in isolation. This thesis integrates selective RAG gating (CAS-RAG) with comparator emotion-classification pipelines for comments associated with videos related to adolescent topics, evaluated against human gold labels under a written non-clinical schema. The novelty lies in retrieval gating, routing decisions, scenario-conditioned classification, and efficiency/accuracy trade-offs—not in teenager psychology or adolescent communication modelling.

1.5.1 Theoretical Contributions

First, the research advances understanding of how multiple LLM providers can be compared for emotion classification and how **offline consensus analysis** supports reliability signalling under the shared Tutoring/Bullying setting (Section 6.7), contributing evidence on agreement, disagreement, and gold-linked proxies for unsupported labels (where "correctness" is label-schema agreement rather than factual QA correctness).

Second, the research extends RAG architectures to emotion reaction classification, demonstrating how selective retrieval can enhance analysis when comments depend on

video context. The development of emotion-oriented gating mechanisms contributes evidence on when retrieval helps and how gating can optimise efficiency–accuracy trade-offs.

Third, the research contributes a clear experimental framing for comments under topic-selected videos: the unit of analysis is comment text, not commenter age, and contributions are framed as routing, gating, and emotion reaction labels rather than age-specific behavioural findings.

1.5.2 Methodological Contributions

The methodology integrates computational evaluation with human gold labels under written guidelines. While AI evaluation often relies only on benchmark accuracy, this thesis reports (i) detailed routing and emotion comparison plus qualitative error analysis on the Tutoring dataset (N=100), and (ii) complementary CAS-RAG Layer 1 / cost checks used to assess behaviour across additional annotated topic datasets. Adolescent participant feedback and developmental-psychology validation are not claimed as completed contributions.

The evaluation methodology—accuracy, reliability where measured, efficiency, and gold-label agreement—provides a template for evaluating selective RAG in domain comment settings without overclaiming equal analysis depth across all topic datasets.

1.5.3 Practical Contributions

Practical applications of the developed systems include more efficient comment analysis pipelines for research and content-moderation-adjacent NLP tasks, where selective retrieval can reduce cost and irrelevant context. This thesis does *not* claim readiness for mental health monitoring, clinical triage, or adolescent wellbeing surveillance.

The emphasis on reliability, transparent gating, and gold-label evaluation addresses practical concerns about deploying emotion reaction classifiers where unsupported confident errors are undesirable.

1.5.4 Scope of Applicability

While developed for YouTube comments under topic-selected videos, the gating mechanisms and evaluation patterns may inform other classification domains (stance, toxicity, intent) where not every input benefits from retrieval. Broader platform or population transfer is future work and is not claimed as demonstrated here.

1.6 Thesis Structure

This thesis is organised into seven chapters:

Chapter 1 (Introduction) establishes the research context, motivation, problem statement, research questions, objectives, and methodology.

Chapter 2 (Literature Review) reviews emotion classification with LLMs, RAG architectures and gating, related social-media NLP literature used to motivate topic selection of videos (wellbeing outcomes are not studied), and hallucination risks.

Chapter 3 (Problem and Solution Overview) presents the overall solution framing and how baseline modelling, selective RAG gating, and evaluation fit together.

Chapter 4 (Comparator Pipelines, Prototypes, and Experimental Methodology) specifies the shared scenario, emotion taxonomy, scenario-conditioned prototypes, Algorithms 1–4, and the CAS-RAG gate under a fair-comparison protocol; the reference software platform (`CAS_RAG_Project`) executes all pipelines with production LLM APIs (OpenAI/ChatGPT, Gemini, or local Ollama/Llama).

Chapter 5 (CAS-RAG / RAG-GATE) presents the two-layer *Context-Aware Selective RAG* framework: semantic three-way classification with seven metrics and spaCy-based retrieval policy, with worked examples and exact Layer 1 scores.

Chapter 6 (Evaluation Across Topic Datasets and Tutoring In-Depth Results) describes six annotated topic datasets collected for diversity, complementary CAS-RAG Layer 1/cost topic checks, and the primary in-depth Tutoring (N=100) algorithm comparison and qualitative analysis.

Chapter 7 (Discussion, Conclusion, and Future Work) synthesises findings, contributions, limitations, ethics, and research directions.

Appendices provide reproducibility notes, pipeline integration remarks, and a numerical/figure reference summary.

1.7 Scope and Delimitations

1.7.1 Scope

This research focuses specifically on emotion reaction detection in comments associated with videos related to adolescent topics, using contemporary LLMs and RAG architectures. The research addresses English-language content primarily. Emotion classification follows a predefined non-clinical label schema used in the experimental chapters, focusing on the reaction categories defined for this study rather than fine-grained clinical affect taxonomies.

The research develops and evaluates selective RAG gating (CAS-RAG) and comparator pipelines, rather than attempting comprehensive coverage of all possible approaches. Human validation is limited to gold labels under written guidelines, not longitudinal wellbeing studies or clinical intervention research.

1.7.2 Delimitations

Several important areas are explicitly excluded from this research scope:

Platform Scope: While methods may have broader applicability, the primary focus is YouTube comments rather than comprehensive coverage of all social media platforms.

Data Scope: Comments under selected videos whose topics relate to adolescents. Commenter demographics (including age) are not collected or modelled. This is not a study of adolescent populations.

Temporal Scope: The research addresses emotional reaction analysis at specific time points rather than longitudinal emotion tracking or temporal dynamics analysis, though

this represents an important direction for future work.

Application Scope: While practical applications are discussed, the research focuses on developing and evaluating detection and gating systems rather than implementing clinical monitoring, wellbeing intervention, or mental-health diagnosis systems.

Cultural Scope: While cultural considerations are acknowledged, the primary focus is on English-language content, with recognition that cross-cultural validation represents important future work.

These delimitations allow for focused investigation of retrieval gating, routing, and emotion reaction classification while acknowledging broader contexts and future research directions.

1.8 Definitions of Key Terms

Emotion Reaction Detection: The automated classification of emotional reactions expressed in comment text under a predefined non-clinical label schema (as used in the experimental chapters), rather than clinical diagnosis of emotional states.

Large Language Models (LLMs): Pre-trained neural network models with billions of parameters, capable of understanding and generating human-like text. Examples include GPT-4, Gemini, and LLaMA.

Hallucination (in LLM classification): Confident but incorrect classifications produced by LLMs, where the model assigns an emotion or routing label that is not supported by the input text or contradicts clear evidence in the content.

Retrieval-Augmented Generation (RAG): An architecture that combines information retrieval from external knowledge bases with language model generation, enhancing model outputs with retrieved relevant information.

Gating Mechanism: A component in RAG systems that determines when retrieval should occur, what information should be retrieved, and how much information is needed—decisions that impact both performance and efficiency.

Consensus-Based Filtering: A method for combining multiple LLM classifications by identifying agreements and disagreements, used to improve reliability and detect potential hallucinations.

Scenario Conditioning: Use of video scenario text, script retrieval (when gated), and scenario-specific prototypes/prompts for routing and emotion reaction classification. This is *not* age-specific modelling of adolescent communication.

Human Gold-Label Validation: The process of validating automated classifications through comparison with trained human annotators under a written schema, ensuring reported metrics reflect agreement with those labels—not validation of emotional meaning as experienced by adolescents.

1.9 Summary

This chapter has established the foundation for research on emotion reaction detection in comments associated with videos related to adolescent topics using contemporary AI technologies. The research addresses gaps regarding multi-LLM reliability, selective RAG for emotion reaction classification, and evaluation of retrieval gating under a shared experimental contract. Through CAS-RAG and comparator pipelines, the thesis advances retrieval gating, routing decisions, emotion reaction classification, and efficiency/accuracy trade-offs. Adolescent-related themes appear only as video-topic context; commenter age is not studied.

The following chapter reviews related literature, establishing technical foundations and identifying research gaps that inform the proposed models and evaluation approach. Methodology emphasises computational evaluation against human gold labels, not social validation of adolescent emotional experience.

CHAPTER 2

LITERATURE REVIEW

Literature review.

2.1 Overview of Existing Research

This chapter provides a comprehensive review of relevant literature across three interconnected domains that inform the development of emotional reaction analysis systems for comments associated with videos related to adolescent topics. The review examines: (1) the evolution of emotion classification approaches, with particular focus on Large Language Models (LLMs); (2) Retrieval-Augmented Generation (RAG) architectures and gating mechanisms; and (3) related social-media NLP literature used to motivate video topic selection (not adolescent population research).

The chapter is structured as follows. Section 2.2 reviews emotion classification methods. Section 2.3 examines RAG architectures. Section 2.4 reviews related social-media NLP and topic context. Section 2.5 discusses hallucination detection. Section 2.6 synthesises findings and identifies research gaps.

2.2 Emotion Reaction Classification: From Traditional Methods to Large Language Models

2.2.1 Traditional Approaches to Emotion Classification

The journey toward automated emotion classification began with lexicon-based methods and traditional machine learning algorithms. Early systems relied on predefined dictionaries of emotion-bearing words to determine sentiment polarity and emotion categories (Mohammad & Turney, 2013). These approaches worked reasonably well for straightforward cases but faltered when confronted with context-dependent expressions, sarcasm, and cultural variations in emotional language—limitations that prove particularly problematic for informal, context-dependent social media comments.

The introduction of machine learning approaches marked a significant step forward. Support Vector Machines (SVM), Random Forests, and Naive Bayes classifiers demonstrated improved performance by learning patterns from labelled datasets (Pang, Lee, et al., 2008). These methods required extensive feature engineering, however, and remained constrained in their ability to capture semantic nuances and long-range dependencies that often carry crucial emotional information in text.

2.2.2 Deep Learning Era: Neural Networks and Transformers

Deep learning brought revolutionary changes to emotion classification. Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks showed that automatically learning hierarchical features from text could achieve superior performance (Hochreiter & Schmidhuber, 1997; Kim, 2014). The real breakthrough came with Transformer architectures. Bidirectional Encoder Representations from Transformers (BERT) and its variants fundamentally changed the field by learning contextualised word representations through bidirectional attention mechanisms (Devlin et al., 2019).

BERT-based models rapidly achieved state-of-the-art performance across various sentiment analysis benchmarks. Fine-tuned BERT models became widely adopted for emotion

classification tasks. RoBERTa (Y. Liu et al., 2019), an optimised variant that refined BERT's training strategies and extended training times, pushed performance even further.

Yet these transformer-based models still face limitations when dealing with domain-specific language. The informal, rapidly evolving communication patterns found in social media comments present particular challenges. Moreover, these models typically require task-specific fine-tuning on labelled datasets, which may not always be available for specialised comment-emotion annotation settings.

2.2.3 Large Language Models in Emotion Classification

The emergence of Large Language Models (LLMs) such as GPT-3, GPT-4, Gemini, and LLaMA opened entirely new possibilities for emotion classification. Unlike traditional fine-tuned models that demand substantial labelled data, these models demonstrate remarkable few-shot and zero-shot learning capabilities, performing emotion classification tasks with minimal or no task-specific training data (Brown et al., 2020). They leverage vast pre-trained knowledge and sophisticated instruction-following abilities to tackle classification tasks that would previously have required extensive domain-specific datasets.

2.2.3.1 GPT Series Applications

OpenAI's Generative Pre-trained Transformer (GPT) series has demonstrated remarkable performance across diverse NLP tasks. GPT-3 showed strong zero-shot performance on sentiment analysis tasks, while GPT-4 achieved even superior results through improved reasoning capabilities and more sophisticated instruction following (Achiam et al., 2023). Recent evaluations of GPT-4's performance on emotion classification tasks report high accuracy on benchmark datasets (Zhao et al., 2023).

One particularly notable advantage of GPT models lies in their ability to handle implicit emotional expressions and context-dependent sentiment—capabilities that prove especially relevant for social media analysis, where emotions are often conveyed indirectly or through cultural references. These strengths come with trade-offs: GPT models exhibit

potential hallucinations, display sensitivity to prompt formulation, and show inconsistent performance across different emotion categories.

2.2.3.2 Gemini and LLaMA Models

Google's Gemini models have shown competitive performance in text classification tasks (Team et al., 2023). Meta's LLaMA models (Touvron et al., 2023) have emerged as powerful open-source alternatives to proprietary LLMs (Grattafiori et al., 2024; Touvron et al., 2023). The open-source nature of LLaMA models makes them particularly attractive for research purposes, allowing for architectural modifications and domain-specific adaptations.

2.2.3.3 Specialised Emotion LLMs

Recent research has shifted toward developing LLMs specifically optimised for emotion-related tasks. EmoLLMs (Z. Liu et al., 2024) represents a series of emotional large language models and annotation tools designed for comprehensive affective analysis. Similarly, SentiXRL presents an advanced framework for multilingual fine-grained emotion classification in complex text environments (Wang et al., 2024).

2.2.3.4 Limitations and Challenges

Despite their impressive capabilities, LLMs face several critical limitations in emotion classification tasks. Hallucination remains a primary concern: LLMs may generate confident but incorrect emotion classifications, particularly for ambiguous or context-dependent expressions. Inconsistency presents another challenge. LLMs may produce different classifications for semantically similar inputs. Interpretability also remains limited, constraining their use in applications requiring explainability.

2.2.4 Gap Analysis: LLMs in Emotion Classification for YouTube Comments

While LLMs have shown considerable promise in emotion classification, several critical gaps exist. Current research primarily evaluates individual LLMs in isolation or compares

them on general emotion classification benchmarks. There is a notable absence of comprehensive comparative studies evaluating multiple LLMs (GPT-4, Gemini, LLaMA) specifically on YouTube comments under topic-selected video settings of the kind used in this thesis.

Additionally, while hallucination detection has been extensively studied for generative tasks, research on detecting and mitigating hallucinations in LLM-based emotion classification remains limited. No specialised hallucination detection frameworks exist for emotion classification systems.

Research Aim 1: To implement and evaluate a lightweight offline multi-LLM consensus audit for emotion reaction detection under the shared experimental setting used in this thesis.

Proposed Model 1: Multi-LLM Ensemble with Consensus-Based Hallucination Filtering (offline audit; Section 6.7).

2.3 Retrieval-Augmented Generation (RAG): A Comprehensive Review

2.3.1 Foundations and Motivation

Retrieval-Augmented Generation (RAG) represents a paradigm shift in natural language processing, elegantly combining the strengths of dense information retrieval with generative language models. The fundamental insight, first formalised by Lewis et al. (2020), addresses a critical limitation of generative LLMs: their reliance on parametric memory, which may contain outdated, incomplete, or incorrect information. RAG systems overcome this limitation by augmenting generation with retrieved relevant documents from an external knowledge base, enabling models to access up-to-date, domain-specific information that extends far beyond their training data.

The architecture operates on a simple yet powerful principle. Given a query, the system first retrieves relevant documents from a corpus, then conditions the language model's

generation on both the query and the retrieved context. This approach has demonstrated remarkable success in knowledge-intensive tasks such as open-domain question answering, fact-checking, and information synthesis.

2.3.2 Evolution of RAG Architectures

2.3.2.1 First-Generation RAG: Naive RAG

The original RAG architecture introduced by Lewis et al. (2020) established the foundational approach, now commonly termed "Naive RAG" or "Retrieve-then-Generate." The architecture consists of two main components: a neural retriever (typically Dense Passage Retrieval) that encodes queries and documents into dense vector representations, and a sequence-to-sequence generator (originally BART) that generates responses conditioned on both the query and retrieved documents.

Naive RAG exhibits several limitations: it includes no mechanism to determine when retrieval is necessary, retrieving documents for every query even when the language model's parametric knowledge would suffice. The system may also retrieve irrelevant documents, introducing noise into generation.

2.3.2.2 Advanced RAG Architectures

Subsequent research focused intensively on improving retrieval quality. REALM (Guu et al., 2020) introduced end-to-end training of retrieval and generation components. Fusion-in-Decoder (FiD) (Izacard & Grave, 2021) enhanced the architecture by independently encoding multiple retrieved passages and fusing them in the decoder. Atlas (Izacard et al., 2023) scaled RAG to larger corpora. REPLUG (Shi et al., 2024) introduced a simpler method that uses retrieval as a plug-in module, enabling easy integration with existing LLMs.

Self-RAG (Asai et al., 2024) represents a significant advancement by introducing a self-reflective mechanism that decides when to retrieve, what to retrieve, and whether to use retrieved information. Corrective RAG (CRAG) (Yoran et al., 2023) introduces a correction mechanism that evaluates the relevance and correctness of retrieved documents.

GraphRAG (Larson & Truitt, 2024) extends RAG by representing knowledge as a graph structure, enabling more sophisticated reasoning over retrieved information.

2.3.3 Gating Mechanisms in RAG Systems

Gating mechanisms represent one of the most critical yet understudied components of RAG architectures. These mechanisms determine when to retrieve, what to retrieve, and how much to retrieve—decisions that directly impact both performance and efficiency.

Several approaches have emerged: confidence-based gating uses the language model's confidence or entropy to decide whether to retrieve; query complexity assessment analyses query characteristics to predict retrieval necessity; trainable gating networks learn optimal retrieval decisions from data.

Current gating mechanisms exhibit important limitations. Most existing mechanisms are designed for knowledge-intensive tasks like question answering and fact-checking, leaving a gap for classification or sentiment analysis tasks. Gating decisions are typically binary (retrieve or not retrieve), with limited granularity. Few mechanisms consider task-specific characteristics such as emotion ambiguity in their gating decisions.

2.3.4 Gap Analysis: RAG in Emotion Classification and Sentiment Analysis

While RAG has demonstrated remarkable success in knowledge-intensive tasks, its application to emotion classification and sentiment analysis remains largely unexplored. Current RAG research focuses almost exclusively on generative, knowledge-intensive tasks. There is a notable absence of RAG architectures specifically designed for emotion classification and sentiment analysis tasks.

Gating mechanisms for emotion tasks should consider emotion ambiguity and context dependency—factors that differ substantially from knowledge-intensive tasks. No existing RAG frameworks address when retrieval is beneficial for emotion versus sentiment tasks.

Research Aim 2: To design and implement a RAG-based framework with intelligent gating mechanisms specifically optimised for emotion reaction detection, where retrieval is selectively activated based on content ambiguity, model confidence, and emotion-specific characteristics, with applications to comments on videos related to adolescent topics.

Proposed Model 2: Gated-RAG for Emotion Detection (GREED) with Adaptive Retrieval Decision Gate.

2.4 Related Social Media NLP and Topic Context (Not a Population Study)

2.4.1 Background Literature Used for Video Topic Selection

Literature on social media use and adolescent mental health has been extensively studied through traditional social science methodologies, including associations with depression, anxiety, self-esteem, and social comparison (Kross et al., 2021; Twenge et al., 2018). **This thesis does not study those outcomes.** Such literature is cited only to explain why videos on education, bullying prevention, safety, and related themes are a relevant *content* setting for collecting YouTube comments for NLP evaluation.

YouTube is widely used among teenagers (Anderson, Jiang, et al., 2018; Vogels et al., 2022). Platform popularity motivates topic selection of videos; it does not establish that comment authors in this study are adolescents, and this thesis does not claim that they are. Public comments under selected videos are text artefacts that may express emotional reactions to video content and therefore form a useful evaluation setting for selective RAG.

Research on YouTube comments has examined patterns in emotional expression across platforms (Mohammad & Turney, 2013; Pang, Lee, et al., 2008). Traditional analysis methods—manual coding or simple keyword matching—often struggle with informal, context-dependent comment text of the kind that appears under these videos.

2.4.2 AI and LLM Applications in Social Media Annotation

The integration of AI and LLM technologies into social science research has opened new possibilities for analysing large-scale social media data while maintaining research rigour.

2.4.2.1 LLM-Assisted Annotation and Coding

Recent research has explored using LLMs to assist or replace human annotators in social science research. Studies have demonstrated that GPT-4 and similar models can achieve inter-annotator agreement levels comparable to human coders for certain coding tasks (Gilardi et al., 2023). For emotion classification in social media content, LLMs offer several advantages beyond consistency: they can handle large volumes of data efficiently and offer potential for fine-grained emotion classification beyond simple sentiment.

Challenges remain. There is a persistent need for validation against human coders to ensure LLM classifications align with gold labels under a written schema. Potential systematic biases in LLM classifications may differentially impact different types of content. Ethical considerations become especially important when automating analysis of comments under sensitive video topics.

2.4.2.2 Privacy and Ethical Considerations

The use of AI and LLMs in analysing public YouTube comments raises ethical and privacy concerns, especially under sensitive topics. Comment authors may not anticipate automated analysis of public posts. Bias and fairness concerns remain important when analysing content from diverse populations. This thesis does not diagnose individuals or measure wellbeing.

2.4.3 Gap Analysis: Selective RAG for Emotion Reaction Classification under Topic-Selected Videos

Despite growing interest in AI applications for social media analysis, significant gaps remain in *selective retrieval* for emotion reaction classification when comments may require

video grounding. Current systems often apply retrieval uniformly or omit retrieval gating entirely. This thesis does not treat adolescent emotional expression patterns as a modelling target; the gap is methodological (when to retrieve) under comments associated with videos related to adolescent topics.

Human gold-label validation under a written schema is essential for reporting accuracy. This is not the same as validating classifications against psychological theories of adolescent emotion.

Research Aim 3: To implement and evaluate CAS-RAG (selective retrieval gating with three-way routing) for emotion reaction analysis on comments associated with videos related to adolescent topics, relative to comparator pipelines, reporting routing accuracy, emotion labels, and efficiency trade-offs.

Proposed Model 3: Scenario-conditioned emotion reaction classification (prompts, prototypes, human gold labels) under the shared experimental contract—not age-specific adolescent modelling.

2.5 Hallucination Detection in LLM-Generated Classifications

2.5.1 Understanding Hallucination in Large Language Models

Hallucination in LLMs refers to the generation of content that is unsupported by the input, training data, or external knowledge. In classification tasks, hallucinations manifest as confident but incorrect classifications, often resulting from overreliance on parametric knowledge, prompt sensitivity, or misunderstanding of context (Huang et al., 2025). Understanding and detecting hallucinations becomes especially important when dealing with subjective emotional content, where "correctness" may be more ambiguous than in factual tasks.

Two primary types of hallucinations are typically distinguished. Intrinsic hallucinations involve content that contradicts information present in the input or retrieved context—for

example, classifying a clearly positive comment as negative. Extrinsic hallucinations involve content that cannot be verified as true or false from available information. In emotion classification, this might involve inferring emotions not clearly expressed in the text.

2.5.2 Hallucination Detection Methods

Several approaches have been developed for detecting hallucinations, though most focus on generative tasks rather than classification. Self-consistency methods involve generating multiple responses and checking for agreement. In classification tasks, this might involve querying the model multiple times with the same input and aggregating results through voting or confidence weighting. High variance in outputs suggests potential hallucination, while consistent outputs increase confidence.

Multi-model consensus compares outputs across multiple LLMs—if models disagree significantly, it may indicate ambiguous or hallucinated content. This approach is particularly relevant for our proposed multi-LLM ensemble framework. Confidence-based detection uses the model's confidence scores or logits to indicate uncertainty, though confidence calibration varies across models and tasks.

2.5.3 Hallucination in Emotion Classification: Specific Challenges

Emotion classification presents unique challenges for hallucination detection that differ from those in factual tasks. Emotions are inherently subjective, making "correctness" difficult to define. What one annotator labels as "sad" another might label as "disappointed," creating ambiguity that complicates hallucination detection. Emotional meaning depends heavily on context, culture, and individual differences. Emotions may be expressed implicitly through sarcasm, humour, or cultural references that models may misinterpret.

2.5.4 Research Gap

As identified in the preceding gap analysis, there is a critical absence of hallucination detection frameworks specifically designed for emotion classification tasks. Current meth-

ods focus on factual errors in generative tasks, while emotion classification requires different approaches that account for subjectivity, ambiguity, and context dependency. This gap is addressed in our proposed Model 1: Multi-LLM Ensemble with Consensus-Based Hallucination Filtering, which leverages multi-model consensus and confidence calibration specifically for emotion classification.

2.6 Synthesis and Research Gaps

2.6.1 Summary of Key Findings

This literature review has synthesised three interconnected research domains, revealing both the substantial progress made in each area and the important gaps that remain. Large Language Models demonstrate remarkable capabilities for emotion classification through zero-shot and few-shot learning, offering powerful tools for analysing emotional content without requiring extensive labelled datasets. GPT-4, Gemini, and LLaMA each offer unique strengths, yet comparative evaluation specifically for YouTube comment emotion reaction classification under the video-topic settings used in this thesis is limited. Critical limitations include hallucination, inconsistency, and limited adaptation to informal, context-dependent comment text.

RAG has revolutionised knowledge-intensive NLP tasks through retrieval-augmented generation, demonstrating how external knowledge can enhance language model capabilities. However, RAG research focuses almost exclusively on generative tasks like question answering and summarisation, with minimal exploration of classification applications. Gating mechanisms, while critical for efficiency, are not designed for emotion classification tasks where ambiguity and context dependency should inform retrieval decisions.

AI and LLMs offer powerful tools for analysing large-scale social media data. Yet selective RAG frameworks that decide *when* retrieval helps for emotion reaction classification—evaluated against human gold labels under a written schema—remain limited. This thesis does not claim a gap in “teenager-specific” psychology modelling.

2.6.2 Identified Research Gaps

Three critical research gaps have been identified, each representing a significant opportunity for advancing emotion reaction analysis for comments associated with videos related to adolescent topics.

Gap 1: Multi-LLM Reliability and Hallucination Detection for Emotion Classification. While LLMs show promise in emotion classification, there is insufficient research on their reliability (hallucination detection) and comparative evaluation across multiple models (GPT-4, Gemini, LLaMA) for YouTube comment emotion reaction classification under the settings used in this thesis.

Research Aim 1: To implement and evaluate a lightweight offline multi-LLM consensus audit for emotion reaction detection on the Tutoring and Bullying datasets, using OpenAI and Gemini (optional Ollama) under the shared experimental setting.

Proposed Model 1: Multi-LLM Ensemble with Consensus-Based Hallucination Filtering (offline audit operationalisation; Section 6.7).

Gap 2: RAG with Specialised Gating for Emotion Detection. RAG has been predominantly applied to knowledge-intensive tasks (QA, summarisation) with limited exploration in emotion classification and sentiment analysis. There is a critical gap in developing gating mechanisms specifically designed for determining when retrieval is beneficial in emotional reaction analysis tasks, particularly for noisy social media content like YouTube comments.

Research Aim 2: To design and implement a RAG-based framework with intelligent gating mechanisms specifically optimised for emotion reaction detection, where retrieval is selectively activated based on content ambiguity, model confidence, and emotion-specific characteristics, with applications to comments associated with videos related to adolescent topics.

Proposed Model 2: Gated-RAG for Emotion Detection (GREED) with Adaptive Retrieval Decision Gate (operationalised as CAS-RAG / RAG-GATE).

Gap 3: Selective RAG Evaluation for Emotion Reaction Classification under Topic-

Selected Videos. Limited work evaluates selective retrieval for emotion reaction classification when comments may depend on video scenario or script context, with transparent cost and retrieval-audit metrics relative to comparator pipelines.

Research Aim 3: To implement and evaluate CAS-RAG for emotion reaction analysis on comments associated with videos related to adolescent topics, reporting routing decisions, emotion reaction labels, and efficiency/accuracy trade-offs.

Proposed Model 3: Scenario-conditioned emotion reaction classification with human gold labels under a non-clinical schema (not teenager-specific population modelling).

2.6.3 Transition to Solution Overview

The identified research gaps establish clear requirements for the proposed solution framework. In the subsequent chapter (Solution Overview), integrated models address these gaps: Model 1 addresses Gap 1 through an offline multi-LLM consensus audit with disagreement-based reliability signalling (Section 6.7)[panel]; Model 2 addresses Gap 2 through Gated-RAG / CAS-RAG with emotion-specific adaptive retrieval; and Model 3 addresses Gap 3 through scenario-conditioned classification and gold-label evaluation. CAS-RAG (Model 2) is the primary implemented and evaluated system; Model 1 and Model 3 components can be evaluated independently for comments associated with videos related to adolescent topics.

CHAPTER 3

PROBLEM AND SOLUTION OVERVIEW

Chapter 2 identified three critical research gaps that limit the effective application of emotional reaction analysis systems to comments on videos related to adolescent topics. This chapter presents a comprehensive solution framework that addresses these gaps through three integrated models.

3.1 Introduction

This chapter presents a comprehensive solution framework that addresses the three gaps identified in the literature review through three integrated models, each designed to tackle specific challenges while contributing to a unified approach for emotion reaction detection in comments associated with videos related to adolescent topics.

The chapter is structured as follows. Section 3.2 provides a consolidated problem statement, synthesising the three gaps. Section 3.3 presents the overall solution framework. Sections 3.4–3.6 detail each of the three proposed models: the Multi-LLM Ensemble with Consensus-Based Hallucination Filtering (Model 1), the Gated-RAG for Emotion Detection (GREED) with Adaptive Retrieval Decision Gate (Model 2; operationalised as CAS-RAG), and Scenario-Conditioned Emotion Reaction Classification with Human Gold Labels

(Model 3). Section 3.7 discusses the integration strategy. Section 3.8 outlines the evaluation approach.

3.2 Consolidated Problem Statement

The literature review revealed three interconnected gaps that collectively limit the effectiveness of current emotional reaction analysis systems for comments associated with videos related to adolescent topics.

3.2.1 Gap 1: Multi-LLM Reliability and Hallucination Detection

Current research evaluates LLMs in isolation or on general emotion classification benchmarks, leaving critical questions unanswered about their performance on YouTube comments under the topic-selected video settings used in this thesis. The absence of comprehensive comparative evaluation across multiple models (GPT-4, Gemini, LLaMA) for this comment setting means that researchers and practitioners lack evidence-based guidance for model selection. More critically, hallucination detection—identifying confident but incorrect classifications—remains underexplored for emotion classification tasks Huang et al., 2025; Zhao et al., 2023.

3.2.2 Gap 2: RAG for Emotion Classification with Specialised Gating

RAG architectures have demonstrated remarkable success in knowledge-intensive tasks such as question answering and document summarisation, yet their application to emotion classification remains largely unexplored Gao et al., 2023; Lewis et al., 2020. Current RAG research focuses almost exclusively on generative tasks, with few specialised architectures designed for emotion or sentiment classification. Critically, existing gating mechanisms are designed for factual tasks and do not consider emotion-specific characteristics such as ambiguity, context dependency, or the potential value of retrieving scenario/script context Asai et al., 2024.

3.2.3 Gap 3: Selective RAG Evaluation for Emotion Reaction Classification under Topic-Selected Videos

Existing emotional reaction analysis systems on social media rarely evaluate selective retrieval for emotion reaction classification when comments may depend on video scenario or script context Anderson, Jiang, et al., 2018. Most systems either retrieve uniformly or omit retrieval gating, with limited transparent cost and retrieval-audit metrics relative to comparator pipelines. This gap is methodological; it is not a claim that adolescent emotional expression patterns must be modelled as a population effect.

3.2.4 Interconnected Nature of the Gaps

These three gaps interact in ways that compound their impact. The absence of reliable hallucination detection (Gap 1) becomes more problematic when retrieval is applied indiscriminately (Gap 2). The lack of selective RAG for emotion classification means systems cannot decide when scenario grounding helps versus when it adds noise. Addressing these gaps requires an integrated approach that improves reliability through multi-model consensus where used, enhances accuracy through intelligent retrieval gating, and reports validity through agreement with human gold labels under a written non-clinical schema.

3.3 Overall Solution Framework

The solution framework addresses the three identified gaps through three integrated models that can operate independently or as components of a unified system.

3.3.1 Framework Architecture

Component 1: Multi-LLM Ensemble with Consensus-Based Hallucination Filtering addresses Gap 1 by comparing multiple LLM providers and analysing agreement to improve reliability signalling. In this thesis it is operationalised as an offline consensus audit (Section 6.7), not as an in-pipeline gate inside CAS-RAG.[panel]

Component 2: Gated-RAG for Emotion Detection (GREED) with Adaptive Retrieval Decision Gate addresses Gap 2 by applying RAG principles to emotion classification with intelligent gating that determines when retrieval is beneficial. This component enhances accuracy for ambiguous or context-dependent cases while minimising computational overhead through selective retrieval activation. In the empirical chapters, this is operationalised as CAS-RAG / RAG-GATE.

Component 3: Scenario-Conditioned Emotion Reaction Classification with Human Gold Labels addresses Gap 3 by using video scenario text, prompts, and prototypes under a shared experimental contract, validated against trained annotator / supervisor gold labels. This component does *not* model adolescent communication styles or validate emotional meaning as experienced by adolescents.

3.3.2 Design Principles

The framework is guided by several key design principles: **Modularity** (each model can operate independently), **Integration Capability** (models integrate seamlessly), **Scalability** (designed for large-scale applications), **Validity Focus** (emphasis on gold-label agreement under a written schema), and **Transparency** (mechanisms for understanding and explaining classifications and gating decisions).

3.3.3 Workflow Overview

The typical workflow proceeds as follows: (1) Input Processing: A YouTube comment is received and preprocessed; (2) Initial Classification: The comment may be classified using a Multi-LLM Ensemble path where applicable; (3) Consensus Analysis: Consensus mechanisms analyse agreement across models where used; (4) Gating Decision: CAS-RAG evaluates whether retrieval would be beneficial; (5) Selective Retrieval: If beneficial, relevant scenario/script context is retrieved; (6) Scenario-Conditioned Classification: apply the selected prompt path (with or without retrieved context) under the video scenario; (7) Final

Classification: a final emotion reaction label is produced; (8) Gold-Label Comparison: classifications are compared against human gold labels under the written schema.

3.4 Model 1: Multi-LLM Ensemble with Consensus-Based Hallucination Filtering

Model 1 addresses Gap 1 by specifying a multi-LLM ensemble framework for improving reliability and detecting unsupported emotion classifications Achiam et al., 2023; Team et al., 2023; Touvron et al., 2023. **Implementation status.** In this thesis, Model 1 is *partially operationalised* as an offline consensus audit (`consensus.py`; Section 6.7) on dual-provider emotion columns from the existing comparison runner. CAS-RAG (Model 2) remains the primary in-pipeline implementation and evaluation focus. Components marked **[implemented offline]**, **[partial]**, or **[conceptual / future]** below.[panel]

3.4.1 Architecture Overview

The Multi-LLM Ensemble consists of four main components: (1) **Multi-Model Classifier [partial]**[panel]—independent emotion classifications from OpenAI and Gemini (optional Ollama/Llama) via the shared `CAS_RAG_Project` provider layer and comparison exports; (2) **Consensus Analyzer [implemented offline]**[panel]—majority vote and agreement ratio on provider labels; (3) **Hallucination Detector [partial]**[panel]—disagreement-based proxy with optional gold-label mismatch (not confidence calibration or repeated-query consistency); and (4) **Aggregation Engine [implemented offline]**[panel]—consensus label as a reported aggregate, not an in-pipeline filter.

3.4.2 Consensus Mechanisms

The consensus analyzer specifies multiple mechanisms: voting-based consensus (simple majority) **[implemented offline]**[panel], weighted consensus (based on validation performance) **[conceptual / future]**[panel], confidence-weighted consensus (weighting by confi-

dence scores) **[conceptual / future]**[panel], and agreement scoring (measuring degree of consensus beyond binary agreement) **[implemented offline]**[panel].

3.4.3 Hallucination Detection Methods

The hallucination detector specifies: disagreement-based detection (flagging when models strongly disagree) **[implemented offline]**[panel], confidence calibration **[conceptual / future]**[panel], consistency checking (repeated queries with slight prompt variations) **[conceptual / future]**[panel], cross-model verification (flagging outlier classifications) **[partial]**[panel] with two providers, and semantic consistency checks **[conceptual / future]**[panel].

3.4.4 Aggregation Strategies

The aggregation engine specifies consensus-first aggregation **[implemented offline]**[panel], hallucination-filtered aggregation (filtering flagged classifications) **[conceptual / future; audit flags only]**[panel], confidence-weighted aggregation **[conceptual / future]**[panel], and fallback mechanisms for low-consensus cases **[partial]**[panel] (`SPLIT` / `NON_LABELLED` reliability tiers). Production experiments use `gpt-4o-mini` (OpenAI) and Gemini; Ollama supports optional local Llama-class models.

3.5 Model 2: Gated-RAG for Emotion Detection (GREED) with Adaptive Retrieval Decision Gate

Model 2 addresses Gap 2 by developing a RAG-based framework specifically designed for emotion classification, with intelligent gating mechanisms that determine when retrieval is beneficial Asai et al., 2024; Lewis et al., 2020.

3.5.1 Architecture Overview

The Gated-RAG for Emotion Detection (GREED) consists of five main components: (1) **Initial Classifier** producing an initial emotion classification and confidence score using a base LLM without retrieval; (2) **Adaptive Retrieval Gate** evaluating whether retrieval would be

beneficial based on confidence, ambiguity, content characteristics, and emotion-specific heuristics; (3) **Emotion-Specific Retriever** retrieving relevant examples when the gate determines retrieval is beneficial; (4) **Augmented Classifier** refining the initial classification using retrieved information; and (5) **Efficiency Monitor** tracking retrieval frequency and optimising gating thresholds.

3.5.2 Adaptive Retrieval Gate Design

The gate considers multiple factors: initial classification confidence (low confidence triggers retrieval), emotion ambiguity detection (mixed indicators, sarcasm, neutral signals), content characteristics (comment length, slang, cultural references), emotion-specific heuristics (complex emotions may benefit more from retrieval), and historical performance (learning from which types benefit from retrieval).

3.5.3 Emotion-Specific Retrieval Strategies

When retrieval is activated, the retriever employs: similar emotional expression retrieval (finding comments with similar emotional patterns), contextual example retrieval (comments from same video or category), ambiguity resolution retrieval (examples that help resolve ambiguity), and hybrid retrieval (combining dense and sparse methods).

3.5.4 Knowledge Base Construction

The knowledge base includes: annotated comment corpus with emotion labels, emotion pattern examples with informal and cultural references where available, contextual information (video scenario, script chunks, metadata), and scenario-conditioned prototypes/prompts.

3.6 Model 3: Scenario-Conditioned Emotion Reaction Classification with Human Gold Labels

Model 3 addresses Gap 3 by developing emotion reaction analysis for **comments associated with videos related to adolescent topics**, using scenario text, prompts, and prototypes under a shared experimental contract, with human gold-label validation Anderson, Jiang, et al., 2018; Gilardi et al., 2023. The framework does *not* diagnose individuals, infer clinical conditions, or model adolescent populations or communication styles.

3.6.1 Architecture Overview

The scenario-conditioned classification stack consists of five main components: (1) **Comment Preprocessor** applying light normalisation for informal comment text (abbreviations, emoji handling) without age-specific slang dictionaries as a scientific claim; (2) **Scenario-Conditioned Analyser** employing LLM prompts with scenario context and optional retrieved script chunks; (3) **Prototype / Similarity Support** using scenario-conditioned prototypes for routing where applicable; (4) **Human Gold-Label Interface** comparing system outputs to trained annotator / supervisor labels; and (5) **Iterative Refinement** incorporating error analysis to refine prompts and prototypes.

3.6.2 Comment Preprocessing

The preprocessor may include: abbreviation and emoji handling, context preservation (capitalisation, repetition, punctuation), recognition of common informal phrases, and light normalisation while preserving intentional variations. These steps address informal YouTube register, not teenager psychology.

3.6.3 Scenario-Conditioned Analysis

The analyser employs: prompt engineering with few-shot examples from the labelled comment sets, optional ensemble components, and rule-based cues where used in compara-

tor pipelines. Adaptation is to the *video scenario and label schema*, not to adolescent age groups.

3.6.4 Human Gold-Label Validation

Validation integrates: coding by trained annotators / supervisors under written guidelines; quantitative agreement with gold labels; qualitative analysis of misclassified cases; and periodic guideline refinement. Adolescent participant feedback and adolescent-psychology expert diagnosis are *not* claimed as completed components of this thesis.

3.7 Integration Strategy

While each model can operate independently, the thesis evaluates CAS-RAG (Model 2) and the comparator pipelines as the primary empirical thread. The integration patterns below are **conceptual design options** for future work; Model 1 consensus metrics are *not* wired into CAS-RAG gating in the submitted implementation. [panel]

While each model can operate independently, they are designed to integrate seamlessly. Several integration strategies are possible:

3.7.1 Sequential Integration

In sequential integration, models are applied in a pipeline: Model 1 $\rightarrow$ Model 2 $\rightarrow$ Model 3. Model 1 produces initial classifications with consensus scores and hallucination flags. Model 2 uses Model 1's confidence and consensus metrics to inform gating decisions. Model 3 applies scenario-conditioned classification (prompts/prototypes under the video scenario) to Model 2's output. This approach allows each model to build upon previous results, though errors may propagate.

3.7.2 Parallel Integration

In parallel integration, all three models process the input simultaneously. An aggregation mechanism then combines results using weighted voting, confidence-weighted combina-

tion, or consensus analysis across all models. Parallel integration enables faster processing and reduces error propagation.

3.7.3 Hybrid Integration

Hybrid integration combines sequential and parallel elements. For example, Model 1 and Model 3 preprocessing operate in parallel, with Model 3 providing scenario-conditioned preprocessing that enhances Model 1's input. Model 2 then uses outputs from both to inform gating decisions.

3.7.4 Selective Integration

Selective integration activates models based on input characteristics. High-confidence, clear expressions may only use Model 1; ambiguous or context-dependent expressions may activate Model 2 (RAG); scenario grounding may trigger Model 3's conditioned prompts; challenging cases may use all three models. This approach optimises efficiency by using models only when beneficial.

3.7.5 Integration Trade-offs

Trade-offs include: accuracy vs. efficiency (more comprehensive integration improves accuracy but increases cost), error propagation (sequential vs. parallel), complexity (sophisticated integration increases maintenance), and deployment flexibility (modular vs. tightly integrated systems).

3.8 Evaluation Approach

This section outlines the evaluation approach employed in subsequent chapters to assess each model and the integrated framework.

3.8.1 Evaluation Framework

The evaluation assesses multiple dimensions: **Classification Accuracy** (precision, recall, F1-score, and accuracy for routing and emotion categories); **Reliability and Consistency** (inter-model agreement where used, agreement with gold labels); **Hallucination Detection Performance** (disagreement-based proxy rates from the offline RQ1 consensus audit; Section 6.7)[panel]; **Efficiency Metrics** (retrieval frequency, overhead, latency, relative cost for Model 2 / CAS-RAG); **Validity Evidence** (agreement with human gold labels under a written schema); and **Error Analysis** (repeated misrouting and mislabelling habits).

3.8.2 Comparative Evaluation

The evaluation includes: baseline comparisons (comparator Algorithms 1–4, individual LLM paths), ablation-style component contrasts (routing gates, RAG on/off), cross-pipeline comparison (different gating policies under a shared experimental contract), and integration analysis where applicable.

3.8.3 Dataset and Ground Truth

Evaluation employs datasets of comments associated with videos related to adolescent topics with human-annotated routing and emotion labels. The datasets include diverse comment types and a range of expressions (clear, ambiguous, context-dependent). Dataset names refer to video topics, not commenter demographics. Ground truth is established through trained annotator / supervisor coding and inter-annotator agreement assessment where available. Adolescent participant validation is not claimed as a completed component.

3.8.4 Experimental Design

The evaluation follows controlled experiments under a shared scenario, label schema, and retrieval corpus where applicable, with detailed error analysis. Quantitative metrics are complemented by qualitative review of misclassified comments where reported.

3.9 Summary

This chapter positioned the research problem and solution space. The empirical thread of the thesis proceeds as follows: Chapter 4 defines the experimental methodology for comparator Algorithms 1–4, scenario-conditioned prototypes, and the CAS-RAG gate; Chapter 5 develops the CAS-RAG implementation (including the `CAS_RAG_Project` platform with OpenAI, Gemini, and Ollama LLM backends) with worked numeric examples; Chapter 6 reports Tutoring and Bullying in-depth emotion evaluations (Algorithms 1–4 versus CAS-RAG), the offline RQ1 multi-LLM consensus audit (Section 6.7), and complementary CAS-RAG Layer 1 / cost topic-diversity checks on six annotated datasets. Chapter 7 discusses contributions, limitations, ethics, and future work. Weighted multi-provider ensembles, in-pipeline consensus filtering, and richer context-enhanced labelling stacks remain optional extensions beyond the submitted offline RQ1 audit.[panel]

CHAPTER 4

COMPARATOR PIPELINES, PROTOTYPES, AND EXPERIMENTAL METHODOLOGY

This chapter specifies the experimental methodology used to compare four LLM- and linguistics-based comparator pipelines (Algorithms 1–4) against the proposed **CAS-RAG** (Context-Aware Selective RAG) gate on a single canonical case study: a hidden-camera tutoring social-experiment video. All pipelines share the same video scenario text, the same non-clinical emotion taxonomy, the same retrieval corpus where applicable, and—where similarity is used—the same scenario-conditioned prototype sets. Architectural differences therefore isolate routing policy and retrieval gating rather than confounding inputs.

Section 4.1 formulates the task and shared inputs. Sections 4.2–4.3.4 define the scenario, emotion labels, prototypes, and retrieval protocol. Sections 4.4–4.7 present Algorithms 1–4 with prompts, routing logic, and pseudocode. Section 4.8 details CAS-RAG's semantic-first two-layer design and its distinction from the comparators. Section 4.9 summarises the fair-comparison matrix and evaluation-metrics linkage. Full architectural walkthroughs appear in Chapter 5. Chapter 6 reports (i) routing evaluation across six annotated datasets, (ii) in-depth emotion evaluation on Tutoring and Bullying, and (iii) com-

parator methodology for Algorithms 1–4 and CAS-RAG.Chapter 6 reports (i) the primary Tutoring in-depth comparison of Algorithms 1–4 and CAS-RAG and (ii) complementary pooled CAS-RAG Layer 1 topic-diversity metrics on six annotated datasets.[panel]

4.1 Experimental Setting and Task Formulation

4.1.1 Research objective

The empirical programme evaluates *when* retrieval should run and *what* non-clinical emotion label a comment expresses under a fixed education / tutoring-video scenario (a video topic often discussed in relation to adolescents). Four comparator pipelines (Algorithms 1–4) answer the emotion-classification question directly, optionally invoking retrieval. CAS-RAG answers a complementary question: whether retrieval is necessary, feasible, or should be suppressed—before any downstream emotion model is applied. Full five-pipeline comparison and qualitative error analysis are reported for Tutoring (Chapter 6); other annotated topic sets support complementary CAS-RAG Layer 1 topic-diversity checks only.

4.1.2 Shared inputs

Every pipeline receives the same tuple $(c, s, \mathscr{D})$:

Comment c A user-generated YouTube-style comment string drawn from the tutoring social-experiment video.

Short Video scenario s A fixed natural-language description of the hidden-camera tutoring setup (Section 4.2).

Script store $\mathscr{D}$ Chunked transcript or script segments indexed for dense retrieval when a pipeline's RAG branch is taken.

4.1.3 Pipeline outputs

Outputs differ by design and must not be conflated when interpreting results (Table 4.1).

Table 4.1: *Primary outputs by pipeline type.*

Pipeline	Primary output
Algorithms 1–4	Non-clinical **emotion labels** in {Shock, Disbelief, Anger, Frustration, Empathy, None}, with designated branches for Ambiguous, or *discarded* cases
CAS-RAG	**Retrieval-routing states**: semantic category $\in$ {explicit, ambiguous, context_dependent}; `rag_needed`; `infeasible`; and retrieval execution decisions prior to downstream emotion classification

4.1.4 Fair-comparison protocol

Across Algorithms 1–4 and CAS-RAG, the following were held constant within each experimental run: the scenario string s, the emotion label definitions (Section 4.2.2), the retrieval corpus $\mathscr{D}$, the embedder checkpoint and top-k retrieval setting, and the fifteen scenario-conditioned prototypes (Section 4.3). Comparator differences arise solely from (i) the first routing gate (LLM reference detection, spaCy linguistics analysis, or semantic similarity), (ii) which prototype subsets participate in similarity, and (iii) whether CAS-RAG performs per-comment LLM gating. Reproducibility requires reporting the exact scenario variant, model identifiers, and retrieval hyperparameters alongside each results table (Section 4.9).

4.2 Shared Video Scenario and Emotion Taxonomy

4.2.1 Canonical tutoring scenario

All prompts, prototype authoring, and spaCy grounding tests use the same scenario description. The full paragraph below was fixed for prototype design and comparator evaluation on the tutoring social-experiment video:

> A hidden-camera social experiment is set up in a public diner where a child actor struggles with basic math while a tutor becomes increasingly rude and verbally harsh. The situation is designed to test how other children nearby will respond. Some hesitate at

> first, but several step in to help the struggling student, offering both academic support and emotional encouragement. The experiment reveals that even young children can recognise unfair treatment and act with empathy, highlighting the importance of kindness, confidence, and standing up against bullying behaviour in educational settings.

A one-sentence header variant may appear in space-constrained prompt templates; any reported result must state which variant was fixed for that run.

4.2.2 Non-clinical emotion taxonomy

Algorithms 1–4 apply the same label inventory through Prompts P2–P4. Models were instructed to use **only** information in c, s, and—when the RAG branch is active—retrieved chunks from $\mathscr{D}$. Clinical or diagnostic labels were excluded by design.

Table 4.2: *Non-clinical emotion labels for comparator pipelines.*

Label	Definition
Shock	Strong surprise, alarm, or emotional disturbance
Disbelief	Difficulty accepting, believing, or trusting what is happening
Anger	Irritation, outrage, hostility, or moral condemnation
Frustration	Annoyance, disappointment, helplessness, or emotional dissatisfaction
Empathy	Compassion, emotional concern, support, kindness, or understanding toward others
None	No dominant emotion from the set above
Ambiguous	Emotional meaning ambiguous, target unclear, or link to scenario cannot be established (no-RAG branch, where enabled)
Ambiguous	Meta-outcome of ambiguity-detection: no reliable emotion assignment

These labels operationalise *affective reaction analysis* for online video comments; they are distinct from CAS-RAG's three-way semantic routing categories (explicit, ambiguous,

context_dependent), which govern retrieval policy rather than final emotion assignment.

4.3 Scenario-Conditioned Prototypes and Retrieval Protocol

4.3.1 Rationale for scenario-conditioned, comment-like prototypes

Semantic similarity in Algorithms 3–4 and Layer 1 of CAS-RAG is anchored by fifteen short sentences—five per class—authored from recurring themes in real comments on the tutoring video (kindness praise, staging ethics, named participants, quoted lines). Prototypes are scenario-conditioned because they reflect lexical and pragmatic patterns that arise *in reaction to this video*, not generic emotion words in isolation. They are comment-like so that embedding-space geometry compares c to plausible viewer utterances rather than abstract category definitions, improving alignment with informal YouTube register.

4.3.2 Prototype inventory

Explicit (self-contained reactions):

(1) The way those kids jumped in proves empathy can beat a bad adult example every time.

(2) This is exactly why kindness and confidence matter more than any worksheet.

(3) Restores a little faith—kids noticing unfairness and refusing to shrug it off.

(4) Great reminder that bullying in learning spaces is never okay, no matter the adult's excuse.

(5) Wholesome watch; we need way more classrooms that normalise standing up for someone struggling.

Ambiguous (unresolved even with full video context):

(1) Cute message but I'm still uneasy about hidden-camera setups in public places.

(2) Mixed feelings—is this stitched like reality TV or genuinely spontaneous?

(3) Inspiring edits, though I wish we knew consent and editing context upfront.

(4) Not sure the lesson needs a diner stunt; feels performative even if the vibe is sweet.

(5) Hard to tell what's heartfelt vs what producers nudged; still liked the takeaway.

Context-dependent (require scene-linked grounding):

(1) Aubrey calmly calling out that tutor felt more composed than half the adults online.

(2) When Harper walked over with the milkshake energy I melted—pure emotional support.

(3) "Everybody's a hero"—that kid's line was the gut-punch moment of the whole bit.

(4) The waitress clocking "oh right he's acting" cracked the fourth wall harder than WWYD usually does.

(5) Rough tutor banging patience at the booth while strangers' kids leaned in—is peak WWYD diner tension.

4.3.3 Prototype usage by pipeline

Algorithms 3 and 4 invoke similarity against the **Explicit** and **Ambiguous** lists only (ten sentences) via Prompt P5 or an embedding surrogate with the same decision boundary. CAS-RAG Layer 1 uses all fifteen prototypes and assigns a third class, `context_dependent`, which has no analogue in the two-class similarity step of Algorithms 3–4. This asymmetry is intentional: comparators test whether binary explicit/ambiguous similarity suffices for emotion routing, whereas CAS-RAG treats context-dependence as a first-class retrieval signal.

4.3.4 Retrieval protocol

When any pipeline triggers retrieval, the query is the comment c (optionally concatenated with salient keywords from s); the top-k chunks from $\mathscr{D}$ populate `{retrieved_context}` for Prompt P2. The same $\mathscr{D}$, dense embedder, and k were used for every RAG-enabled branch within a given experimental run. LLM temperature was set low (0–0.2) where configurable. Each classification prompt required a fixed response template (`Label: + Explanation:`) to support automatic parsing and audit.

4.4 Algorithm 1: LLM–LLM Comparator Pipeline

4.4.1 Purpose and relation to other methods

Algorithm 1 is the pure-LLM comparator: every routing decision before emotion classification is made by a language model. It establishes an upper bound on LLM-mediated gating cost and serves as the reference against which spaCy-first (Algorithm 2), similarity-augmented (Algorithms 3–4), and semantic-first CAS-RAG alternatives are judged under identical s, $\mathscr{D}$, and label definitions.

4.4.2 Architecture overview

As illustrated in Figure 4.1, the pipeline comprises three conceptual stages: (i) unknown-reference detection, (ii) RAG versus no-RAG emotion classification, and (iii) semantic-ambiguity resolution when the no-RAG path yields `Ambiguous`, with human verification before discard or finalised.

4.4.3 Routing logic

1. **Unknown reference (P1).** If the LLM judges that c contains unresolved referents relative to s, retrieve from $\mathscr{D}$ and apply Prompt P2.

2. **No reference (P1 = No).** Apply Prompt P3 without retrieval.

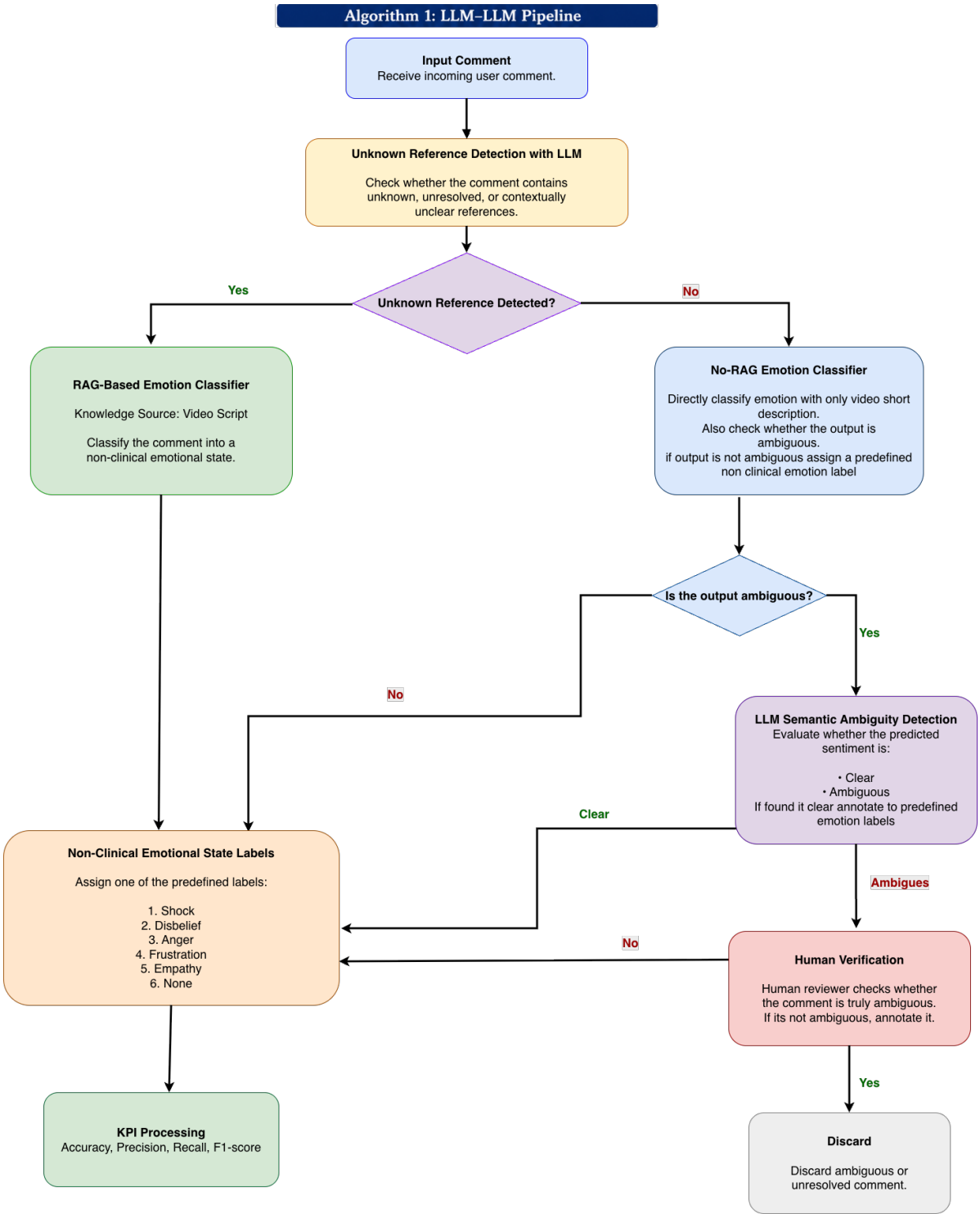


Figure 4.1: *Algorithm 1 end-to-end flow: Prompt P1 gates retrieval; Prompts P2–P3 classify emotion; Prompt P4 resolves ambiguity on the no-RAG branch when* `Ambiguous` *is returned.*

3. **Ambiguous on no-RAG path.** If P3 returns `Ambiguous`, invoke Prompt P4; if the outcome is `Ambiguous`, route to required human verification (discard if the comment remains ambiguous, else assign a predefined non-clinical emotion label); if P4 returns a clear label, assign that label directly.

4.4.4 Prompt P1 — Unknown reference detection

```
You are an unknown reference detection system for non-clinical emotional
reaction analysis.

Your task is to determine whether the following user comment contains any
unknown, unresolved, or contextually unclear references.

Use ONLY:
1. The explicitly available information in the comment.
2. The provided video scenario.

Do NOT infer missing identities, unstated contextual links, or external
context beyond the provided scenario.

Short Video Scenario:
A hidden camera social experiment in a public diner shows a tutor becoming
verbally harsh toward a struggling child while nearby children react by
offering empathy, emotional support, encouragement, and academic help.
A reference should be considered unknown or unresolved when:
• the identity of a person, group, or target is unclear,
• a pronoun cannot be reliably linked to a known entity,
• the emotional target is implicit but not identifiable,
• or the comment depends on missing contextual information that is not
```

```
available in the comment or scenario.

Example:
Comment: ‘‘She handled that really well.’’
Unknown Reference:
Yes

Explanation:
The identity of ‘‘she’’ is unclear.

Instructions:
• Check whether the comment contains unclear pronouns, unnamed
entities, implicit targets, or unresolved references.
• Focus only on detecting unknown references.
• Do not classify emotions or sentiment.
• If the comment contains an unresolved or unclear reference,
output "Yes".
• If all references can be reliably and explicitly resolved using
the comment and provided scenario, output "No"
• Provide a concise explanation for the result.

Input Comment:
{comment}

Respond ONLY in the following format:
Unknown Reference: <Yes / No>
```

```
Explanation:
```

4.4.5 Prompt P2 — RAG-based emotion classification

```
You are an emotion classification system designed to identify non-clinical
emotional reactions
in user comments related to RAG knowledge source of video script.
Your task is to classify the emotional reaction expressed in the user comment
using ONLY:
1. The explicitly available information in the comment.
2. The provided video scenario.
3. The retrieved video script context.
Do NOT infer hidden intentions, unstated emotions, or external context beyond
the provided scenario and retrieved context.

Video Scenario:
A hidden camera experiment is set up in a public diner where a child actor
struggles with basic math while a tutor becomes increasingly rude and verbally
harsh.
The situation is designed to test how other children nearby will respond.
Some hesitate at first, but several step in to help the struggling student,
offering both academic support and emotional encouragement.
The experiment reveals that even young children can recognise unfair treatment
and act with empathy,
highlighting the importance of kindness, confidence, and standing up against
bullying behaviour in educational settings.

Predefined non-clinical emotional state labels:
```

- Shock
- Disbelief
- Anger
- Frustration
- Empathy
- None

Label Definitions:

Shock:
A strong emotional reaction of surprise, alarm, or emotional disturbance.

Disbelief:
A reaction expressing difficulty accepting, believing, or trusting what is happening.

Anger:
A reaction expressing irritation, outrage, hostility, or moral condemnation.

Frustration:
A reaction expressing annoyance, disappointment, helplessness, or emotional dissatisfaction.

Empathy:
A reaction expressing compassion, emotional concern, support, kindness, or understanding toward others.

```
None:
Use this label when the comment's emotional reaction does not belong to any
of the categories below: Shock, Disbelief, Anger, Frustration, Empathy.
This label is included to avoid forcing the model to assign an emotional
category when no meaningful emotion is explicitly expressed.

Instructions:
• Use only explicitly available evidence.
• Do not assume missing context beyond the retrieved context.
• Focus only on non-clinical emotional reactions.
• If multiple emotions appear, choose the dominant explicitly expressed emotion.
• Use the retrieved context to resolve unclear references or contextual ambiguity.
• Provide a concise explanation for the classification result.
Input Comment:
{comment}

Retrieved Context:
{retrieved context}

Respond ONLY in the following format:
Emotion Label: <Shock / Disbelief / Anger / Frustration / Empathy / None>

Explanation:
```

4.4.6 Prompt P3 — No-RAG emotion classification

```
You are an emotion classification system designed to identify non-clinical
emotional reactions in user comments related to short video scenarios.
Your task is to classify the emotional reaction expressed in the user
```

```
comment using ONLY:
1. The explicitly available information in the comment.
2. The provided short video scenario.

Do NOT infer hidden intentions, unstated emotions, or external context
beyond the provided scenario.

Video Scenario:
A hidden camera social experiment in a public diner shows a tutor becoming
verbally harsh toward a struggling child while nearby children react by
offering empathy, emotional support, encouragement, and academic help.

Predefined non-clinical emotional state labels:
• Shock
• Disbelief
• Anger
• Frustration
• Empathy
• None
• Ambiguous

Label Definitions:
Shock:
A strong emotional reaction of surprise, alarm, or emotional disturbance.

Disbelief:
A reaction expressing difficulty accepting, believing, or trusting what
```

is happening.

Anger:
A reaction expressing irritation, outrage, hostility, or moral condemnation.

Frustration:
A reaction expressing annoyance, disappointment, helplessness, or emotional dissatisfaction.

Empathy:
A reaction expressing compassion, emotional concern, support, kindness, or understanding toward others.

None:
Use this label when the comment's emotional reaction does not belong to any of the categories below: Shock, Disbelief, Anger, Frustration, Empathy. This label is included to avoid forcing the model to assign an emotional category when no meaningful emotion is explicitly expressed.

Ambiguous: Use this label when:
- the emotional meaning is ambiguous,
- the emotional target is unclear,
- the comment is too generic or weakly expressed,
- or the connection between the comment and the video scenario cannot be reliably established.

Examples of Ambiguous comments:

- "I love it"
- "Amazing"
- "Wow"
- "So good"

These comments may contain a positive reaction, but they do not provide enough contextual evidence to determine a reliable emotional interpretation related to the video scenario.

Instructions:

- Use only explicitly available evidence.
- Do not assume missing context.
- Focus only on non-clinical emotional reactions.
- If multiple emotions appear, choose the dominant explicitly expressed emotion.
- Provide a concise explanation for the classification result.

Input Comment:

{comment}

Respond ONLY in the following format:

Emotion Label: <Shock / Disbelief / Anger / Frustration / Empathy / None / Ambiguous>

Explanation:

4.4.7 Prompt P4 — Semantic ambiguity-aware classification

Applied when Prompt P3 returns `Ambiguous`; a second LLM runs the verification prompt below.

```
You are a verification system designed to verify the user non-clinical
emotional  reactions comments related to social experiment short video
scenarios is ambiguous or not.


Your task is to:
1. Determine whether the emotional meaning expressed in the user comment is
clear or ambiguous.
2. If the emotional meaning is clear, classify the dominant emotional reaction
using the predefined labels below.

Use ONLY:
1. The explicitly available information in the comment.
2. The provided video scenario.

Do NOT infer hidden intentions, unstated emotions, or external context beyond
the provided scenario.

Video Scenario:
A hidden camera social experiment in a public diner shows a tutor becoming
verbally harsh toward a struggling child while nearby children react by
offering empathy, emotional support, encouragement, and academic help.

Predefined non-clinical emotional state labels:
• Shock
• Disbelief
• Anger
```

- Frustration
- Empathy
- None

Label Definitions:

Shock:
A strong emotional reaction of surprise, alarm, or emotional disturbance.

Disbelief:
A reaction expressing difficulty accepting, believing, or trusting what is happening.

Anger:
A reaction expressing irritation, outrage, hostility, or moral condemnation.

Frustration:
A reaction expressing annoyance, disappointment, helplessness, or emotional dissatisfaction.

Empathy:
A reaction expressing compassion, emotional concern, support, kindness, or understanding toward others.

None:
Use this label when the comment's emotional reaction does not belong to any of the categories below: Shock, Disbelief, Anger, Frustration, Empathy.

This label is included to avoid forcing the model to assign an emotional category when no meaningful emotion is explicitly expressed.

A comment should be considered Ambiguous when:

- the emotional meaning is unclear,
- the emotional target is unclear,
- the comment is too generic or weakly expressed,
- or the connection between the comment and the video scenario cannot be reliably established.

Examples of ambiguous comments:

- "I love it"
- "Amazing"
- "Wow"
- "So good"

These comments may contain a positive reaction, but they do not provide enough contextual evidence to determine a reliable emotional interpretation related to the video scenario.

Instructions:

- Use only explicitly available evidence.
- Do not assume missing context.
- Focus only on non-clinical emotional reactions.
- If multiple emotions appear, choose the dominant explicitly expressed emotion.
- If the emotional meaning cannot be reliably interpreted, return

```
"Ambiguous".
• If the emotional meaning is clear, classify the emotion using the
predefined labels.
• Provide a concise explanation for the result.

Input Comment:
{comment}

Respond ONLY in the following format:
Emotion Label: <Shock / Disbelief / Anger / Frustration / Empathy /
None / Ambiguous>

Explanation:
```

4.4.8 Pseudocode

```
Algorithm 1: LLM-LLM Emotion Pipeline
Input: comment c, scenario s, script store D, human reviewer H
Output: emotion label y or DISCARDED

1:  r <- LLM(P1, c, s)
2:  if r.unknown_reference = Yes then
3:      tau <- RETRIEVE(D, query = c)
4:      return LLM(P2, c, s, tau)
5:  else
6:      y <- LLM(P3, c, s)
7:      if y = Ambiguous then
8:          a <- LLM(P4, c, s)
9:          if a = Ambiguous then
```

```
10:             if H(c) confirms ambiguous then
11:                 return DISCARDED
12:             else
13:                 y <- emotion label from H(c)
14:                     // Shock, Disbelief, Anger, Frustration,
15:                     // Empathy, or None
16:             end if
17:         else
18:             y <- a  // P4 returned a predefined non-clinical label
19:         end if
20:     end if
21:     return y
22: end if
```

Algorithm 2 reuses Prompts P2–P4 unchanged; Algorithm 3 adds semantic similarity before routing; Algorithm 4 applies similarity without spaCy as the first gate. CAS-RAG replaces the entire per-comment LLM gating chain with embedding-based Layer 1 (Section 4.8).

4.5 Algorithm 2: spaCy–LLM Comparator Pipeline

4.5.1 Purpose and architectural substitution

Algorithm 2 isolates the contribution of linguistic preprocessing by replacing Prompt P1 with spaCy-based referential analysis. Emotion classification remains LLM-driven via Prompts P2–P4, preserving fair comparison with Algorithm 1 on all post-routing stages. The research question is whether structured NLP features can approximate LLM reference detection with lower cost and comparable retrieval exposure.

4.5.2 Architecture overview

Figure 4.2 shows that the first gate is no longer an LLM call but a spaCy feature extractor followed by a deterministic `LINGUISTIC_NEEDS_RAG` rule. All subsequent branches match Algorithm 1.

4.5.3 Linguistic features and routing rule

For comment c, spaCy extracts noun phrases, pronoun tokens from a closed set (e.g. *this, that, it, they, he, she*), and person-denoting noun phrases. A mention is treated as an **unknown referent** when a generic person-denoting phrase (e.g. *guy, kid, lady*) lacks modifiers that disambiguate it against the scenario text s. Internal scoring may penalise pronouns without local descriptive context.

Routing rule (conceptual): if an unresolved person-like reference remains set `needs_rag = true` and execute Prompt P2 after retrieval; otherwise follow the no-RAG path with Prompt P3 and, when `Ambiguous` is returned, Prompt P4 with required human verification on `Ambiguous` outcomes (as in Algorithm 1). Prompt P1 is **not** invoked.

4.5.4 Prompts

Prompts P2, P3, and P4 are **identical** to Section 4.4. No additional prompt family is introduced at the routing stage.

4.5.5 Pseudocode

```
Algorithm 2: spaCy-LLM Emotion Pipeline
Input: comment c, scenario s, script store D, spaCy model nlp,
       human reviewer H
Output: emotion label y or DISCARDED

1:  L <- SPACY_ANALYSE(c, nlp)
```

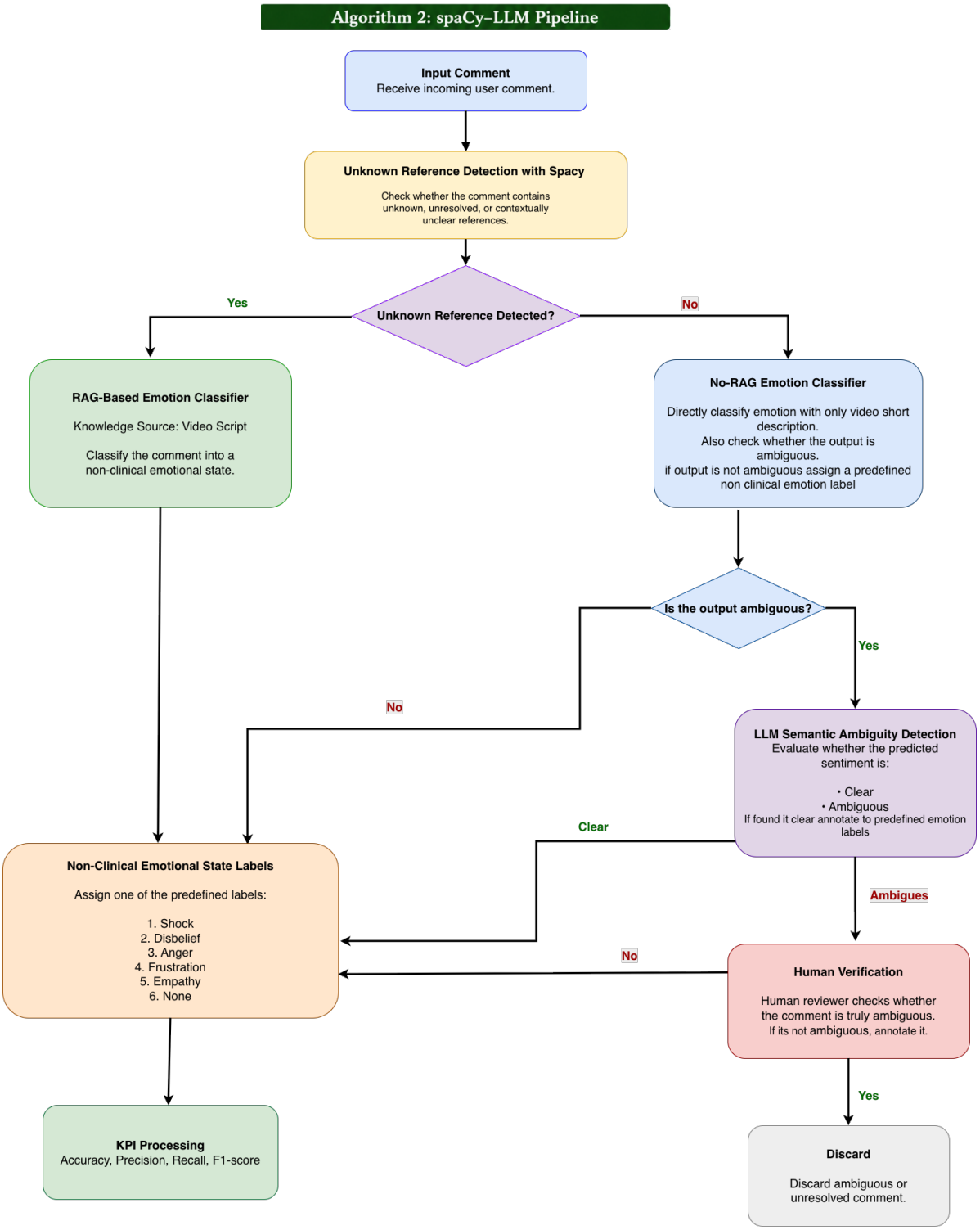


Figure 4.2: *Algorithm 2 replaces Prompt P1 with spaCy linguistic analysis; Prompts P2–P4 are unchanged from Algorithm 1.*

```
2:  needs_rag <- LINGUISTIC_NEEDS_RAG(L, s)
3:  if needs_rag then
4:      tau <- RETRIEVE(D, c)
5:      return LLM(P2, c, s, tau)
6:  else
7:      y <- LLM(P3, c, s)
8:      if y = Ambiguous then
9:          a <- LLM(P4, c, s)
10:         if a = Ambiguous then
11:             if H(c) confirms ambiguous then
12:                 return DISCARDED
13:             else
14:                 y <- emotion label from H(c)
15:             end if
16:         else
17:             y <- a
18:         end if
19:     end if
20:     return y
21: end if
```

Relative to Algorithm 1, the only architectural delta is line 1–2; relative to Algorithm 3, spaCy precedes rather than accompanies semantic similarity.

4.6 Algorithm 3: spaCy–Semantic Similarity Comparator Pipeline

4.6.1 Purpose

Algorithm 3 tests whether explicit/ambiguous prototype similarity, combined with spaCy structural cues, improves routing before expensive LLM emotion calls. The reference implementation in `CAS_RAG_Project` uses `SemanticSimilarityEngine` (SentenceTransformer + seven-metric fusion) for the two-way label σ; Prompt P5 documents the equivalent LLM behaviour. Non-retrieval branches use Prompt P7 and semantic ambiguity-aware Prompt P6.

4.6.2 Architecture overview

Figure 4.3 depicts parallel signals: spaCy features ℓ and semantic label $\sigma \in \{\texttt{Explicit}, \texttt{Ambiguous}\}$ from the ten prototype sentences in Section 4.3. The function `ROUTE_RAG` fuses ℓ, σ, and s to decide retrieval.

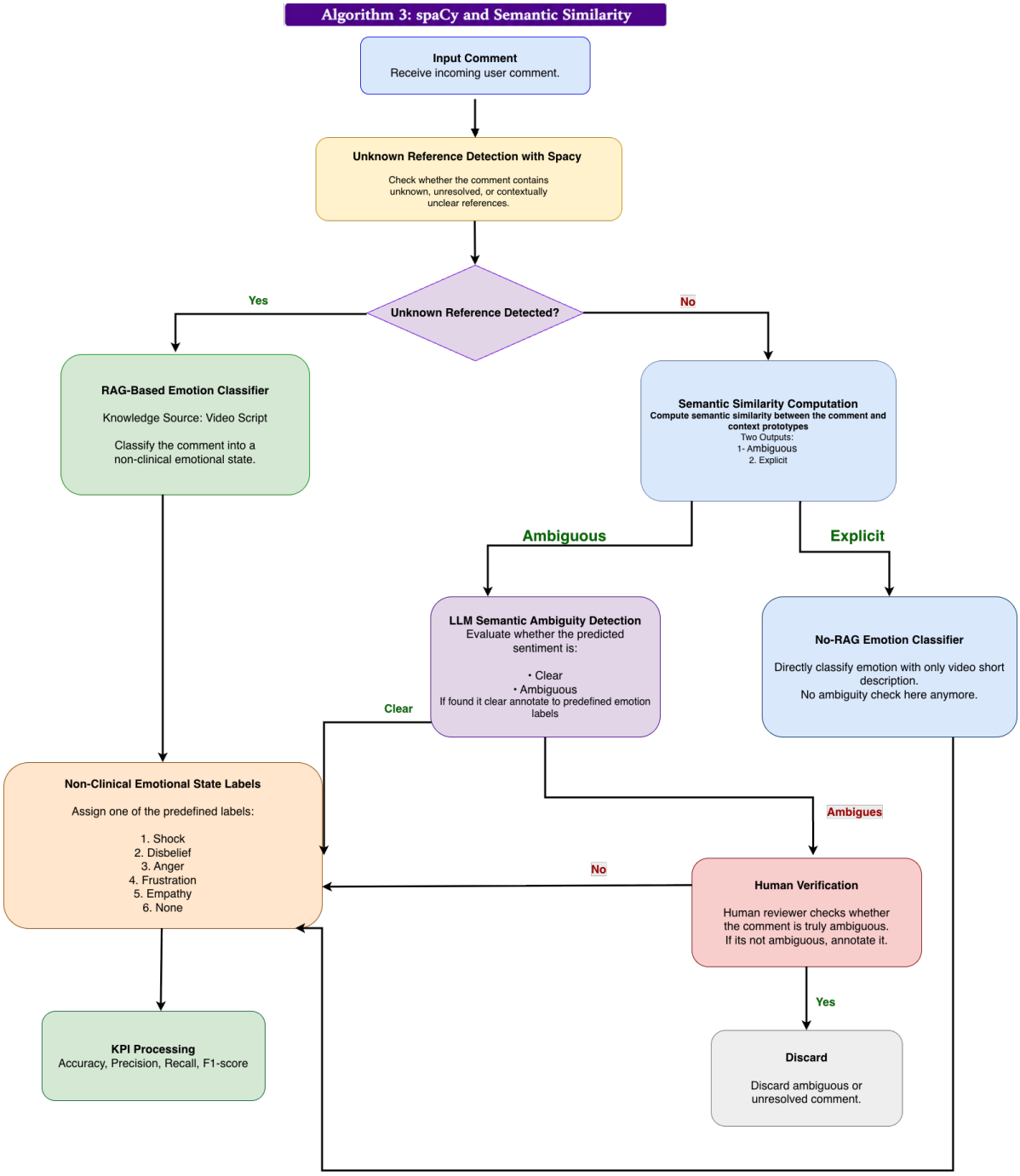


Figure 4.3: *Algorithm 3: spaCy analysis and Explicit/Ambiguous similarity jointly determine retrieval; emotion labelling uses Prompts P2, P6, and P7.*

4.6.3 Prompt P5 — Semantic similarity (Explicit vs Ambiguous)

Prompt P5 operationalises membership in the explicit or ambiguous prototype manifold using only c, s, and the ten-sentence lists. Implementations may substitute cosine similarity between embed(c) and class centroids; the prompt below defines the intended LLM behaviour.

```
You are a semantic similarity computation system for non-clinical
emotional reaction analysis in user comments related to short video
scenarios.

Your task is to determine whether the semantic meaning expressed
in the user comment is more semantically aligned with:
1. Explicit emotional reaction prototypes
or
2. Ambiguous emotional reaction prototypes

Use ONLY:
1. The explicitly available information in the comment.
2. The provided video scenario.
3. The provided context prototypes.

Do NOT infer hidden intentions, unstated emotions,
or external context beyond the provided scenario and prototypes.

Video Scenario:
A hidden camera social experiment in a public diner shows a tutor becoming
verbally harsh toward a struggling child while nearby children react
by offering empathy, emotional support, encouragement, and academic help.
```

Definitions:

Explicit:

A comment should be considered Explicit when the emotional meaning, emotional target, or emotional reaction is clearly and directly expressed and can be reliably interpreted from the comment and scenario context.

Ambiguous:

A comment should be considered Ambiguous when:

- the emotional meaning is weak, mixed, generic, indirect, or unclear,
- the emotional target is not identifiable,
- the reaction is only loosely implied,
- the comment focuses on uncertainty, production authenticity, editing concerns, or contextual doubt,
- or the connection between the comment and the video scenario cannot be reliably established.

Explicit prototypes:

(1) "The way those kids jumped in proves empathy can beat a bad adult example every time."

(2) "This is exactly why kindness and confidence matter more than any worksheet."

(3) "Restores a little faith-kids noticing unfairness and refusing to shrug it off."

(4) "Great reminder that bullying in learning spaces is never okay, no matter the adult's excuse."

(5) "Wholesome watch; we need way more classrooms that normalise standing up for someone struggling."

Ambiguous prototypes:

(1) "Cute message but I'm still uneasy about hidden-camera setups in public places."

(2) "Mixed feelings-is this stitched like reality TV or genuinely spontaneous?"

(3) "Inspiring edits, though I wish we knew consent and editing context upfront."

(4) "Not sure the lesson needs a diner stunt; feels performative even if the vibe is sweet."

(5) "Hard to tell what's heartfelt vs what producers nudged; still liked the takeaway."

Instructions:

- Compare the semantic meaning of the input comment against the provided prototype categories.
- Focus only on semantic similarity between the comment and the prototype groups.
- Use only explicitly available evidence.
- Do not assume missing context.
- Do not classify specific emotions such as anger, empathy, or frustration.
- If the comment is more semantically aligned with clear, direct, emotionally interpretable reactions similar to the Explicit prototypes, output ‘‘Explicit’’.
- If the comment is more semantically aligned with uncertain, mixed,

```
  indirect, weak, or contextually unclear reactions similar to the
  Ambiguous prototypes, output ‘‘Ambiguous’’.
  • Provide a concise explanation for the result.

Input Comment:
{comment}

Respond ONLY:
Semantic Similarity Label: <Explicit | Ambiguous>

Explanation: <text>
```

4.6.4 Prompt P6 — Semantic Ambiguity-Aware Emotion Classification Prompt

```
You are a verification system designed to verify the user
non-clinical emotional reactions comments related to social
experiment short video scenarios is ambiguous or not.

Your task is to:
1. Determine whether the emotional meaning expressed in the user
comment is clear or ambiguous.
2. If the emotional meaning is clear, classify the dominant
emotional reaction using the predefined labels below.

Use ONLY:
1. The explicitly available information in the comment.
2. The provided video scenario.
```

Do NOT infer hidden intentions, unstated emotions, or external context beyond the provided scenario.

Video Scenario:
A hidden camera social experiment in a public diner shows a tutor becoming verbally harsh toward a struggling child while nearby children react by offering empathy, emotional support, encouragement, and academic help.

Predefined non-clinical emotional state labels:

- Shock
- Disbelief
- Anger
- Frustration
- Empathy
- None

Label Definitions:

Shock:
A strong emotional reaction of surprise, alarm, or emotional disturbance.

Disbelief:
A reaction expressing difficulty accepting, believing, or trusting what is happening.

```
Anger:
A reaction expressing irritation, outrage, hostility, or moral
condemnation.

Frustration:
A reaction expressing annoyance, disappointment, helplessness, or
emotional dissatisfaction.

Empathy:
A reaction expressing compassion, emotional concern, support,
kindness, or understanding toward others.

None:
Use this label when the comment's emotional reaction does not belong
to any of the categories below:
Shock, Disbelief, Anger, Frustration, Empathy.
This label is included to avoid forcing the model to assign an
emotional category when no meaningful emotion is explicitly expressed.

A comment should be considered Ambiguous when:
• the emotional meaning is unclear,
• the emotional target is unclear,
• the comment is too generic or weakly expressed,
• or the connection between the comment and the video scenario
cannot be reliably established.

Examples of ambiguous comments:
```

- ‘‘I love it’’
- ‘‘Amazing’’
- ‘‘Wow’’
- ‘‘So good’’

These comments may contain a positive reaction, but they do not provide enough contextual evidence to determine a reliable emotional interpretation related to the video scenario.

Instructions:

- Use only explicitly available evidence.
- Do not assume missing context.
- Focus only on non-clinical emotional reactions.
- If multiple emotions appear, choose the dominant explicitly expressed emotion.
- If the emotional meaning cannot be reliably interpreted, return ‘‘Ambiguous’’.
- If the emotional meaning is clear, classify the emotion using the predefined labels.
- Provide a concise explanation for the result.

Input Comment:

{comment}

Respond ONLY in the following format:

Emotion Label: <Shock / Disbelief / Anger / Frustration / Empathy /

None / Ambiguous>

Explanation:

4.6.5 Prompt P7 — No-RAG Emotion Classification Prompt (without checking ambiguity)

You are an emotion classification system designed to identify non-clinical emotional reactions in user comments related to short video scenarios.
Your task is to classify the emotional reaction expressed in the user comment using ONLY:
1. The explicitly available information in the comment.
2. The provided short video scenario.

Do NOT infer hidden intentions, unstated emotions, or external context beyond the provided scenario.
Video Scenario:
A hidden camera social experiment in a public diner shows a tutor becoming verbally harsh toward a struggling child while nearby children react by offering empathy, emotional support, encouragement, and academic help.
Predefined non-clinical emotional state labels:

- Shock
- Disbelief
- Anger
- Frustration
- Empathy
- None

Label Definitions:

Shock: A strong emotional reaction of surprise, alarm, or emotional disturbance.

Disbelief: A reaction expressing difficulty accepting, believing, or trusting what is happening.

Anger: A reaction expressing irritation, outrage, hostility, or moral condemnation.

Frustration: A reaction expressing annoyance, disappointment, helplessness, or emotional dissatisfaction.

Empathy: A reaction expressing compassion, emotional concern, support, kindness, or understanding toward others.

None: Use this label when the comment's emotional reaction does not belong to any of the categories below: Shock, Disbelief, Anger, Frustration, Empathy. This label is included to avoid forcing the model to assign an emotional category when no meaningful emotion is explicitly expressed.

Instructions:

- Use only explicitly available evidence.
- Do not assume missing context.
- Focus only on non-clinical emotional reactions.
- If multiple emotions appear, choose the dominant explicitly expressed emotion.
- Provide a concise explanation for the classification result.

Input Comment: {comment}

```
Respond ONLY in the following format:
Emotion Label: <Shock / Disbelief / Anger / Frustration /
Empathy /None>
Explanation:
```

4.6.6 Routing and remaining prompts

Prompt P2 follows Section 4.4. Prompt P1 is not used. For no-RAG paths, Algorithm 3 uses Prompt P7 when σ = `Explicit` (without `Ambiguous` as an output label). If σ = `Ambiguous`, Prompt P6 determines whether the emotional meaning is `Clear` (assign a predefined non-clinical label) or `Ambiguous` (required human verification: discard if the comment remains ambiguous, else assign a valid label). Prompt P3 is reserved for Algorithms 1–2 and is not used in Algorithm 3.

4.6.7 Pseudocode

```
Algorithm 3: spaCy-Semantic Similarity Pipeline
Input: c, s, D, nlp, prototype sets P_exp, P_vag, human reviewer H
Output: y or DISCARDED

1:  L <- SPACY_ANALYSE(c, nlp)
2:  sigma <- SEMANTIC_SIMILARITY(c, s, P_exp, P_vag)  // P5 or embed
3:  needs_rag <- LINGUISTIC_NEEDS_RAG(L, s) OR ROUTE_RAG(sigma, L)
4:  if needs_rag then
5:      tau <- RETRIEVE(D, c)
6:      return LLM(P2, c, s, tau)
7:  else
8:      if sigma = Ambiguous then
9:          a <- LLM(P6, c, s)
```

```
10:         if a = Ambiguous then
11:             if H(c) confirms ambiguous then
12:                 return DISCARDED
13:             else
14:                 return emotion label from H(c)
15:             end if
16:         else
17:             return a  // Clear: predefined non-clinical label
18:         end if
19:     else
20:         return LLM(P7, c, s)  // sigma = Explicit
21:     end if
22: end if
```

Algorithm 4 removes spaCy from the first gate and applies σ before RAG routing (Section 4.7). CAS-RAG instead performs three-class semantic routing with seven metrics and defers spaCy to context-dependent comments only (Section 4.8).

4.7 Algorithm 4: LLM–Semantic Similarity Comparator Pipeline

4.7.1 Purpose

Algorithm 4 evaluates hybrid **LLM + prototype similarity** routing without spaCy as the first gate. Semantic similarity (Prompt P5 or embeddings) precedes RAG/no-RAG emotion classification, testing whether explicit/ambiguous alignment alone can substitute for both LLM reference detection (Algorithm 1) and spaCy-first analysis (Algorithm 2–3).

4.7.2 Architecture overview

As shown in Figure 4.4, σ is computed first. Comments classified as `Ambiguous` by similarity are handled by Prompt P6: a `Clear` outcome yields a predefined non-clinical label; an `Ambiguous` outcome proceeds to required human verification (discard if still ambiguous, else assign a valid label). Comments with σ = `Explicit` follow `ROUTE_TO_RAG` and Prompts P2 or P7.

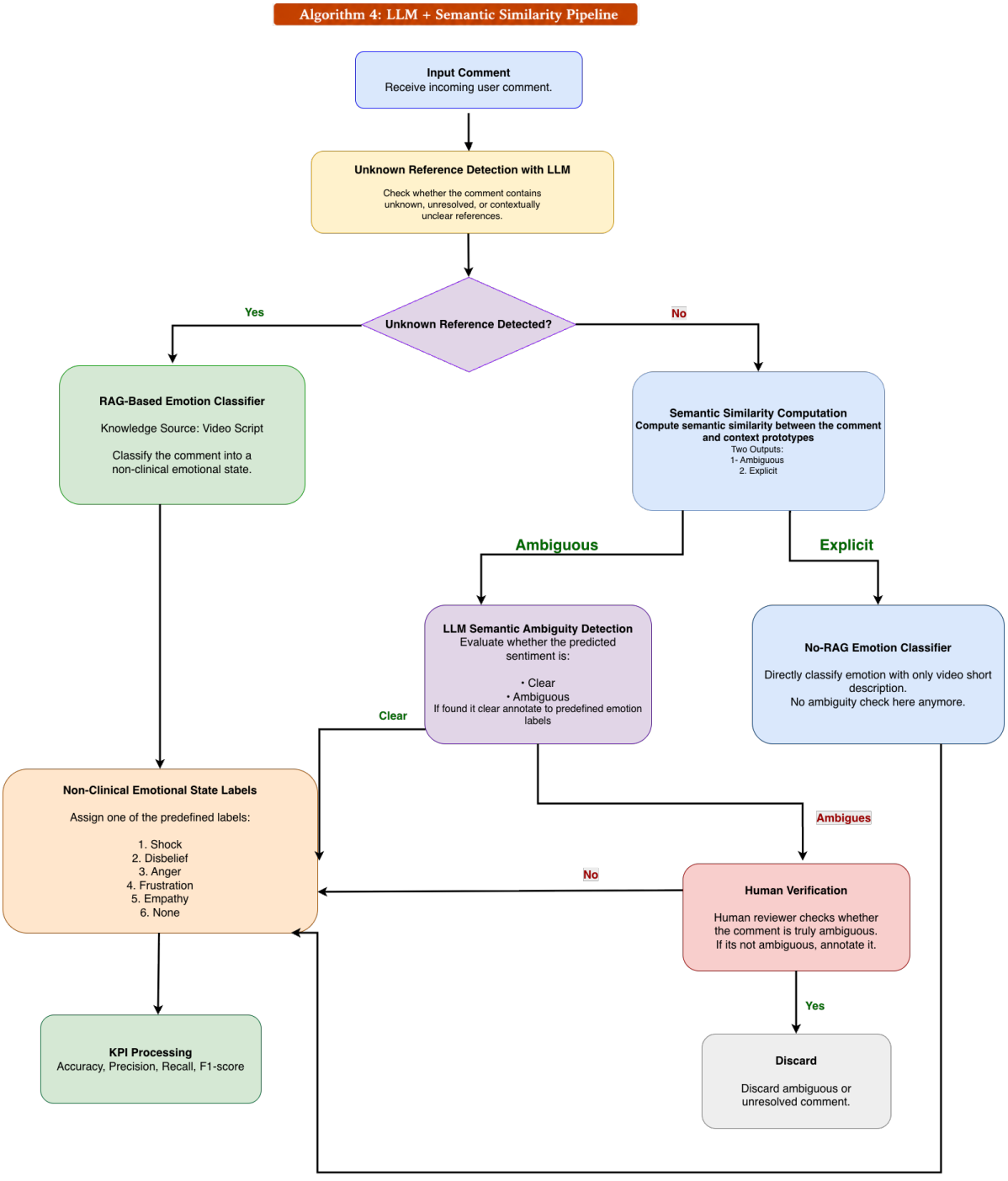


Figure 4.4: *Algorithm 4 applies Explicit/Ambiguous similarity before emotion LLM calls; Prompt P1 is omitted in the reference specification.*

4.7.3 Prompts

- **P5** — Section 4.6 (Explicit vs Ambiguous prototypes).
- **P2** — Section 4.4.
- **P6–P7** — Section 4.6.
- **P3** — used in Algorithms 1–2 only; not used in Algorithm 4.
- **P1** — omitted in the reference specification; hybrid implementations may optionally retain P1 as a secondary check.

4.7.4 Pseudocode

```
Algorithm 4: LLM-Semantic Similarity Pipeline
Input: c, s, D, P_exp, P_vag, human reviewer H
Output: y or DISCARDED

sigma <- SEMANTIC_SIMILARITY(c, s, P_exp, P_vag)  // P5 or embed
if sigma = Ambiguous then
    a <- LLM(P6, c, s)
    if a = Ambiguous then
        if H(c) confirms ambiguous then
            return DISCARDED
        else
            return emotion label from H(c)
        end if
    else
        return a  // Clear: predefined non-clinical label
    end if
end if
```

```
14: if ROUTE_TO_RAG(c, s, sigma) then
15:     tau <- RETRIEVE(D, c)
16:     return LLM(P2, c, s, tau)
17: else
18:     return LLM(P7, c, s)  // sigma = Explicit
19: end if
```

This ordering contrasts with CAS-RAG, which never applies comparator routing prompts (P1, P5) at the gate and uses three prototype classes rather than two (Section 4.8).

4.8 CAS-RAG: Context-Aware Selective Retrieval (Proposed Method)

CAS-RAG (Context-Aware Selective RAG) routes each incoming comment through **semantic classification** (`Explicit`, `Ambiguous`, or `Context-Dependent`), then applies **selective retrieval**, **ambiguity detection**, or **human verification** as required, so that retrieval is used only when it is necessary and resolvable. Figure 4.5 summarises the pipeline.

4.8.1 Input Stage

1. Receive the incoming user comment c.

4.8.2 Semantic Similarity Computation

2. Compute the semantic similarity between the input comment c and the predefined semantic prototypes.

3. Assign the comment to one of the following categories:

    - **Explicit**
    - **Ambiguous**
    - **Context-Dependent**

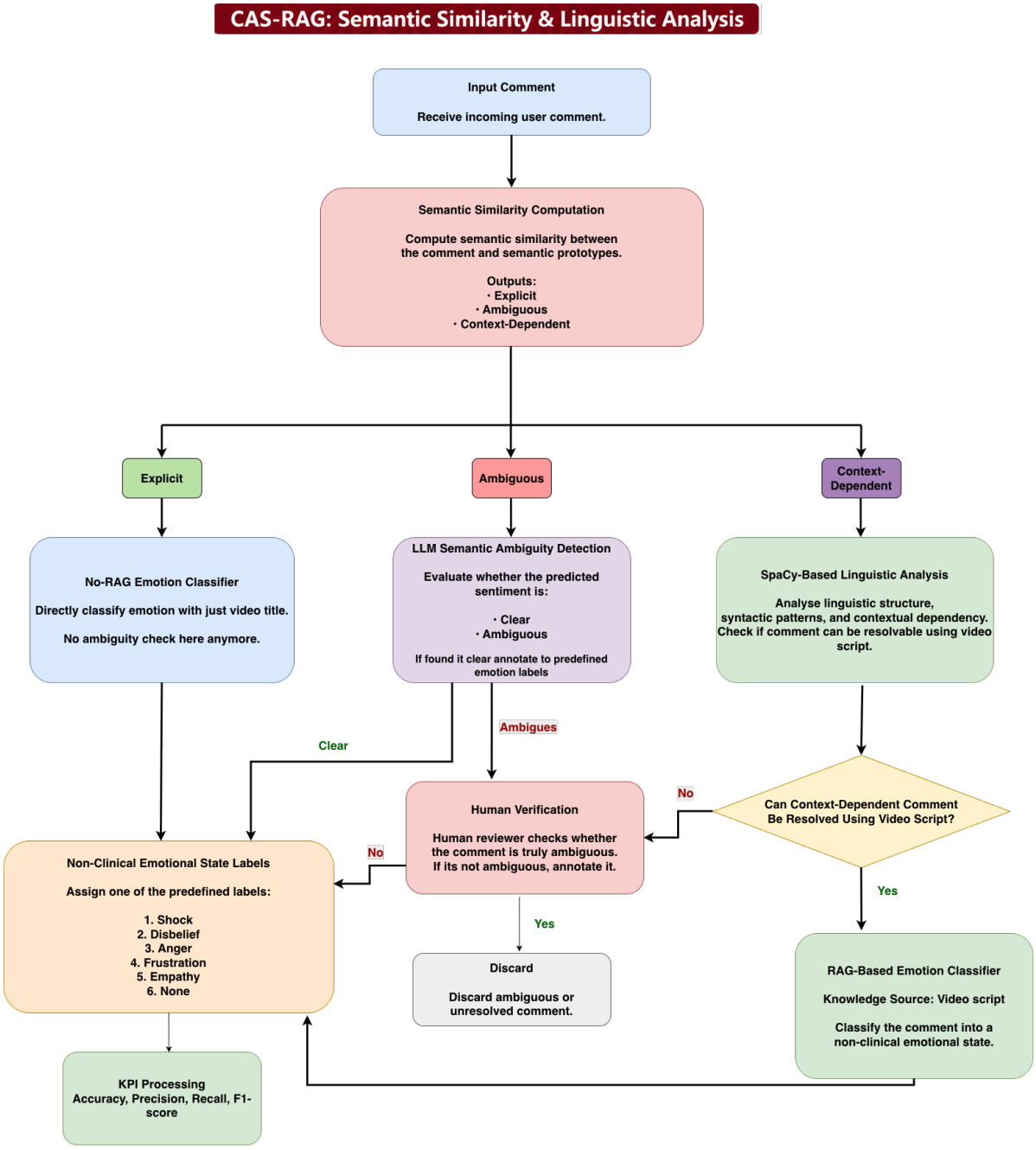


Figure 4.5: *CAS-RAG pipeline: semantic routing, selective RAG (Prompt P2), no-RAG classification (Prompt P7), semantic ambiguity detection (Prompt P6), human verification, and KPI processing.*

4.8.3 Branching Based on Semantic Category

The pipeline branches according to the assigned semantic category. The predefined non-clinical emotion labels used on all labelling paths are: **Shock**, **Disbelief**, **Anger**, **Frustration**, **Empathy**, and **None**.

4.8.3.1 Explicit Branch

4. If the comment is classified as **Explicit**:

5. Apply the **No-RAG Emotion Classifier** (**Prompt P7**):

    - Classify the emotion directly using only the comment and the video scenario (no retrieval is used).
    - No ambiguity verification is performed at this stage.

6. Assign one of the predefined non-clinical emotion labels listed above.

7. Send the result to **KPI Processing**.

4.8.3.2 Ambiguous Branch

8. If the comment is classified as **Ambiguous**:

9. Apply **LLM-based Semantic Ambiguity Detection** (**Prompt P6**):

    - Determine whether the emotional meaning is **Clear**, or **Ambiguous**.

Case 1: Clear.

10. If the comment is **Clear**:

11. Assign one of the predefined non-clinical emotion labels (same label set as above).

12. Send the result to **KPI Processing**.

Case 2: Ambiguous.

13. If the comment is **Ambiguous**:

14. Send the comment to **Human Verification**.

Human Verification.

15. A human reviewer evaluates whether the comment is truly ambiguous.

    - If the comment is **not actually ambiguous**:
      - Assign a valid emotion label.
      - Send the result to **KPI Processing**.
    - If the comment **remains ambiguous**:
      - **Discard** the comment (no label is assigned).

4.8.3.3 Context-Dependent Branch

16. If the comment is classified as **Context-Dependent**:

17. Apply **spaCy-based Linguistic Analysis**:

    - Analyse linguistic structure and syntactic patterns.
    - Detect contextual dependencies.
    - Determine whether the comment can be grounded using the video script.

Decision: Can the comment be resolved using the video script?

Case 1: YES (Resolvable).

18. If the comment **can be resolved**:

19. Apply the **RAG-based Emotion Classifier** (**Prompt P2**):

- Retrieve relevant context from the video script.
- Use the retrieved information to classify the emotion.

20. Assign one of the predefined non-clinical emotion labels.

21. Send the result to **KPI Processing**.

Case 2: NO (Not Resolvable).

22. If the comment **cannot be resolved** using the video script:

23. Send the comment to **Human Verification**.

Human Verification.

24. A human reviewer checks the comment:

    - If resolvable:
      - Assign an emotion label → KPI Processing
    - If not resolvable:
      - **Discard** the comment

4.8.4 KPI Processing

25. For all successfully labelled comments, compute evaluation metrics:

    - Accuracy
    - Precision
    - Recall
    - F1-score

4.8.5 CAS-RAG emotion classification prompts

Semantic routing (Steps 2–3) and spaCy-based linguistic analysis (Step 17) do **not** invoke any LLM prompt. Emotion labelling after routing uses exactly **three** prompts, distinct from comparator routing prompts (P1, P3–P5). Comparator Prompt P3 (with `Ambiguous` as an output label) is **not** used in CAS-RAG.

Table 4.3: *CAS-RAG prompt mapping: pipeline stage → prompt.*

Pipeline stage	Prompt	Role
Explicit branch: No-RAG Emotion Classifier	P7	Classify emotion from c and scenario s only; no ambiguity check
Ambiguous branch: LLM Semantic Ambiguity Detection	P6	Return `Clear` (emotion label) or `Ambiguous`
Context-dependent branch: RAG Emotion Classifier	P2	Classify emotion using c, s, and retrieved script context τ

4.8.5.1 Prompt P2 — RAG-based emotion classification

Used when a context-dependent comment is resolvable (Steps 18–20). Full prompt text:

```
You are an emotion classification system designed to identify
non-clinical emotional reactions in user comments related to RAG
knowledge source of video script.
Your task is to classify the emotional reaction expressed in the user
comment using ONLY:
1. The explicitly available information in the comment.
2. The provided video scenario.
3. The retrieved video script context.
```

```
Do NOT infer hidden intentions, unstated emotions, or external
context beyond the provided scenario and retrieved context.

Video Scenario:
A hidden camera experiment is set up in a public diner where a child
actor struggles with basic math while a tutor becomes increasingly
rude and verbally harsh.
The situation is designed to test how other children nearby will
respond.
Some hesitate at first, but several step in to help the struggling
student, offering both academic support and emotional encouragement.
The experiment reveals that even young children can recognise unfair
treatment and act with empathy,
highlighting the importance of kindness, confidence, and standing up
against bullying behaviour in educational settings.

Predefined non-clinical emotional state labels:
```

- Shock
- Disbelief
- Anger
- Frustration
- Empathy
- None

```
Label Definitions:

Shock:
```

```
A strong emotional reaction of surprise, alarm, or emotional
disturbance.

Disbelief:
A reaction expressing difficulty accepting, believing, or trusting
what is happening.

Anger:
A reaction expressing irritation, outrage, hostility, or moral
condemnation.

Frustration:
A reaction expressing annoyance, disappointment, helplessness, or
emotional dissatisfaction.

Empathy:
A reaction expressing compassion, emotional concern, support,
kindness, or understanding toward others.

None:
Use this label when the comment's emotional reaction does not belong
to any of the categories below: Shock, Disbelief, Anger,
Frustration, Empathy.
This label is included to avoid forcing the model to assign an
emotional category when no meaningful emotion is explicitly
expressed.
```

```
Instructions:
• Use only explicitly available evidence.
• Do not assume missing context beyond the retrieved context.
• Focus only on non-clinical emotional reactions.
• If multiple emotions appear, choose the dominant explicitly
expressed emotion.
• Use the retrieved context to resolve unclear references or
contextual ambiguity.
• Provide a concise explanation for the classification result.
Input Comment:
{comment}

Retrieved Context:
{retrieved context}

Respond ONLY in the following format:
Emotion Label: <Shock / Disbelief / Anger / Frustration / Empathy /
None>

Explanation:
```

4.8.5.2 Prompt P7 — No-RAG emotion classification (without ambiguity checking)

Used on the explicit branch (Steps 5–6). Full prompt text:

```
You are an emotion classification system designed to identify
non-clinical emotional reactions in user comments related to short
video scenarios.
Your task is to classify the emotional reaction expressed in the user
```

comment using ONLY:

1. The explicitly available information in the comment.

2. The provided short video scenario.

Do NOT infer hidden intentions, unstated emotions, or external context beyond the provided scenario.

Video Scenario:

A hidden camera social experiment in a public diner shows a tutor becoming verbally harsh toward a struggling child while nearby children react by offering empathy, emotional support, encouragement, and academic help.

Predefined non-clinical emotional state labels:

- Shock
- Disbelief
- Anger
- Frustration
- Empathy
- None

Label Definitions:

Shock: A strong emotional reaction of surprise, alarm, or emotional disturbance.

Disbelief: A reaction expressing difficulty accepting, believing, or trusting what is happening.

Anger: A reaction expressing irritation, outrage, hostility, or moral condemnation.

Frustration: A reaction expressing annoyance, disappointment,

```
helplessness, or emotional dissatisfaction.
Empathy: A reaction expressing compassion, emotional concern,
support, kindness, or understanding toward others.
None: Use this label when the comment's emotional reaction does not
belong to any of the categories below: Shock, Disbelief, Anger,
Frustration, Empathy. This label is included to avoid forcing the
model to assign an emotional category when no meaningful emotion is
explicitly expressed.

Instructions:
• Use only explicitly available evidence.
• Do not assume missing context.
• Focus only on non-clinical emotional reactions.
• If multiple emotions appear, choose the dominant explicitly
expressed emotion.
• Provide a concise explanation for the classification result.

Input Comment: {comment}

Respond ONLY in the following format:
Emotion Label: <Shock / Disbelief / Anger / Frustration / Empathy /
None>
Explanation:
```

4.8.5.3 Prompt P6 — Semantic ambiguity-aware emotion classification

Used on the ambiguous branch for LLM-based semantic ambiguity detection (Steps 9–14). Full prompt text:

```
You are a verification system designed to verify the user
non-clinical emotional reactions comments related to social
experiment short video scenarios is ambiguous or not.

Your task is to:
1. Determine whether the emotional meaning expressed in the user
comment is clear or ambiguous.
2. If the emotional meaning is clear, classify the dominant
emotional reaction using the predefined labels below.

Use ONLY:
1. The explicitly available information in the comment.
2. The provided video scenario.

Do NOT infer hidden intentions, unstated emotions, or external
context beyond the provided scenario.

Video Scenario:
A hidden camera social experiment in a public diner shows a tutor
becoming verbally harsh toward a struggling child while nearby
children react by offering empathy, emotional support,
encouragement, and academic help.

Predefined non-clinical emotional state labels:
• Shock
• Disbelief
• Anger
```

- Frustration
- Empathy
- None

Label Definitions:

Shock:
A strong emotional reaction of surprise, alarm, or emotional disturbance.

Disbelief:
A reaction expressing difficulty accepting, believing, or trusting what is happening.

Anger:
A reaction expressing irritation, outrage, hostility, or moral condemnation.

Frustration:
A reaction expressing annoyance, disappointment, helplessness, or emotional dissatisfaction.

Empathy:
A reaction expressing compassion, emotional concern, support, kindness, or understanding toward others.

None:

Use this label when the comment's emotional reaction does not belong to any of the categories below:
Shock, Disbelief, Anger, Frustration, Empathy.

This label is included to avoid forcing the model to assign an emotional category when no meaningful emotion is explicitly expressed.

A comment should be considered Ambiguous when:

- the emotional meaning is unclear,
- the emotional target is unclear,
- the comment is too generic or weakly expressed,
- or the connection between the comment and the video scenario cannot be reliably established.

Examples of ambiguous comments:

- "I love it"
- "Amazing"
- "Wow"
- "So good"

These comments may contain a positive reaction, but they do not provide enough contextual evidence to determine a reliable emotional interpretation related to the video scenario.

Instructions:

- Use only explicitly available evidence.

```
• Do not assume missing context.
• Focus only on non-clinical emotional reactions.
• If multiple emotions appear, choose the dominant explicitly
expressed emotion.
• If the emotional meaning cannot be reliably interpreted, return
"Ambiguous".
• If the emotional meaning is clear, classify the emotion using the
predefined labels.
• Provide a concise explanation for the result.

Input Comment:
{comment}

Respond ONLY in the following format:

Emotion Label: <Shock / Disbelief / Anger / Frustration / Empathy /
None / Ambiguous>

Explanation:
```

When Prompt P6 returns `Ambiguous`, the comment is sent to human verification (Step 15); comparator Prompt P4 is not used in CAS-RAG.

4.8.6 Pseudocode

```
Algorithm CAS-RAG: Context-Aware Selective Retrieval (per comment)
Input: comment c, video scenario s, video script corpus D,
       semantic prototypes P, spaCy nlp
Output: emotion label y or DISCARDED
```

```
1:  receive c
2:  compute semantic_similarity(c, P)
3:  category <- assign {Explicit, Ambiguous,
                        Context-Dependent}
4:  if category = Explicit then
5:      return LLM(P7, c, s)  // No-RAG; no ambiguity verification
6:  else if category = Ambiguous then
7:      a <- LLM(P6, c, s)
8:      if a = Ambiguous then
9:          if HumanVerification_Ambiguous(c) = not actually ambiguous then
10:             return emotion label from human review
11:         else
12:             return DISCARDED  // remains ambiguous
13:         end if
14:     else
15:         return a  // Clear: predefined non-clinical label
16:     end if
17: else if category = Context-Dependent then
18:     resolvable <- spaCy_LinguisticAnalysis(c, D, nlp)
19:     if resolvable = YES then
20:         tau <- RETRIEVE(D, c)
21:         return LLM(P2, c, s, tau)
22:     else
23:         if HumanVerification_Context(c) = resolvable then
24:             return emotion label from human review
25:         else
```

```
26:             return DISCARDED
27:         end if
28:     end if
29: end if

Algorithm CAS-RAG-KPI: Evaluation metrics (batch)
Input: labelled set L = {(c_i, y_i)} for all successfully labelled comments
Output: Accuracy, Precision, Recall, F1-score

33: compute Accuracy(L), Precision(L), Recall(L), F1(L)
```

4.9 Comparative Summary and Implementation

4.9.1 Cross-pipeline comparison

Table 4.4 consolidates architectural differences under the shared experimental contract (Section 4.1). None of the comparators performs three-way semantic routing with multi-metric fusion; none of the comparators suppresses per-comment LLM calls at the gate in the reference CAS-RAG implementation.

Table 4.4: *Methodological comparison of routing components (shared scenario, corpus, and labels).*

Component	Alg. 1	Alg. 2	Alg. 3	Alg. 4	CAS-RAG
First routing gate	LLM (P1)	spaCy	spaCy + sim.	Similarity (P5)	7-metric embed.
Prototype classes	—	—	Exp., Vag.	Exp., Vag.	Exp., Vag., Ctx.
Emotion LLM prompts	P2–P4	P2–P4	P2, P6–P7	P2, P6–P7	P2, P6–P7 (post-gate)
Primary output	Emotion	Emotion	Emotion	Emotion	RAG gate

4.9.2 Evaluation metrics linkage

When comparator pipelines are executed through `CAS_RAG_Project`, reported columns include standard classification metrics (accuracy, macro/weighted F1, per-class precision/recall) and gate-oriented diagnostics: retrieval exposure rate, cost-saving percentage, wrong-retrieval counts, and confusion between retrieval categories. Batch JSON exports land in `data/evaluation_outputs/`; the five-algorithm Excel comparison is produced by `scripts/run_five_algorithm_comparison.py`. These metrics align gate behaviour with emotion accuracy without conflating the two output types in Table 4.1.

4.9.3 Software reference implementation

All five pipelines (Algorithms 1–4 and CAS-RAG) share one `PipelineContext`: scenario text, script corpus, per-video prototypes, versioned prompts, a pluggable LLM provider (`openai`, `gemini`, or `ollama`), dense RAG retrieval, and a persistent human-verification queue. Algorithm selection is a single function dispatch (`orchestration/engine.py`); emotion stages call production LLMs through prompts P1–P7 with structured response parsing. Full platform documentation appears in Section 5.10.

4.9.4 Analytical comparison of baseline algorithms and CAS-RAG

Table 4.4 summarises components; this subsection analyses *behavioural* differences—routing, retrieval, context handling, and expected strengths/weaknesses—using Tutoring (Section 6.3) and Bullying (Section 6.5) in-depth evidence. [panel]

Algorithm 1 (LLM–LLM). **Routing.** P1 (unknown-reference detection) is the sole gate. On Tutoring, routing accuracy is 51.0% (49/100 mistakes)—mid-ranked among comparators. The LLM gate tends to over-trigger retrieval when comments contain pronouns resolvable from the scenario, and under-trigger when scene-specific names (e.g., Harper, Aubrey) require script grounding. **Retrieval.** Every P1=`Yes` path invokes dense retrieval and P2; there is no three-way semantic bypass. Retrieval exposure is therefore high relative to CAS-RAG.

Context handling. P2 conditions on retrieved chunks; strength depends on corpus quality (transcript coverage). Post-refinement emotion accuracy reaches 79.6% (highest among five pipelines on Tutoring) when retrieval fires appropriately, but routing errors inject irrelevant context in many rows. **Expected strengths:** Flexible reference reasoning when P1 is correct; strong emotion prompts after refinement. **Expected weaknesses:** Cost (two LLM calls common); routing instability; sensitivity to P1 scenario examples. On Bullying, three-way routing falls to 49.0% (emotion 73.9% on 88 labels)—stronger on emotion than on the gate. panel

Algorithm 2 (spaCy–LLM). **Routing.** Linguistic heuristics (NER, noun phrases, pronoun patterns) replace P1. Tutoring routing: 53.0% (47 mistakes)—similar to Algorithm 1 but with different error profile (rule misses on informal referents). **Retrieval.** Same P2/P3/P4 emotion stack as Algorithm 1 after the spaCy gate. **Context handling.** spaCy resolves explicit named entities but struggles with sarcasm, ellipsis, and scene-dependent nicknames. **Expected strengths:** Lower LLM cost at gate; transparent linguistic features. **Expected weaknesses:** Brittle on social-media informality; no three-way ambiguity class at Layer 1. On Bullying, spaCy over-labels Context-dependent (36.0% three-way routing; 43% binary). panel

Algorithm 3 (spaCy + two-way semantic). **Routing.** spaCy gate plus P5 two-way similarity (Explicit vs. Ambiguous prototypes only). Tutoring routing: 32.0% (68 mistakes)—weakest routing; many explicit comments routed to retrieval. **Retrieval.** Over-retrieval is the dominant failure mode: explicit personal stories and opinions frequently enter the RAG path. **Context handling.** Lacks a dedicated Context-Dependent prototype class; scene-specific comments are forced into Explicit/Ambiguous bins ill-suited to retrieval decisions. **Expected strengths:** Simple deployment when comments are mostly self-contained. **Expected weaknesses:** Poorest routing in Tutoring comparison (32.0%) and jointly weakest on Bullying with Algorithm 2 (36.0% three-way); conflates ambiguity with context-

dependence. panel

Algorithm 4 (LLM P1 + two-way semantic). **Routing.** Combines P1 with P5 two-way similarity on the no-P1-retrieval branch. Tutoring routing: 66.0% (34 mistakes)—second-best after CAS-RAG among comparators. **Retrieval.** More conservative than Algorithm 3 because P1 filters some references; still lacks three-way Layer 1 semantics. **Context handling.** Better on unresolved pronouns than Algorithm 3; worse than CAS-RAG on borderline explicit/context-dependent boundaries. **Expected strengths:** Balanced LLM + similarity; strong routing among baselines (Tutoring 66.0%; Bullying 71.0% three-way, 75% binary). **Expected weaknesses:** No explicit Ambiguous→RAG-infeasible policy; two-way gate cannot encode context-dependent retrieval economics. panel

CAS-RAG. **Routing.** Seven-metric semantic fusion over three prototype classes (Explicit, Ambiguous, Context-Dependent) plus margin rules (m, b). Three-way routing is **72.0%** on both Tutoring (28 mistakes) and Bullying (28 mistakes)—best in each five-pipeline comparison. **Retrieval.** Layer 2 spaCy policy on Context-Dependent only; Explicit→no RAG; Ambiguous→RAG-infeasible. Pooled six-dataset audit: 66.8% avoided retrieval, 44.6% cost saving, 51/199 wrong retrievals (Section 6.2). **Context handling.** Designed for emotion-specific gating: distinguishes "cannot resolve even with script" (Ambiguous) from "may resolve with script" (Context-Dependent). Qualitative Tutoring review (Section 6.3.3) shows residual false context-dependency and missed scene references. **Expected strengths:** Best routing accuracy; explicit retrieval economics; human-verification hooks for low-confidence paths. **Expected weaknesses:** Emotion accuracy (77.6% post-refinement) slightly below Algorithm 1 (79.6%) on Tutoring—the rank reverses on Bullying (CAS-RAG 75.0% vs. Algorithm 1 73.9% on 88 labels); threshold sensitivity; prototype quality dependence; residual false context-dependency on both videos. panel

Table 4.5: *Analytical summary: routing gate, retrieval policy, and observed three-way routing accuracy on Tutoring (post-refinement) and Bullying. Emotion ranks differ; see Tables 6.3 and 6.6.*

Pipeline	Layer 1 gate	Retrieval policy	Tutoring routing	Bullying routing	Primary risk
Alg. 1	LLM P1	Retrieve if P1=Yes	51.0%	49.0%	Over-retrieval; P1
Alg. 2	spaCy	Retrieve if linguistic flag	53.0%	36.0%	Informal language;
Alg. 3	spaCy + P5 (2-way)	Similarity-triggered	32.0%	36.0%	Severe over-retri
Alg. 4	P1 + P5 (2-way)	Hybrid	66.0%	71.0%	No 3-way ambiguit
CAS-RAG	7-metric 3-way	Selective Layer 2	**72.0%**	**72.0%**	False context-depe

CHAPTER 5

CAS-RAG: CONTEXT-AWARE SELECTIVE RETRIEVAL GATING

This chapter presents **CAS-RAG** (Context-Aware Selective RAG), also referred to as **RAG-GATE** in the codebase: a two-layer pipeline that classifies comments on videos related to adolescent topics as *explicit*, *ambiguous*, or *context-dependent*, and gates retrieval accordingly. The design uses SentenceTransformer embeddings with seven complementary similarity metrics at Layer 1 and spaCy-based retrieval policy at Layer 2. Worked examples with exact numeric Layer 1 scores demonstrate reproducibility. Section 5.10 documents the `CAS_RAG_Project` software platform, including integration with production LLM APIs (OpenAI/ChatGPT, Gemini, and optional local Ollama/Llama). Chapter 4 specifies the shared experimental contract, comparator Algorithms 1–4, and the methodological role of CAS-RAG relative to those pipelines. Chapter 6 reports the Tutoring primary in-depth evaluation and complementary Layer 1 topic-diversity metrics across six annotated datasets.

5.1 Design Rationale: Three-Category Taxonomy

Deciding when to trigger retrieval for **context-aware emotional reaction analysis** requires distinguishing (i) a semantic three-way routing signal in Layer 1 and (ii) a **three-case RAG**

outcome committed in Layer 2, consistent with retrieval design principles (Gao et al., 2023; Lewis et al., 2020):

Explicit (self-contained affect) Comments that are self-contained (Lewis et al., 2020)—clear affective wording that does not depend on which specific person or moment in the video is meant. Example: *"I love how the kids stood up to the bully"*.

Ambiguous (affective meaning not resolved by context) Comments that remain inherently ambiguous (Gao et al., 2023): even with full video context, a single interpretation is not justified. Retrieval does not resolve ambiguity. Example: *"Not sure if this is inspiring or manipulative"*.

Context-dependent (may require grounding) Comments that refer to specific people or moments (Lewis et al., 2020) and may become interpretable with the script. Example: *"That kid is so brave"*.

Layer 2 RAG outcomes (three cases). Independent of the Layer 1 labels, the gate maps each comment to one of:

RAG-resolvable Retrieval plausibly clarifies emotional meaning (e.g., unclear reference or moment cues).

RAG-unnecessary Affective meaning is clear without retrieval, or context-dependent cues do not warrant retrieval.

RAG-infeasible Retrieval should not run: either semantic ambiguity (ambiguous), or **video mismatch**—the comment appears to require grounding, but key referents are not supported by the available scenario text (so retrieval would not connect the comment to this video).

5.2 System Architecture Overview

Input: `video_scenario` (short description) and `comment`.

Output: `category` ∈ {`explicit`, `ambiguous`, `context_dependent`}, `rag_needed`, `infeasible`, `rationale`, plus a discrete **RAG outcome** (`rag_unnecessary` / `rag_resolvable` / `rag_infeasible`) and, when applicable, an `infeasible_reason` (`semantic_ambiguity` vs. `video_mismatch`).

1. **Layer 1—Semantic 3-way routing:** The comment and **prototype sentences** (five per category), either manual or scenario-generated (Reimers & Gurevych, 2019), are encoded with SentenceTransformer. Seven similarity metrics compare the comment embedding to each category's prototypes; a weighted combination assigns the category. No dataset-specific phrase lists are required.

2. **Layer 2—Retrieval policy:** `explicit` → **RAG-unnecessary**; `ambiguous` → **RAG-infeasible** (semantic ambiguity). For `context_dependent` comments, spaCy (Honnibal & Montani, 2017; Honnibal et al., 2023) checks **video mismatch** (referents not supported by the scenario) before unclear person/entity and temporal cues, yielding the three-case RAG mapping above.

5.3 Layer 1: Semantic Three-Way Classification

5.3.1 Embeddings and prototypes

The comment is encoded using SentenceTransformer (`all-MiniLM-L6-v2`) (Reimers & Gurevych, 2019), producing a 384-dimensional embedding. Fixed prototype sets (five sentences per category per video) are stored in `resources/datasets/Experiments-data/<video_slug>/prototypes_semantic.json` and loaded by `SemanticSimilarityEngine` (`src/cas_rag/semantic/similarity.py`).

5.3.2 Scenario-conditioned prototype generation (optional)

For new videos, an LLM may read the same `video_scenario` string and output JSON with five informal strings per category; outputs are validated and cached alongside the video folder for reproducibility.

5.3.3 Seven similarity metrics

Metrics (normalised to $[0, 1]$, higher = more similar) include cosine, Euclidean, dot product, Manhattan, Dice, Pearson, and exp-Euclidean. Per-category scores use a weighted combination (cosine 0.22, Euclidean 0.14, dot 0.12, Manhattan 0.10, Dice 0.18, Pearson 0.14, exp-Euclidean 0.10).

Figure 5.1 visualises metric-by-metric scores for the three walkthrough comments in Section 5.9.

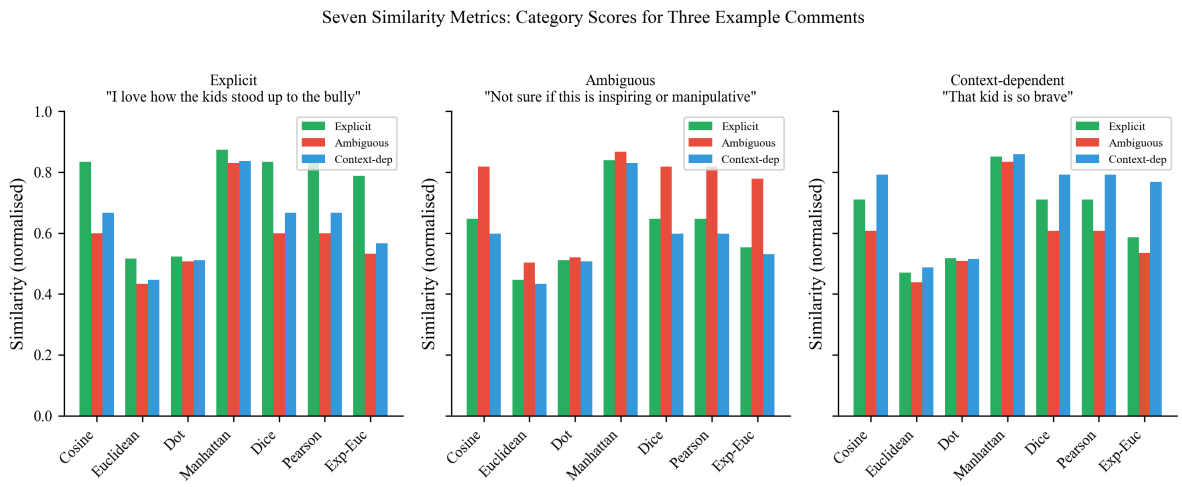


Figure 5.1: *Seven similarity metrics: category scores for three illustrative comments (explicit, ambiguous, context-dependent walkthrough).*

5.3.4 Decision rules

Let s_{exp}, s_{amb}, and s_{cd} be the Layer 1 combined scores. The initial winner is $\hat{k}_0 = \arg\max\{s_{\text{exp}}, s_{\text{amb}}, s_{\text{cd}}\}$. For the Explicit–Ambiguous boundary define $\Delta = s_{\text{amb}} - s_{\text{exp}}$. With

margins $m = 0.12$ and $b = 0.08$ (defaults in `config/default.yaml`), post-processing in `classify_three_way` applies: (1) if $\hat{k}_0 =$ `explicit` and $s_{\text{amb}} > s_{\text{exp}} + m$, set `ambiguous` (rarely triggered given $\hat{k}_0 =$ `explicit` $\Rightarrow s_{\text{exp}} \geq s_{\text{amb}}$ among three classes); (2) if $\hat{k}_0 =$ `ambiguous` and $s_{\text{exp}} \geq s_{\text{amb}} - b$, set `explicit`. Rule (2) is the main operational safeguard against over-predicting `ambiguous`. The policy is motivated by reject-option / selective prediction theory (Chow, 1970; Geifman & El-Yaniv, 2017) and asymmetric costs in cost-sensitive classification (Duda et al., 2001; Elkan, 2001); calibration of score magnitudes is discussed by (Guo et al., 2017).

Illustrative score-pair examples (pedagogy). Using only $(s_{\text{exp}}, s_{\text{amb}})$: Case A $(0.60, 0.65)$ gives $\Delta = 0.05 \leq b$—if `ambiguous` were the provisional winner, rule (2) flips to `explicit`. Case B $(0.40, 0.55)$ gives $\Delta = 0.15 > m$, favouring `ambiguous` when it leads strongly. Case C $(0.52, 0.60)$ gives $\Delta = 0.08$, a borderline band aligned with presentation slides. Full pipeline scores use three-way argmax; Section 5.9 gives reproduced numbers.

This is a reject-option design: rather than committing to `ambiguous` under marginal evidence, we require a stronger likelihood that "Ambiguous" is correct. Decision-theoretically, selective prediction thresholds depend on the relative cost of the two error types (e.g., over-predicting `ambiguous` vs. under-predicting it). While our fused metric scores are normalised to $[0, 1]$, the thresholding operates in the same way as Bayes-optimal decisions under calibrated (or monotonic) confidence estimates (Guo et al., 2017).

Risk objective and threshold selection. Let D_{val} be the pooled development set with gold Layer 1 labels. For any threshold parameters (m, b) (here m=0.12, b=0.08), the empirical risk is

$$\mathscr{R}(m, b) = \frac{1}{|D_{val}|} \sum_{i \in D_{val}} C\big(y_i, \hat{y}_i(m, b)\big), \tag{5.1}$$

where $C(\cdot,\cdot)$ encodes misclassification costs. Restricting to Explicit/Ambiguous confusions, this reduces to

$$\mathscr{R}(m,b) = \frac{1}{|D_{val}|}\left(C_{E\to A}\cdot\#\{i : y_i = E, \hat{y}_i = A\} + C_{A\to E}\cdot\#\{i : y_i = A, \hat{y}_i = E\}\right). \tag{5.2}$$

Worked validation example (using pooled counts). For the complementary pooled six-dataset Layer 1 topic-diversity counts (n=600), gold counts are $|E|$=279 and $|A|$=60 as reported in Fig. 6.2. The pooled confusion indicates 0 Explicit comments were predicted as Ambiguous, and 19 Ambiguous comments were predicted as Explicit. Therefore, for the chosen thresholds $(m,b) = (0.12, 0.08)$:

$$\mathscr{R}(0.12, 0.08) = \frac{C_{E\to A}\cdot 0 + C_{A\to E}\cdot 19}{600}. \tag{5.3}$$

With a less conservative explicit→ambiguous override (smaller m), the expected effect is to increase $\#\{y_i = E, \hat{y}_i = A\}$ (false Ambiguous), which increases risk proportionally to $C_{E\to A}$. This is exactly the trade-off formalised by reject-option / selective prediction theory (Chow, 1970; Geifman & El-Yaniv, 2017).

5.4 Layer 2: Retrieval Policy and Linguistic Grounding

When Layer 1 assigns `explicit`, the gate returns **RAG-unnecessary** (`rag_needed = false`). When Layer 1 assigns `ambiguous`, the gate returns **RAG-infeasible** (semantic ambiguity) (`rag_needed = false`, `infeasible = true`). For `context_dependent` comments only, the layer tests **scenario support**: if salient noun lemmas in the comment (e.g., person-like or salient object references) do not appear in the scenario text (with lightweight lemma aliases), the system marks **RAG-infeasible (video mismatch)**—retrieval should not run because the available scenario does not ground the comment. Otherwise spaCy (`en_core_web_sm`) performs NER, noun-phrase extraction, disambiguation checks against the scenario, and temporal/moment cues (*that part*, *when he said*, etc.). If `context_dependent_score` > 0.4 and references remain under-specified, `other_context_needs` can trigger RAG (**RAG-resolvable**).

The semantic fallback threshold τ=0.4 is again a cost-sensitive reject-option boundary: retrieving is useful only when the likelihood of true context-dependence is high enough to justify the retrieval cost and the risk of injecting irrelevant context. Let $FP(\tau)$ be the number of comments where RAG is triggered but the gold label is not context-dependent (wrong retrieval), and let $FN(\tau)$ be the number of context-dependent gold comments where retrieval is missed. We minimise the empirical validation risk

$$\mathscr{R}(\tau) = \frac{1}{|D_{val}|} \left(C_{FP} \cdot FP(\tau) + C_{FN} \cdot FN(\tau)\right), \tag{5.4}$$

where C_{FP} and C_{FN} weight (i) retrieval/computation cost under the stated cost model and (ii) the impact of unnecessary vs. missed context injection. Using pooled results, RAG is triggered for 199 comments and 51 of them are wrong retrievals (see Table 6.2); among the 261 gold context-dependent comments (Fig. 6.2), this implies FN = 113 for the final decision, yielding

$$\mathscr{R}(0.4) \propto 51C_{FP} + 113C_{FN}. \tag{5.5}$$

Context-dependent comments with scenario mismatch → **RAG-infeasible** (video mismatch). Otherwise context-dependent: `rag_needed` if unclear person or `other_context_needs` (**RAG-resolvable**), else **RAG-unnecessary**.

5.5 Design Principles and Efficiency

Routing avoids data-derived keyword rules, improving generalisation (Brown et al., 2020; Mohammad & Turney, 2013). Layer 2 runs only on context-dependent predictions; under the complementary six-topic Layer 1/cost check (Chapter 6), 199 of 600 pooled comments triggered RAG (33.2%).

5.6 Algorithm Pseudocode

The canonical end-to-end procedure and emotion prompts (P2, P6, P7) are specified in Section 4.8 (Steps 1–25; Table 4.3). The `CAS_RAG_Project` implementation realises the

same control flow in `pipelines/cas_rag.py`; Layer 1 semantic scoring and Layer 2 spaCy checks instantiate `SemanticSimilarityEngine` and `SpacyLinguisticAnalyzer` respectively (Sections 5.3–5.4; software details in Section 5.10).

```
Algorithm CAS-RAG: Context-Aware Selective Retrieval (per comment)
Input: comment c, video scenario s, video script corpus D,
       semantic prototypes P, spaCy nlp
Output: emotion label y or DISCARDED

receive c
compute semantic_similarity(c, P)
category <- assign {Explicit, Ambiguous, Context-Dependent}
if category = Explicit then
    return LLM(P7, c, s)
else if category = Ambiguous then
    a <- LLM(P6, c, s)
    if a = Ambiguous then
        if HumanVerification_Ambiguous(c) = not actually ambiguous then
            return emotion label from human review
        else
            return DISCARDED
        end if
    else
        return a
    end if
else if category = Context-Dependent then
    resolvable <- spaCy_LinguisticAnalysis(c, D, nlp)
    if resolvable = YES then
```

```
20:         tau <- RETRIEVE(D, c)
21:         return LLM(P2, c, s, tau)
22:     else
23:         if HumanVerification_Context(c) = resolvable then
24:             return emotion label from human review
25:         else
26:             return DISCARDED
27:         end if
28:     end if
29: end if
```

```
Algorithm CAS-RAG-KPI: Evaluation metrics (batch)
Input: labelled set L = {(c_i, y_i)} for all successfully labelled comments
Output: Accuracy, Precision, Recall, F1-score
```

```
32: compute Accuracy(L), Precision(L), Recall(L), F1(L)
```

5.7 Routing Categories: Theoretical Basis and Operational Definitions

Layer 1 routing categories are not ad hoc labels; they operationalise three distinct *retrieval necessity* states for emotion reaction classification, aligned with RQ2–3 and Gap 2 (emotion-specific gated RAG). Each category implies a different epistemic status of the comment relative to the video scenario s and script corpus $\mathscr{D}$. [panel]

5.7.1 Explicit (self-contained affect)

Theoretical basis. Drawing on selective prediction and cost-sensitive classification (Chow, 1970; Elkan, 2001), Explicit comments carry sufficient illocutionary force to assign an emo-

tion label without consulting $\mathscr{D}$. The affective target and reaction type are recoverable from (c, s) alone. **Operational definition.** A comment is Explicit when: (i) referents link to scenario entities without unresolved pronouns; (ii) the emotional reaction is stated or strongly implied; and (iii) no specific scene moment or unnamed character is required. **Example (bullying scenario).** *"Bullying a girl at a bus stop is disgusting"* — moral outrage is explicit; no script retrieval needed. **Routing decision.** Layer 1→`explicit`; Layer 2→RAG-unnecessary; emotion via P7 (no retrieval). **Retrieval implication.** Retrieving $\mathscr{D}$ risks injecting spurious scene detail (wrong-retrieval cost in Table 6.2). [panel]

5.7.2 Ambiguous (affective meaning not resolved by context)

Theoretical basis. Reject-option theory treats low-confidence regions as distinct from errors (Geifman & El-Yaniv, 2017). Ambiguous comments are those where *even with* full script access, a single emotion label is not justified. **Operational definition.** Weak, generic, mixed, or authenticity-questioning comments where the emotional target or intensity cannot be reliably fixed. **Example.** *"Not sure if this is inspiring or staged"* — meta-commentary on production, not a stable emotion toward video actors. **Routing decision.** Layer 1→`ambiguous`; Layer 2→RAG-infeasible (semantic ambiguity); P6 may confirm; human verification if still ambiguous. **Retrieval implication.** Retrieval cannot disambiguate; running RAG wastes cost and may hallucinate connections. [panel]

5.7.3 Context-Dependent (may require grounding)

Theoretical basis. Grounded language understanding requires linking underspecified references to discourse antecedents (Lewis et al., 2020). Context-Dependent comments presuppose information in $\mathscr{D}$ not entailed by s alone. **Operational definition.** Comments referencing specific individuals, gestures, or moments identifiable only from the transcript (e.g., *"When Harper walked over I melted"* in the tutoring video; *"That last guy though"* in bullying). **Routing decision.** Layer 1→`context_dependent`; Layer 2 tests scenario support and spaCy cues; if resolvable, RAG→P2. **Retrieval implication.** This is the only class where

retrieval is *potentially* beneficial; false positives here dominate wrong-retrieval counts in the six-dataset audit. [panel]

5.7.4 Alignment with research questions

RQ2 asks which gating strategies optimise accuracy and retrieval overhead: the three-way taxonomy directly encodes overhead (Explicit→skip, Ambiguous→skip, Context-Dependent→conditional). RQ3 evaluates whether CAS-RAG's semantic-first implementation of this taxonomy outperforms LLM-only (Alg. 1/4) and linguistics-only (Alg. 2/3) gates on routing and emotion metrics. The six-dataset Layer 1 evaluation tests whether category *boundaries* generalise across video topics; emotion-depth evaluation tests label quality on Tutoring and Bullying only (Chapter 6). [panel]

5.8 Semantic Similarity in CAS-RAG: Metrics, Thresholds, and Retrieval Decisions

Semantic similarity is the computational core of CAS-RAG Layer 1. Unlike keyword gates, embedding-based prototype comparison supports informal, non-lexical paraphrase—critical for YouTube comments. This section expands the role of semantic retrieval in routing, threshold calibration, and downstream context selection (RQ2–3). [panel]

5.8.1 Embedding space and prototype banks

Each comment c and prototype sentence $p_k^{(j)}$ (category $j \in \{\text{Exp}, \text{Amb}, \text{CD}\}$, index k) is encoded by SentenceTransformer `all-MiniLM-L6-v2` into $\mathbb{R}^{384}$ (Reimers & Gurevych, 2019). Prototypes are scenario-conditioned: five strings per category per video, stored in `prototypes_semantic.json`. Similarity is not computed against the full comment corpus at Layer 1; $\mathscr{D}$ is reserved for Layer 2 retrieval after a Context-Dependent win. [panel]

5.8.2 Seven-metric fusion

Seven normalised metrics (cosine, Euclidean, dot, Manhattan, Dice, Pearson, exp-Euclidean) are fused with fixed weights (Section 5.3.3) to reduce single-metric blind spots. Cosine and Dice emphasise directional overlap; Manhattan and Euclidean penalise magnitude differences; the ensemble score $s_{\text{exp}}, s_{\text{amb}}, s_{\text{cd}}$ feeds the three-way argmax. Figure 5.1 illustrates separation for walkthrough comments. [panel]

5.8.3 Thresholds and margin rules

Raw argmax over three scores would over-predict `ambiguous` on borderline explicit comments. Margin rules with m=0.12 and b=0.08 implement asymmetric costs: false Ambiguous predictions block cheap P7 paths; false Explicit predictions trigger unnecessary retrieval. The semantic fallback threshold τ=0.4 in Layer 2 governs spaCy-assisted RAG triggers for Context-Dependent comments (Section 5.4). These values were set by empirical risk minimisation on development-style decisions (Eqs. in Section 5.3), not by dataset-specific phrase lists—supporting cross-topic routing evaluation on six datasets. [panel]

5.8.4 Semantic similarity vs. dense retrieval

Layer 1 similarity answers: *which routing class*? It compares c to *category prototypes* (class exemplars). **Layer 2 / RAG retrieval** answers: *which script chunks*? It compares c to $\mathscr{D}$ via the same embedder in `RAGEngine.retrieve`. Confusing these stages—treating prototype similarity as script relevance—is a design error the architecture explicitly prevents: retrieval runs only after Context-Dependent classification and Layer 2 resolvability checks. [panel]

5.8.5 Role in routing and context selection

1. **Explicit win:** High s_{exp} relative to other classes → skip $\mathscr{D}$; emotion from scenario s only (P7).

2. **Ambiguous win:** High s_{amb} → mark RAG-infeasible; P6 preserves uncertainty; human queue if needed.

3. **Context-Dependent win:** High s_{cd} → Layer 2; if resolvable, top-k chunks from $\mathscr{D}$ feed P2.

4. **Comparator Algorithms 3–4:** P5 uses two-way Explicit/Ambiguous prototypes only—a strict subset of CAS-RAG's similarity machinery, explaining poorer Context-Dependent discrimination (Section 4.9.4).

5.8.6 Significance for the study

Semantic similarity operationalises Gap 2: emotion-specific *when-to-retrieve* decisions without hand-built phrase lists. Its significance is threefold: (i) efficiency—66.8% retrieval avoided in the six-dataset audit; (ii) interpretability—prototype sentences and seven metric scores are inspectable per comment; (iii) generalisation—the same mechanism runs on all six video packages with scenario-specific prototypes only. Limitations (prototype coverage, threshold transfer, embedding domain shift) are discussed in Section 7.3. [panel]

5.9 Worked Examples with Exact Numerical Outputs

This section reproduces Layer 1 scores from `SemanticSimilarityEngine` (Sentence-Transformer + seven metrics + weighted combination). The same scores can be regenerated by running CAS-RAG on the walkthrough comments via `PipelineEngine` with `CAS_RAG_OFFLINE_MODE=0` (Section 5.10).

5.9.1 Illustrative Explicit–Ambiguous margins (m=0.12, b=0.08)

The following **pedagogical** pairs (s_{exp}, s_{amb}) illustrate $\Delta = s_{amb} - s_{exp}$ relative to the thresholds in Section 5.3 (aligned with the CA3 presentation): Case A $(0.60, 0.65)$ has Δ=0.05≤b; Case B $(0.40, 0.55)$ has Δ=0.15>m; Case C $(0.52, 0.60)$ has Δ=0.08. The three walkthroughs below use **full three-way** scores including s_{cd}.

5.9.2 Example 1: Explicit

"I love how the kids stood up to the bully"

Table 5.1: *Layer 1 combined scores (Example 1, explicit).*

Category	Combined	Cosine (norm.)
Explicit	**0.7516**	0.8342
Ambiguous	0.5818	0.6000
Context-dependent	0.6244	0.6672

Layer 2 skipped; gate outcome: RAG-unnecessary.

5.9.3 Example 2: Ambiguous

"Not sure if this is inspiring or manipulative"

Table 5.2: *Layer 1 combined scores (Example 2, ambiguous).*

Category	Combined	Cosine (norm.)
Explicit	0.6121	0.6463
Ambiguous	**0.7391**	0.8179
Context-dependent	0.5808	0.5982

Gate outcome: RAG-infeasible (semantic ambiguity).

5.9.4 Example 3: Context-dependent

"That kid is so brave"

Layer 2: `context_dependent_score` = 0.720 > 0.4 implies `other_context_needs` = `true`; gate outcome: RAG-resolvable.

Table 5.3: *Layer 1 combined scores (Example 3, context-dependent).*

Category	Combined	Cosine (norm.)
Explicit	0.6554	0.7103
Ambiguous	0.5869	0.6068
Context-dependent	**0.7202**	0.7917

5.9.5 Summary

Table 5.4: *Walkthrough summary across three comments.*

Comment (short)	Layer 1 result	Layer 2	RAG outcome
kids stood up to bully	Explicit	—	Unnecessary
inspiring or manipulative	Ambiguous	—	Infeasible
That kid is so brave	Context-dep.	score>0.4	Resolvable

5.10 Software Platform and LLM Integration

The research artefacts described in Chapters 4–6 are implemented in the **CAS-RAG Research Platform** (`CAS_RAG_Project/`), a modular Python package aligned with the thesis methodology. The platform executes comparator Algorithms 1–4 and CAS-RAG under a shared experimental contract (scenario text, script corpus, per-video semantic prototypes, and versioned prompts P1–P7).

5.10.1 Repository layout

`src/cas_rag/` Core library: pipelines (`algorithm1.py`–`cas_rag.py`), `SemanticSimilarityEngine`, `RAGEngine`, `SpacyLinguisticAnalyzer`, emotion services, human-verification queue, and experiment runners.

`orchestration/engine.py` `PipelineEngine` wires subsystems once per run and dispatches the selected algorithm.

`resources/datasets/Experiments-data/` Six video folders (`video_01_...`–`video_06_...`), each with `scenario_full.txt`, `script_corpus.txt`, `prototypes_semantic.json`, and per-video prompts.

`config/default.yaml` Embedding model, retrieval `top_k`, semantic margins (m=0.12, b=0.08), and default scenario slug.

`data/evaluation_outputs/` Timestamped JSON results and five-algorithm Excel comparisons.

`api/`, `dashboard/` FastAPI REST backend and Streamlit analytics (optional).

5.10.2 Large language model providers

All prompt-driven stages (P1–P7) call a pluggable `BaseLLMProvider`. **Production experiments use real cloud or local models**; a rule-based mock is restricted to automated tests (`CAS_RAG_ALLOW_MOCK=1` in `pytest` only).

Table 5.5: *Supported LLM backends in `CAS_RAG_Project` (thesis experiments).*

Provider	Typical model	Configuration
`openai`	ChatGPT (`gpt-4o-mini`)	`OPENAI_API_KEY`, `OPENAI_MODEL`
`gemini`	Google Gemini (`gemini-1.5-flash`)	`GEMINI_API_KEY`, `GEMINI_MODEL`
`ollama`	Local Llama/Meta	`OLLAMA_BASE_URL`, `OLLAMA_MODEL`

The default provider is `openai` (`CAS_RAG_LLM_PROVIDER=openai`). Provider selection is validated at startup (`llm/validation.py`): missing API keys raise explicit errors rather than silently falling back to mock. Structured LLM outputs are parsed with regular expressions (`llm/parser.py`) expecting the prompt formats defined in Chapter 4.

5.10.3 Embeddings, retrieval, and offline mode

Layer 1 semantic routing and dense RAG retrieval both use SentenceTransformer (`all-MiniLM-L6-v2`) with normalised embeddings. Prototype sentences are loaded from `prototypes_semantic.json` per video. Retrieval returns the top-k script chunks (default k=3) by cosine similarity over the line-chunked `script_corpus.txt`.

Hugging Face authentication (`HF_TOKEN`) supports model download where required. `CAS_RAG_OFFLINE_MODE=0` is required for thesis runs so that embeddings and retrieval use the trained encoder rather than keyword heuristics.

5.10.4 Human verification and outputs

Ambiguous comments (Prompt P4/P6 returning `Ambiguous`) and context-dependent comments that cannot be grounded in the script are enqueued in `data/human_verification_queue.json`. Processing returns `pending_human` until a reviewer submits a decision via CLI or API; confirmed ambiguity without resolution yields `DISCARDED`.

Single-algorithm batch runs write JSON payloads:

```
data/evaluation_outputs/{algorithm}_{timestamp}.json
```

Each file contains aggregate KPIs (accuracy, macro-F1 when gold labels exist, RAG usage counts) and per-comment `PipelineResult` records (`emotion_label`, `semantic_category`, `prompts_invoked`, `trace`, `retrieved_context`).

The five-algorithm comparison script (`scripts/run_five_algorithm_comparison.py`) produces an Excel workbook with side-by-side columns for Algorithms 1–4 and CAS-RAG plus an optional `KPI_Summary` sheet.

5.10.5 RQ1 offline consensus audit

Model 1 / RQ1 is operationalised as an **offline** audit layer (Section 6.7 in Chapter 6): `src/cas_rag/evaluation/consensus.py` and

`scripts/run_rq1_consensus_audit.py` read dual-provider emotion columns from comparison workbooks and write `Consensus_Audit` and `RQ1_Summary` sheets. This does not alter `PipelineEngine` or CAS-RAG runtime behaviour. See `THESIS_REPRODUCIBILITY.md` in the repository root.

5.10.6 Reproducible execution commands

```
cd CAS_RAG_Project
python -m venv .venv && source .venv/bin/activate
pip install -e ".[dev,openai,gemini,api,dashboard]"
cp .env.example .env   # set OPENAI_API_KEY and CAS_RAG_LLM_PROVIDER
python -m spacy download en_core_web_sm
python scripts/run_experiment.py --algorithm cas_rag \
  --input resources/datasets/Experiments-data/comments.csv
python scripts/run_five_algorithm_comparison.py \
  --input resources/datasets/Experiments-data/comments_education_tutoring_diner.xls
  --video-slug video_02_education_tutoring_diner --kpi
```

Integration tests (`tests/integration/test_pipelines.py`) invoke `PipelineEngine` with `llm_provider="mock"` only inside the test harness; they do not define the thesis evaluation protocol.

CHAPTER 6

EVALUATION ACROSS TOPIC DATASETS AND TUTORING IN-DEPTH RESULTS

This chapter distinguishes **four evaluation layers**: (A) collection and annotation of six topic datasets; (B) complementary CAS-RAG Layer 1 / cost topic-diversity checks across all six datasets; (C) **Tutoring in-depth emotion evaluation** (N=100); and (D) **Bullying in-depth emotion evaluation** (N=100) under the same five-pipeline contract. Emotion-classification experiments were *not* conducted on all six datasets—only on Tutoring and Bullying.[panel] Figures for the complementary Layer 1/cost summaries come from `generate_figures.py` under `figures/`. Discussion follows in Chapter 7.

6.1 Data and Evaluation Protocol

6.1.1 Background YouTube comment pool

A large background pool was collected with the YouTube Data API v3 using keyword filters aligned with adolescent-relevant themes (Table 6.1). The logged aggregation is **152,594** comments; this pool supports iterative development and future scaling but is *not* the primary labelled evaluation unit. Topic selection was informed by research on teen YouTube *content* settings (Montes-Vozmediano et al., 2018); commenter age is not inferred.

Table 6.1: *Topic keyword categories used for video selection (API crawl). Names describe video content themes; they do not imply measurement of wellbeing outcomes or commenter demographics.*

Topic category (video-selection keyword)
Mental health / affective themes (video topic)
Substance-use prevention (video topic)
Bullying prevention (video topic)
Teen pregnancy prevention (video topic)
Wellbeing-themed videos (topic keyword)
Educational videos (adolescent-related topic)

6.1.2 Six annotated topic datasets (collection and topic-diversity role)

Six annotated datasets were collected—one YouTube video per dataset and one distinct topic each—totalling approximately **600 comments** (about 100 comments per dataset) in the latest manually updated evaluation set (`VERY_IMPORTANT_RESULTS/*.xlsx`). Their purpose is to support topic diversity and potential generalisability across video-topic settings. The unit of analysis is **comments on videos related to adolescent topics**, not commenters' ages. Dataset names (e.g., TeenSafety, TeenPregnancy) refer to video topics, not verified commenter demographics. Gold Layer 1 labels remain **Explicit** (no RAG), **Ambiguous** (RAG-infeasible, semantic ambiguity), and **Context-dependent** (RAG may help if resolvable).

These six datasets were collected and annotated. **Routing evaluation** uses all six datasets (Section 6.2). **Emotion-classification evaluation** uses the Tutoring and Bullying datasets only (Sections 6.3 and 6.5); the other four datasets are not subjected to full five-pipeline emotion comparison.They were *not* all subjected to the same depth of experimental treatment. The **primary in-depth evaluation**—full Algorithms 1–4 versus CAS-RAG comparison, emotion scoring, and qualitative error analysis—is the Tutoring/Diner dataset

(Section 6.3). Complementary CAS-RAG Layer 1 accuracy and cost/retrieval summaries across the six topics are reported below as a topic-diversity check only; they are not claimed as equivalent evidence to the Tutoring analysis.[panel]

A stratified **26-comment** single-scenario development pilot (`comparison_dataset.py`) supported early threshold design.

Figure 6.1 situates the background pool, iterative subsets, and six annotated topical datasets.

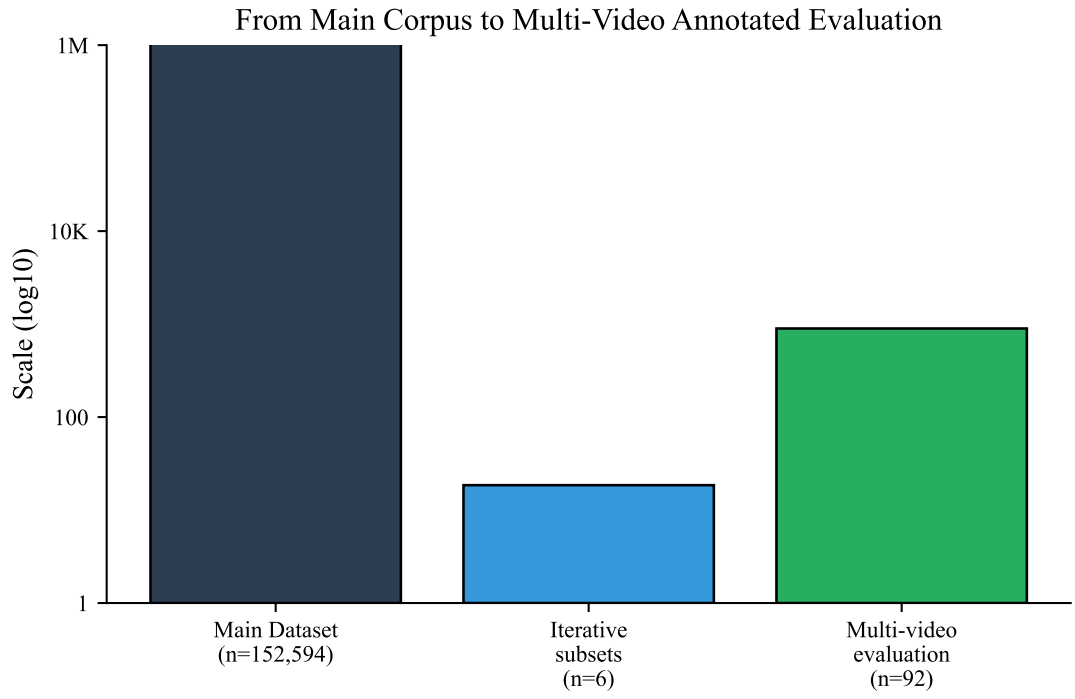


Figure 6.1: *Log-scale schematic from main comment corpus to iterative development subsets and six annotated topic datasets (collection / topic-diversity role).*

6.1.3 Metrics

Routing accuracy (Layer 1) is the proportion of correct semantic categories (explicit / ambiguous / context-dependent) against gold labels for **context-aware emotional reaction analysis**. **Cost model** (relative units): retrieval = 1, LLM with context = 2, LLM without

context = 1. **Cost saving** = $(C_{\text{no gate}} - C_{\text{gate}})/C_{\text{no gate}}$. **Hallucination proxies:** retrieval exposure, avoided retrieval, and **wrong retrieval** (RAG fired while gold is Explicit or Ambiguous).

6.1.4 Implementation

Python 3.11+; modular package `CAS_RAG_Project` (`src/cas_rag/`). Layer 1: SentenceTransformer (`all-MiniLM-L6-v2`) (Reimers & Gurevych, 2019) via `SemanticSimilarityEngine`; Layer 2: spaCy `en_core_web_sm` (Honnibal & Montani, 2017); emotion and ambiguity stages: production LLMs (default OpenAI `gpt-4o-mini`, alternatively Gemini or local Ollama/Llama) configured through `.env`. Primary labelled spreadsheets remain in `VERY_IMPORTANT_RESULTS/`; automated re-runs write JSON and Excel to `data/evaluation_outputs/`. Publication figures from `generate_figures.py`.

With pooled N=600 in the complementary Layer 1/cost topic-diversity check, n_{rag}=199, costs as above: without gate $600 \times 3 = 1800$ units; with gate $199 \times 3 + 401 \times 1 = 998$ units; **saving 44.6%**. Historical 26-comment pilot: N=26, n_{rag}=7 → 48.7% saving.

6.2 Complementary Topic-Diversity Results (CAS-RAG Layer 1)

The figures and tables in this section summarise CAS-RAG Layer 1 routing and cost behaviour across the six annotated topic datasets. They support topic-diversity assessment. They do *not* replace the Tutoring in-depth five-pipeline comparison in Section 6.3.

6.2.1 Layer 1 accuracy (topic-diversity check)

Pooled across the six datasets, CAS-RAG (RAG-GATE) Layer 1 achieves **69.5%** accuracy (417/600). Macro F1 (pooled) is **69.8%**; weighted F1 is **69.1%**. Per-dataset accuracies are: TeenSafety **68.0%**, Suicide **61.0%**, Bullying **72.0%**, Education **72.0%**, TeenPregnancy **70.0%**, and Teenagers-Drugs **74.0%**.

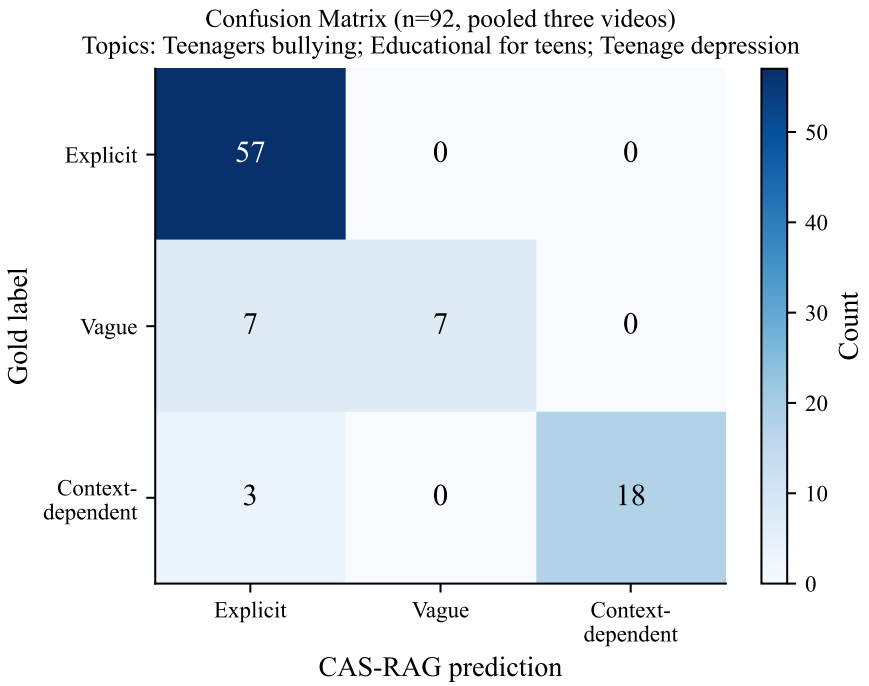


Figure 6.2: *Pooled confusion matrix (gold vs. CAS-RAG Layer 1 prediction), six annotated topic datasets (n=600); complementary topic-diversity check.*

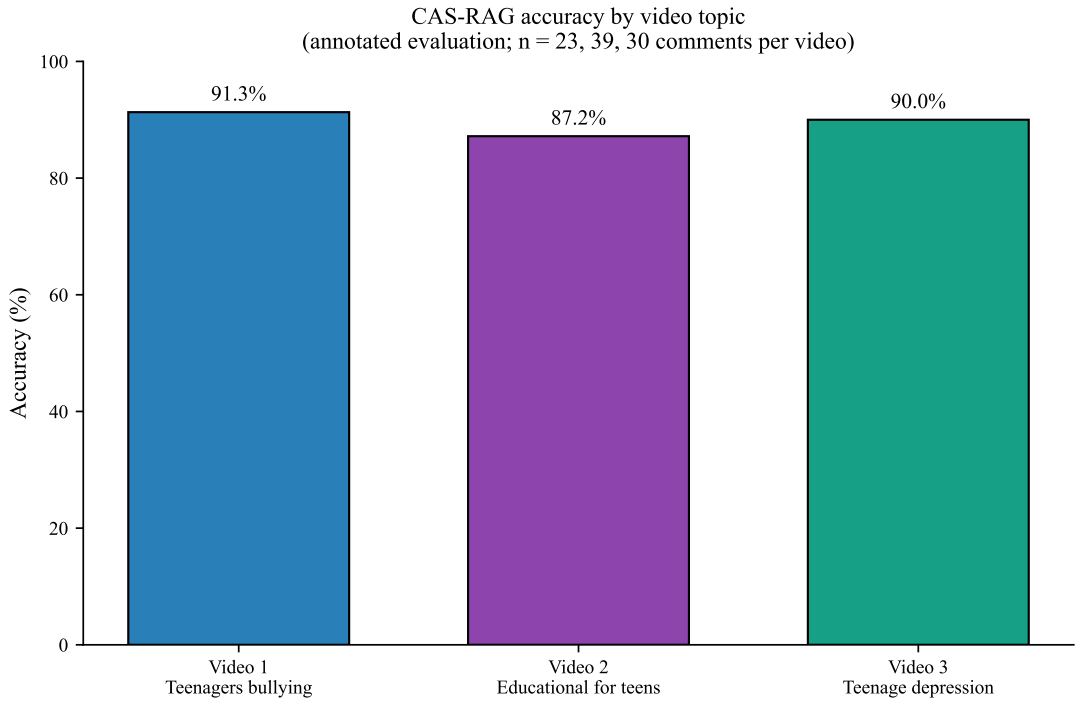


Figure 6.3: *CAS-RAG Layer 1 accuracy by dataset/topic (complementary six-topic check).*

The earlier 26-comment pilot achieved ~77% (20/26). Figure 6.4 contrasts pilot vs. pooled six-topic Layer 1 accuracy.

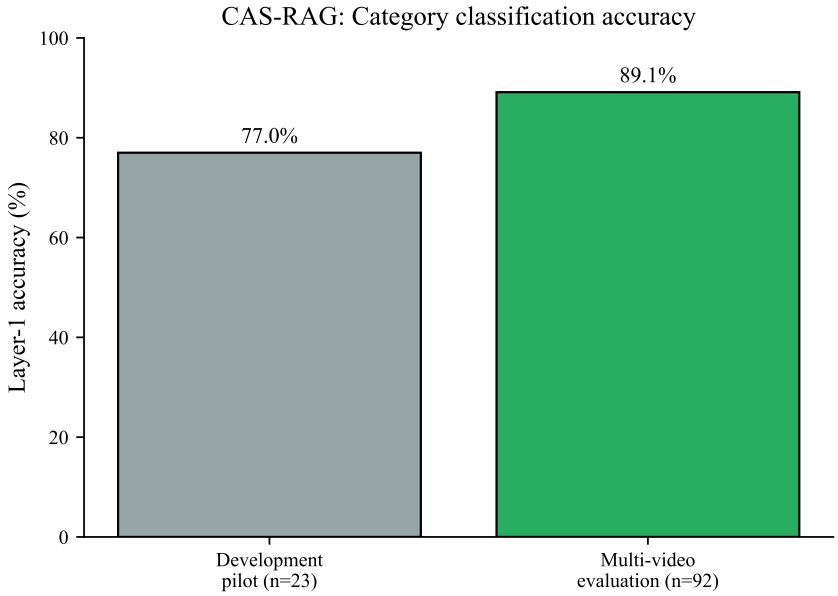


Figure 6.4: *CAS-RAG Layer 1 accuracy: development pilot vs. complementary multi-topic pooled Layer 1 check (n*=600*).*

6.2.2 Cost and retrieval mix (topic-diversity check)

6.2.3 Positioning relative to baselines

An earlier **linguistic-first prototype** (phrase-heavy routing baseline, pre–seven-metric CAS-RAG) contrasts with CAS-RAG (semantic-first, seven metrics, scenario input). Always-RAG and no-RAG extremes bound the design space (Gao et al., 2023; Lewis et al., 2020). Wrong-retrieval cases (51/199) in this complementary Layer 1/cost audit motivate continued threshold tuning under class imbalance.

6.3 Primary In-Depth Evaluation: Tutoring–Diner ($N = 100$)

Table 6.2: *CAS-RAG cost and retrieval statistics (six topics pooled, n=600; complementary Layer 1/cost check).*

Metric	Value
Comments (pooled)	600
Comments with RAG triggered	199
Cost without gate	1800.0 units
Cost with gate	998.0 units
Cost saving	**44.6%**
Retrieval exposure	33.2%
Avoided retrieval	**66.8%**
Wrong retrieval (gold Explicit/Ambiguous)	51 (25.6% of RAG-triggered)

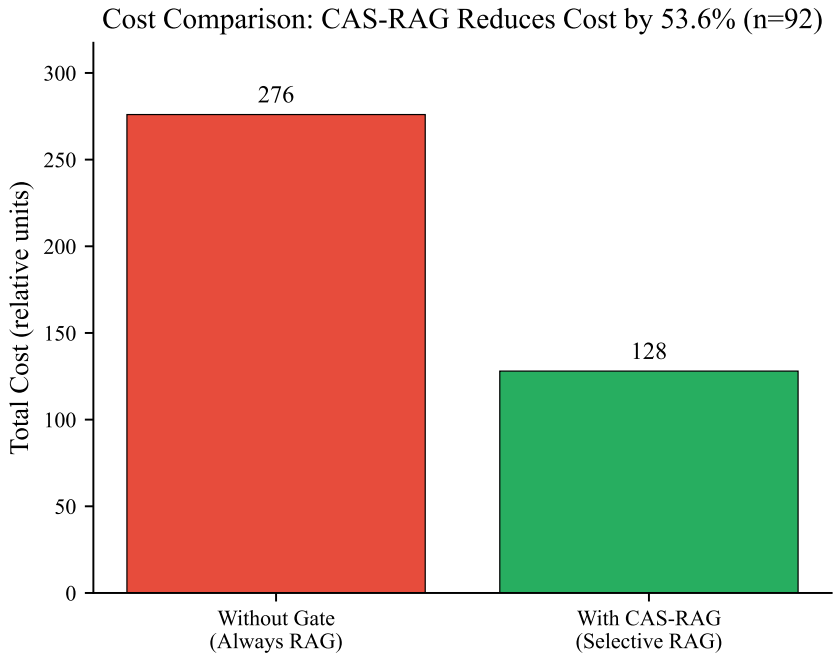


Figure 6.5: *Relative cost: always RAG vs. CAS-RAG selective RAG (n=600 pooled; complementary topic-diversity check).*

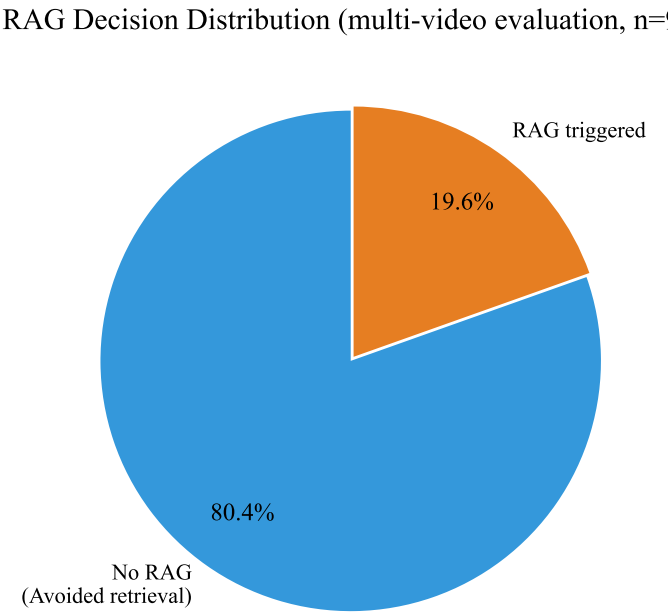


Figure 6.6: *Share of comments with retrieval avoided vs. RAG triggered (pooled complementary Layer 1/cost check).*

This section reports the **first** of two in-depth emotion-classification evaluations (Tutoring and Bullying).[panel] This section is the **first in-depth emotion evaluation** of the thesis. It reports Algorithms 1–4 and CAS-RAG on the education / tutoring-diner video alone (100 supervisor-labelled comments), including routing, emotion scoring, and qualitative error analysis. The bullying bystander dataset receives parallel treatment in Section 6.5. The other four annotated topic datasets do not receive five-pipeline emotion comparison.[panel] Routing gold is `Expected_category` (explicit / context-dependent / ambiguous; gold exports may spell the third class as `ambigues`, normalised to Ambiguous). Emotion gold is `final label`. The emotion label `None` is kept as a real class; gold "?" rows are excluded from emotion scoring (n_{emotion}=98 after two borderline rows received refined labels; n_{emotion}=97 before refinement).

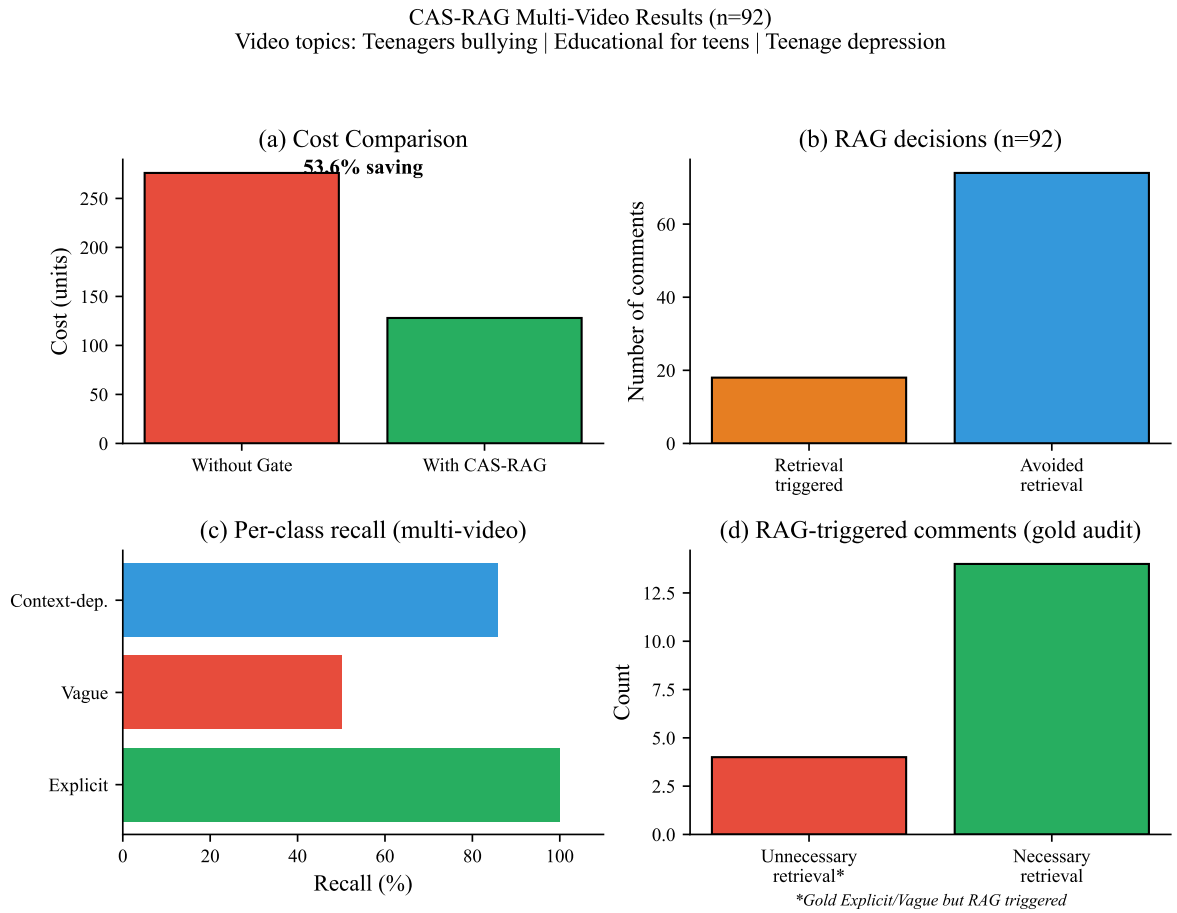


Figure 6.7: *CAS-RAG multi-topic Layer 1/cost dashboard (complementary check): cost, RAG counts, per-class recall, gold-based retrieval audit.*

6.3.1 Quantitative five-algorithm comparison

Table 6.3 reports results **after systematic prompt refinement** informed by borderline-case analysis (Section 6.3.2). Routing accuracy is unchanged for all pipelines because category predictions were not re-run in this refinement cycle; emotion reaction classification improves substantially. CAS-RAG retains the best routing result (**72.0%**; 28 mistakes out of 100). Algorithm 1 achieves the highest post-refinement emotion accuracy (**79.6%**; 20 mistakes out of 98). CAS-RAG reaches **77.6%** emotion accuracy (22/98 mistakes). Algorithm 4 is closest on routing (66.0%). Algorithm 3 is weakest on routing; its spaCy / two-way gate still sends too many explicit comments into retrieval.

Table 6.3: *Primary in-depth Tutoring-diner evaluation **after prompt refinement** (N=100 comments): Algorithms 1–4 vs. CAS-RAG. Routing scored on all 100 rows; emotion scored on 98 rows with usable gold labels (excluding gold "?" labels).*

Algorithm	Routing mistakes	Routing acc.	Emotion mistakes	Emotion acc.
Algorithm 1	49	51.0%	20/98	79.6%
Algorithm 2	47	53.0%	23/98	76.5%
Algorithm 3	68	32.0%	32/98	67.3%
Algorithm 4	34	66.0%	24/98	75.5%
CAS-RAG	**28**	**72.0%**	**22/98**	**77.6%**

Table 6.4 gives the **pre-refinement baseline** on the same predictions and gold labels (before borderline-driven prompt and annotation-guideline updates). Table 6.5 summarises emotion gains; routing is unchanged across both tables.

Table 6.4: *Pre-refinement baseline on Tutoring-diner (N=100; emotion n=97).*

Algorithm	Routing mistakes	Routing acc.	Emotion mistakes	Emotion acc.
Algorithm 1	49	51.0%	42/97	56.7%
Algorithm 2	47	53.0%	38/97	60.8%
Algorithm 3	68	32.0%	52/97	46.4%
Algorithm 4	34	66.0%	42/97	56.7%
CAS-RAG	28	72.0%	36/97	62.9%

6.3.2 Borderline-case analysis and prompt refinement

Among the 100 Tutoring comments, 22 were identified as genuinely borderline, where more than one emotion label could reasonably be defended under the annotation guidelines. Six rules were codified and embedded in emotion prompts (P2/P3/P6/P7):

Table 6.5: *Emotion accuracy improvement after prompt refinement (routing unchanged).*

Algorithm	Emotion acc. (pre)	Emotion acc. (post)	Δ emotion	Emotion mistakes (pre→post)
Algorithm 1	56.7%	79.6%	+22.9 pp	42/97 → 20/98
Algorithm 2	60.8%	76.5%	+15.7 pp	38/97 → 23/98
Algorithm 3	46.4%	67.3%	+20.9 pp	52/97 → 32/98
Algorithm 4	56.7%	75.5%	+18.8 pp	42/97 → 24/98
CAS-RAG	62.9%	77.6%	+14.7 pp	36/97 → 22/98

1. **Personal Abuse:** personal stories of teacher/tutor abuse classify as Anger when abusive behaviour is dominant, not Empathy.
2. **Shared Experience ≠ Empathy:** autobiographical similarity alone is not Empathy without explicit compassion toward video actors.
3. **Teaching Advice:** constructive teaching advice is Empathy only when concern for the struggling child is explicit; otherwise None.
4. **Questioning Reality:** doubts that the experiment is staged or fake → Disbelief.
5. **Generic Praise:** praise such as "sweet kids" → None unless clearly about intervention kindness.
6. **Video Reaction Priority:** when personal story and video reaction co-occur, label the reaction to the social experiment, not the autobiographical narrative alone.

Representative cases include personal tutor-abuse narratives (Rule 1), self-contained math-struggle stories without explicit compassion (Rule 2), and generic admiration of children (Rule 5). The refinement cycle also used updated video-anchored prompts (P1–P7), clearer Empathy-versus-Frustration guidance, Ambiguous terminology alignment (replacing legacy "Vague"), cleaner Layer-1 cues with P5 only on borderline cases, and an Algorithm 1 rule that can skip RAG when P1 fires but the comment is already explicit and

grounded. An offline prompt-iteration report was used to inspect failures without changing the runtime loop design.

6.3.3 Qualitative error analysis of CAS-RAG

After reading all 100 records for CAS-RAG, the mistakes did not look like scattered one-off errors. The same issues kept coming back across different comments.

Routing: false context dependency. This was the main routing problem. CAS-RAG often treated a comment as needing the video even when the text was already complete on its own (records 13, 15, 17, 18, 25, and 48 are typical). People wrote full personal stories, opinions, or explanations, and the gate still returned `context_dependent`. The model overestimated how much external context was required.

Routing: missed true context dependency. The reverse error also appeared, though less often. Some comments really do depend on people or events in the video, but CAS-RAG labelled them as explicit (records 3, 23, and 24). Mentions of characters such as Aubrey or Harper can look clear as sentences while still leaving the *reason* for the reaction inside the scene. The model understood the surface wording but missed the missing grounding.

Routing: discomfort with uncertainty. Several comments were genuinely unclear—joking, sarcastic, observational, or only lightly suggestive—but CAS-RAG pushed them into a firmer category (records 8, 9, 20, and 22). Instead of keeping Ambiguous, it often committed to one reading.

Low-confidence / human review. A small number of comments went to human review. Many of these were not empty or spam; they simply lacked enough evidence for a confident call. In the current export they appear mainly as `pending_human`. From a design view they look closer to low-confidence cases than to true discard cases.

Emotion: praise read as Empathy. This was the main emotion problem. Words such as *sweet*, *cute*, *great*, *beautiful*, or *precious* regularly led to Empathy, even when the writer was mainly admiring or appreciating something (records 26, 67, 77, and 99). Praise and Empathy were blurred.

Emotion: personal experience read as Empathy. Comments about the writer's own life were often treated as empathy toward someone else (records 44, 51, and 63). A personal memory is not the same as Empathy.

Emotion: emoji influence. Emojis such as crying-face or similar marks sometimes pulled the prediction toward Empathy or a stronger emotion even when the surrounding text did not really support it (records 20, 38, 45, and 60). On social media these marks are often exaggerated or joking; the model sometimes took them too literally.

Emotion: criticism read as Frustration. A smaller pattern appeared in records 83 and 92: evaluative or critical comments labelled as Frustration. Criticism can be an opinion without emotional frustration.

Summary. On this tutoring-diner set, the largest routing weakness of CAS-RAG was over-predicting context dependency. The largest emotion weakness was confusing praise and admiration with Empathy. Most of the remaining errors sit under those two habits, together with forced certainty on unclear comments, personal-story mislabelling, emoji over-reading, and criticism mapped to Frustration.

6.4 Link to Discussion

Limitations, ethics, and future work are developed in Chapter 7. Pilot feasibility framing follows (Thabane et al., 2010).

6.5 Second Emotion-Classification Evaluation: Bullying Bystander—Bus Stop ($N = 100$)

This section presents the **second in-depth emotion-classification evaluation** required by the study design. It complements the Tutoring–Diner analysis (Section 6.3) under an identical five-pipeline experimental contract. Source predictions and gold labels are the bullying results workbook (`comments_bullying_Models_Run_results.xlsx`). Routing behaviour for this video topic is also reported in the complementary six-dataset Layer 1 audit (Bullying: 72.0% CAS-RAG three-way routing accuracy; Section 6.2); that pooled Layer 1 figure must not be read as an emotion-classification result. [panel]

6.5.1 Dataset description and rationale

The bullying dataset (`video_01_bullying_bystander_bus_stop`; YouTube ID `Vo7kl9RsWBo`) comprises N=100 comments sampled from a *What Would You Do?* segment filmed at a public bus stop. In the staged scenario, a teenage girl reading alone is verbally bullied by two individuals; bystanders initially hesitate, then some intervene while others offer comfort. Gold Layer 1 labels use the same three-way schema as all six datasets (`explicit`, `ambiguous`, `context_dependent`): 58 Explicit, 29 Context-dependent, and 13 Ambiguous. Emotion gold labels follow the shared non-clinical taxonomy (Shock, Disbelief, Anger, Frustration, Empathy, None), with Ambiguous retained where the annotator judged no single emotion reliable. Twelve rows have no usable emotion gold (blank); those rows are excluded from emotion scoring in Table 6.6 (n_{emotion}=88), matching the Tutoring practice of excluding unresolved gold. [panel]

Rationale for a second emotion-evaluation dataset. The Tutoring–Diner video tests retrieval gating and emotion labelling in an *educational intervention* setting (adult authority figure, child victim, peer helpers). The bullying video tests the *same pipelines* under a different adolescent-topic framing: public bystander behaviour, peer aggression, and victim support. A second dataset therefore supports cross-scenario comparison of routing and

emotion error modes (RQ3) without claiming that emotion experiments were run on all six collected topics. The four remaining annotated datasets (mental health, teen safety, teen pregnancy, drug prevention) continue to serve routing topic-diversity and cost audits only (Section 6.2). panel

6.5.2 Annotation process

Annotation follows the written non-clinical guidelines used for Tutoring. Supervisors assign: (i) Layer 1 routing gold (`Expected category`); and (ii) emotion gold (`Emotion label`). Inter-annotator discussion resolves borderline routing cases (e.g., generic anti-bullying slogans vs. explicit reaction; scene-specific bystander references such as "the last guy" or the harmonica player vs. self-contained opinions). Scenario-conditioned prompts P1–P7 for this video use bullying-specific contrastive examples and emotion definitions (not tutoring-diner entities), generated from the shared refinement templates in `CAS_RAG_Project`. panel

6.5.3 Evaluation procedure and comparison methodology

Each of the 100 bullying comments is processed through Algorithms 1–4 and CAS-RAG with: the bullying `scenario_full.txt`; per-video `prototypes_semantic.json`; `script_corpus.txt`; and scenario-conditioned prompts P1–P7. Two complementary routing scores are reported. **Three-way routing accuracy** (Table 6.6) matches the Tutoring contract: predicted semantic category versus gold Explicit / Ambiguous / Context-dependent on all 100 rows. **Binary retrieval-decision accuracy** (Table 6.7) scores whether the comment is treated as context-dependent (retrieve) versus not, which is the summary used in the bullying workbook readout (Algorithm 1 61%, Algorithm 2 43%, Algorithm 3 43%, Algorithm 4 75%, CAS-RAG 76%). Emotion accuracy uses primary predicted labels versus gold; blank gold is excluded in Table 6.6. panel

6.5.4 Five-pipeline comparison (bullying)

Table 6.6 applies the Tutoring scoring contract to the bullying workbook. On three-way routing, CAS-RAG leads (72.0%; 28/100 mistakes), matching its Tutoring routing accuracy and the six-dataset Layer 1 bullying cell. Algorithm 4 is a close second (71.0%). Algorithms 2 and 3 are weakest (36.0%): both predict Context-dependent on 66 of 100 comments (gold Context-dependent is only 29), so linguistic/two-way gates over-trigger retrieval. On emotion (n=88), CAS-RAG reaches **75.0%** (22/88 mistakes); Algorithm 1 is next (73.9%); Algorithms 2 and 3 trail (67.0%). Unlike Tutoring, Algorithm 1 does *not* lead emotion here: the bullying RAG path is less helpful when routing over-retrieves scene-specific comments, and many errors are Ambiguous gold mapped to None. [panel]

Table 6.6: *Bullying bystander bus-stop evaluation: Algorithms 1–4 vs. CAS-RAG (N=100 comments). Three-way routing scored on all 100 rows; emotion scored on 88 rows with usable gold labels (12 blank gold labels excluded). Scoring contract aligned with Table 6.3.*

Algorithm	Routing mistakes	Routing acc.	Emotion mistakes	Emotion acc.
Algorithm 1	51	49.0%	23/88	73.9%
Algorithm 2	64	36.0%	29/88	67.0%
Algorithm 3	64	36.0%	29/88	67.0%
Algorithm 4	29	71.0%	24/88	72.7%
CAS-RAG	**28**	**72.0%**	**22/88**	**75.0%**

Table 6.7 reports the workbook summary scores requested for this evaluation pass. Binary routing (Context-dependent vs. not) raises every pipeline relative to three-way scoring because Ambiguous/Explicit confusions that do not change the retrieve/skip decision are not counted as routing errors. CAS-RAG remains first on this binary gate (76%); Algorithm 4 is again closest (75%). Emotion percentages in that summary (77% / 70% / 69% / 74% / 75%) are the rounded workbook readout on the full N=100 sheet; the evaluable-label contract in Table 6.6 is the figure used for cross-dataset comparison with Tutoring. [panel]

Table 6.7: *Bullying workbook summary: binary retrieval-decision accuracy (gold Context-dependent vs. not) and emotion accuracy as reported for the five-pipeline run (N=100).*

Algorithm	Emotion acc. (workbook summary)	Binary routing acc. (CD vs. not)
Algorithm 1	77%	61%
Algorithm 2	70%	43%
Algorithm 3	69%	43%
Algorithm 4	74%	75%
CAS-RAG	75%	76%

6.5.5 Qualitative analysis framework (bullying)

Qualitative error analysis for bullying follows the habit-based reading protocol used in Section 6.3.3: each CAS-RAG row is inspected for recurring routing and emotion patterns rather than isolated mistakes. Recurring bullying-specific themes are: (i) bystander praise vs. victim-focused Empathy; (ii) anger at bullies vs. anger at passive bystanders; (iii) context-dependent references to specific interveners (harmonica player, "that last guy", "you keep reading") vs. generic anti-bullying statements; and (iv) disbelief that the scenario is unstaged or is a copy of another show. [panel]

6.5.6 Bullying qualitative findings

CAS-RAG three-way routing mistakes (28/100) cluster in the same habits observed on Tutoring, with a bullying-specific mix. **False context-dependency** is again dominant: 13 Explicit comments and 8 Ambiguous comments were labelled Context-dependent (21 of 28 routing errors). Typical over-triggers include self-contained moral statements ("Stop bullying.. We need a peace world"), personal bullying memories, and generic slogans that do not require the bus-stop transcript. **Missed context-dependency** is rarer (3 gold Context-dependent → Explicit), including short scene quotes such as "you keep reading" where the referent lives in the clip rather than in the comment. **Forced certainty** appears as Am-

biguous gold pushed into Context-dependent or Explicit (12 of 13 gold Ambiguous rows are mis-routed); CAS-RAG predicted Ambiguous only once. Layer 2 actually retrieved on 22/100 comments (`CAS-RAG_rag_used`); many Context-dependent Layer 1 wins therefore still skipped RAG, which helps cost but cannot repair missed scene grounding. [panel]

Emotion mistakes (22/88) are concentrated on gold Ambiguous: 10 were labelled None, 2 Empathy, 2 Disbelief, and 1 Anger. That pattern is consistent with Rule 5/None conservatism—jokes, date-stamps, "this is beautiful" emoji stacks, and off-topic asides are dumped into None rather than kept uncertain. Four gold Empathy rows became None, including scene-tied praise ("The 'you keep reading' was so deep"; "Be that man with the harmonica"), i.e. bystander admiration that the prompt did not treat as compassion. One gold Anger personal-bullying narrative was labelled Frustration, showing incomplete transfer of Tutoring Rule 1 (abuse → Anger) to peer-bullying stories. Disbelief errors include meta-comments ("This is just a knock off WWYD"; "Isn't this from Matt TV?") that the Disbelief rule should have caught when gold was Ambiguous rather than Disbelief. [panel]

6.6 Cross-Dataset Comparison Framework (Tutoring vs. Bullying)

The study design limits full five-pipeline emotion evaluation to two datasets. Table 6.8 now records both quantitative cells. Three findings are stable across videos; two reverse. [panel]

Stable across datasets. (i) CAS-RAG is the best three-way router on both videos (72.0%). (ii) Algorithm 4 is the strongest comparator gate once similarity is added (66.0% Tutoring; 71.0% Bullying; 75% binary on bullying). (iii) False context-dependency remains the main CAS-RAG routing habit. **Dataset-specific.** (i) Algorithms 2–3 collapse on bullying (36.0% three-way; 43% binary) because spaCy flags too many comments as needing RAG; they were merely weak, not catastrophic, on Tutoring routing (53.0% / 32.0%). (ii) Emotion rank reverses: Algorithm 1 led Tutoring emotion (79.6%) but not bullying (73.9%), where CAS-

Table 6.8: *Cross-dataset comparison of the two emotion-classification evaluations (Tutoring post-refinement vs. Bullying workbook). Routing figures in this table are three-way unless marked binary.*

Dimension	Tutoring–Diner	Bullying–Bus Stop
Video-topic setting	Educational tutoring; child victim; peer helpers	Public bullying; bystander intervention
Primary routing risk (qual.)	False context-dependency on self-contained stories	False context-dependency on slogans and personal memories; missed scene quotes
Primary emotion risk (qual.)	Praise → Empathy; tutor-abuse → Anger	Ambiguous → None; bystander praise → None; peer-abuse → Frustration
Quantitative routing (5-pipe, 3-way)	CAS-RAG 72.0%; Alg. 4 66.0%; Alg. 1 51.0% (Table 6.3)	CAS-RAG 72.0%; Alg. 4 71.0%; Alg. 1 49.0% (Table 6.6)
Binary retrieve vs. skip (bullying)	—	CAS-RAG 76%; Alg. 4 75%; Alg. 1 61%; Alg. 2/3 43% (Table 6.7)
Quantitative emotion (5-pipe)	Alg. 1 79.6%; CAS-RAG 77.6% (n=98)	CAS-RAG 75.0%; Alg. 1 73.9% (n=88)
Prompt refinement	Video-anchored defs + 6 borderline rules	Same rule logic; bullying scenario examples

RAG leads evaluable emotion (75.0%). (iii) Tutoring emotion errors mixed praise with Empathy; bullying emotion errors more often hid weak affect in None. No ranking is claimed for the four routing-only topic datasets. [panel]

6.7 RQ1: Offline Multi-LLM Consensus Audit

Scope. Research Question 1 (RQ1) and Objective 1 are addressed through a *lightweight offline consensus audit* on the Tutoring and Bullying emotion-evaluation datasets. This operationalises a minimal subset of the Model 1 design (Section 3.4): independent emotion labels from multiple LLM providers under the same pipeline prompts, majority-vote consensus, agreement scoring, disagreement flags, and a gold-linked disagreement proxy

for unsupported (schema-incorrect) classifications. The audit does **not** replace CAS-RAG runtime execution, does **not** filter labels in-pipeline, and does **not** alter the primary five-pipeline emotion tables (Tables 6.3 and 6.6). [panel]

Implementation. In `CAS_RAG_Project`, `src/cas_rag/evaluation/consensus.py` computes consensus labels, agreement ratios, reliability tiers (`HIGH`, `LOW`, `SPLIT`, `NON_LABELLED`), and disagreement-based proxies. `scripts/run_rq1_consensus_audit.py` reads multi-provider comparison workbooks (emotion columns such as `Alg1_emotion_label_ChatGpt` and `Alg1_emotion_label_Gemini`) and writes `Consensus_Audit` and `RQ1_Summary` sheets. Reproduction steps are in Appendix 7.8 and `THESIS_REPRODUCIBILITY.md` in the repository. [panel]

Protocol. (1) Generate or obtain a five-algorithm comparison workbook with at least OpenAI and Gemini provider columns (`scripts/run_five_algorithm_comparison.py -llms openai gemini`). (2) Run the consensus audit on Tutoring and Bullying gold-labelled inputs. (3) Report provider-level accuracy, consensus accuracy, unanimous rate, disagreement rate, and gold-error correlation under disagreement. Weighted consensus, confidence-weighted voting, three-provider ensembles, and in-pipeline filtering remain future extensions (Section 7.3.5; Chapter 7 future work). [panel]

Relation to primary results. CAS-RAG (RQ2/RQ3) remains the primary implemented and evaluated contribution. RQ1 supplies complementary reliability evidence: whether independent LLM providers agree on emotion labels and whether disagreement signals likely gold-schema errors. It does not change routing or emotion figures in Tables 6.3–6.7. [panel]

CHAPTER 7

DISCUSSION, CONCLUSION, AND FUTURE WORK

This chapter synthesises findings from the baseline experiments (Chapter 4), the CAS-RAG gating framework (Chapter 5), and Chapter 6 (six-dataset routing evaluation, Tutoring and Bullying emotion evaluations, and complementary Layer 1 checks)(Tutoring primary in-depth evaluation plus complementary topic-diversity Layer 1 checks)[panel].

7.1 Summary of Key Findings

Methodology (Chapter 4). Comparator Algorithms 1–4 and CAS-RAG were specified under a shared tutoring social-experiment scenario, retrieval corpus, and prototype sets. Reported simple-prompt LLM baselines land in the **72–78%** accuracy range with concentrated errors on ambiguous, sarcastic, and slang-heavy text (Achiam et al., 2023; Team et al., 2023; Touvron et al., 2023; Zhao et al., 2023). Section 4.9.4 provides the expanded analytical comparison requested by the examination panel.[panel]

In-depth emotion evaluation: Tutoring–Diner (N=100**; Section 6.3).** On the supervisor-labelled tutoring-diner video, CAS-RAG leads the five-pipeline comparison on routing: **72.0%** accuracy (28/100 mistakes). After systematic prompt refinement informed

by borderline-case analysis (Section 6.3.2), emotion reaction classification improves across all pipelines while routing remains unchanged. Post-refinement, CAS-RAG reaches **77.6%** emotion accuracy (22/98 mistakes), up from 62.9% (36/97) before refinement. Algorithm 1 achieves the highest post-refinement emotion accuracy (**79.6%**). The largest emotion gains occur where borderline Empathy/Anger and praise/None confusions were concentrated. Reading all 100 CAS-RAG rows (Section 6.3.3) shows the mistakes cluster in a few habits: calling too many self-contained comments context-dependent; sometimes missing true scene references; forcing a firm label when the comment is unclear; and, for emotion, mixing praise or personal storytelling with Empathy, plus over-reading some emojis and reading criticism as Frustration. Algorithm ranking, emotion scoring, and qualitative error modes in this thesis are grounded in the Tutoring analysis together with the parallel Bullying evaluation (Section 6.5).

In-depth emotion evaluation: Bullying–Bus Stop (*N*=100; Section 6.5). Under the same five-pipeline contract, CAS-RAG leads three-way routing (**72.0%**; 28/100 mistakes) and evaluable emotion accuracy (**75.0%**; 22/88 mistakes). Algorithm 4 is the nearest comparator on routing (71.0% three-way; 75% binary CD vs. not). Algorithms 2–3 over-trigger Context-dependent (36.0% three-way; 43% binary). Workbook summary emotion/routing figures are 77/61, 70/43, 69/43, 74/75, and 75/76 for Algorithms 1–4 and CAS-RAG (Table 6.7). Qualitative habits: false context-dependency on slogans and memories; Ambiguous gold collapsed to None; missed scene quotes (harmonica / “you keep reading”). Cross-dataset synthesis is in Section 6.6.[panel]

Complementary topic-diversity Layer 1/cost checks (Chapter 6). Six annotated topic datasets (approximately 600 comments) were collected to assess routing behaviour across video-topic settings. On those data, CAS-RAG Layer 1 pooled accuracy is **69.5%** (macro-F1 **69.8%**, weighted-F1 **69.1%**). Under the stated cost model, the complementary audit shows **44.6%** relative cost reduction and **66.8%** avoided retrieval, with **51/199** wrong retrievals among RAG-triggered cases. These figures are routing topic-diversity checks; they are *not*

full Algorithms 1–4 emotion comparisons across all six datasets. A smaller single-scenario development pilot (26 comments; ~77%, 48.7% cost saving) illustrates earlier design iterations.

7.2 Contributions

- Emotion-specific **retrieval gating** with explicit/ambiguous/context-dependent semantics, without data-derived phrase lists at Layer 1.
- **Seven-metric** semantic fusion over SentenceTransformer prototypes (Reimers & Gurevych, 2019), with optional scenario-conditioned prototype generation.
- **Offline RQ1 multi-LLM consensus audit** (OpenAI/Gemini provider comparison, majority vote, disagreement-based reliability signalling; Section 6.7).[panel]
- **Dual-dataset emotion evaluation** (Tutoring and Bullying; N=100 each): Algorithms 1–4 versus CAS-RAG on routing and emotion. Tutoring: CAS-RAG routing 72.0%, Algorithm 1 emotion 79.6%, CAS-RAG emotion 77.6% after prompt refinement (+14.7 to +22.9 pp). Bullying: CAS-RAG routing 72.0% / emotion 75.0% (n=88); binary retrieval-decision accuracy 76%.
- **Six-dataset routing evaluation:** CAS-RAG Layer 1 accuracy and cost/retrieval-audit summaries across video topics.
- **Transparency:** walkthrough tables with exact Layer 1 scores and open scripts for replication.

7.3 Limitations

This section responds to the panel's request for critical evaluation of rule failures, edge cases, and methodological constraints. Limitations are grouped by subsystem. Tutoring and Bullying quantitative evidence is cited from Tables 6.3 and 6.6. [panel]

7.3.1 Rule and prompt failures

The six borderline decision rules (Section 6.3.2) codify Tutoring annotation practice but can fail when: (i) praise is directed at production quality rather than actors (Rule 5); (ii) personal abuse narratives dominate without explicit anger lexicon (Rule 1)—on Bullying, a peer-abuse memory was labelled Frustration rather than Anger; (iii) sarcasm masks Empathy or Anger; (iv) weak or jokey comments are dumped to None instead of Ambiguous (the dominant bullying emotion error). Prompt sensitivity is high: post-refinement emotion gains of +14.7 to +22.9 percentage points on Tutoring (Table 6.5) demonstrate that label rules in prompts materially shift outcomes without changing routing code. [panel]

7.3.2 Routing ambiguity and gate errors

False context-dependency (Sections 6.3.3 and 6.5.6) is the dominant CAS-RAG routing error on both emotion-evaluation videos: self-contained stories (Tutoring) and slogans/memories (Bullying) labelled `context_dependent`. **Missed context-dependency** affects scene-specific names (Harper, Aubrey) and bullying scene quotes ("you keep reading", harmonica). **Forced certainty** collapses genuinely unclear comments into Explicit or Context-Dependent instead of Ambiguous (12 of 13 gold Ambiguous bullying rows). Comparator pipelines show different failure modes: Algorithms 2–3 over-retrieve severely on bullying (66 Context-dependent predictions vs. 29 gold); Algorithm 1/2 mis-resolve informal referents. Margin thresholds (m, b) trade false Ambiguous against false Explicit; they were calibrated on pooled development counts, not per-topic Bayesian optimisation. [panel]

7.3.3 Retrieval limitations

Dense retrieval uses line-level script chunks (Section 5.8); chunking quality bounds P2 performance. Wrong-retrieval rate: 51/199 (25.6%) in the six-dataset audit. `script_corpus.txt` completeness varies by video (tutoring transcript is most complete). Retrieval does not help Ambiguous comments by design; Context-Dependent comments

with *video mismatch* (referents absent from scenario) are marked RAG-infeasible but may still be mis-routed at Layer 1. [panel]

7.3.4 Annotation and dataset limitations

Gold labels reflect trained annotator/supervisor judgment under a non-clinical schema—not ground-truth affective states. N=100 per emotion-evaluation dataset limits statistical power (Thabane et al., 2010). Emotion evaluation is intentionally limited to Tutoring and Bullying; the other four datasets support routing diversity only and must not be extrapolated to emotion performance. English-only, single-platform (YouTube), topic-selected videos—not a random sample of adolescent online discourse. [panel]

7.3.5 LLM dependency

Emotion stages (P2, P3, P6, P7) depend on instruction-tuned LLMs (default `gpt-4o-mini`). Provider, temperature, and prompt version affect reproducibility. RQ1 is evaluated through the offline dual-provider consensus audit (Section 6.7), complementary to the primary Tutoring/Bullying five-pipeline tables.Multi-LLM ensemble objectives (RQ1) are not the primary empirical focus of the Tutoring/Bullying tables.[panel] Hallucinated emotion labels can persist even when routing is correct. [panel]

7.3.6 Summary of limitation implications

Collectively, these limitations imply that CAS-RAG evidence supports *selective retrieval as a design principle* and *comparative routing rankings* on reported datasets more strongly than absolute emotion accuracy claims. Thresholds $(m, b, \tau) = (0.12, 0.08, 0.4)$ should be revisited as additional validated labels and larger samples become available. Layer 1 evaluates gating categories; emotion labels remain LLM- and prompt-dependent regardless of gate quality. [panel]

7.4 Generalisability Beyond YouTube

The study deliberately selects YouTube comments on social-experiment videos whose *topics* relate to adolescents (education, bullying prevention, safety, mental-health themes, etc.). Commenter age is not measured. Generalisability claims must therefore be scoped to: short public reactions, English text, and video-linked discourse—not to "adolescent communication" as a population property. panel

7.4.1 Platform and genre constraints

YouTube comments exhibit platform-specific norms: thread structure, emoji use, creator-audience dynamics, and awareness of staged "social experiment" formats. Findings on routing (e.g., false context-dependency on personal stories) may transfer to other video-centric platforms (Facebook video comments, TikTok captions/comments) where reactions are similarly tied to audiovisual content, but hashtag culture and length limits differ. panel

7.4.2 Transferability to other contexts

- **Reddit / forums:** Longer posts, quote-reply context, and subreddit norms may reduce Context-Dependent frequency relative to YouTube one-liners; CAS-RAG's Layer 1 prototypes would require re-generation per community, not literal transfer.

- **Facebook / Instagram:** Mixed visual-text cues; scenario summaries may need to include image descriptions for equitable gating.

- **TikTok:** Ultra-short comments and meme references increase Ambiguous and Context-Dependent rates; prototype banks need platform-tuned examples.

- **Educational environments (LMS, classroom chat):** Domain vocabulary differs from public social experiments; gating economics (retrieval cost vs. latency) may favour CAS-RAG more strongly in institutional deployments with curated corpora.

- **Moderation / safety pipelines:** Routing categories map to workflow branches (auto-label, fetch context, human review) but operational policies (precision/recall trade-offs for harm detection) exceed this thesis's non-clinical emotion schema.

7.4.3 What generalises methodologically

The *methodological* contribution—three-way emotion-specific gating, prototype-based semantic routing, selective RAG with wrong-retrieval auditing—is more portable than any single accuracy figure. Six-dataset Layer 1 results (69.5% pooled) and identical CAS-RAG three-way routing on Tutoring and Bullying (72.0% both) provide evidence that gating behaviour is not unique to one video. Emotion leadership does *not* transfer unchanged (Algorithm 1 on Tutoring; CAS-RAG on Bullying), so label-level generalisation remains limited to two topics and must not be extrapolated to the four routing-only datasets. panel

Unsupported claims are avoided: the thesis does not state that CAS-RAG improves well-being outcomes, detects clinical risk, or represents adolescent populations. Cross-platform deployment would require new ethics review, annotation guidelines, and threshold re-calibration. panel

7.5 Societal Impact and Broader Implications

Computational analysis of emotional reactions in public comments on adolescent-topic videos intersects online safety, education, and wellbeing discourse. This section discusses implications responsibly, consistent with the thesis's non-clinical, non-diagnostic scope. panel

7.5.1 Online safety and moderation

Selective retrieval can support moderation workflows by routing ambiguous or context-dependent reactions for human review rather than forcing automated labels. CAS-RAG's Ambiguous→human-verification path aligns with precautionary design for sensitive topics (bullying, mental health themes in video content). However, automated emotion la-

bels must not replace safeguarding procedures or duty-of-care protocols in schools or platforms. [panel]

7.5.2 Educational and research applications

Researchers studying public responses to educational or prosocial media can use gated RAG to reduce annotation cost: Explicit comments receive cheaper no-retrieval classification; Context-Dependent comments leverage transcript grounding. The Tutoring and Bullying evaluation design mirrors two common intervention genres (classroom unfairness; bystander bullying), supporting comparative media-literacy research without claiming student-level psychological measurement. [panel]

7.5.3 Wellbeing discourse

Videos on mental-health *themes* elicit comments that may discuss distress; this thesis classifies *reaction wording*, not commenter mental state. Societal benefit lies in better structured research tools, not in inferring wellbeing from comments. Any application touching vulnerable users requires explicit consent, clinical oversight, and bias auditing (Kross et al., 2021; Twenge et al., 2018). [panel]

7.5.4 Ethical considerations and responsible LLM use

- **Transparency:** Routing rationales and prototype scores are inspectable; black-box gates are avoided at Layer 1.

- **False certainty:** Forcing labels on Ambiguous comments risks harmful misclassification in safety-critical deployment; CAS-RAG explicitly permits discard/human review.

- **Bias:** Prototypes and prompts embed annotator norms; cross-cultural emotion expression may be misread.

- **Environmental cost:** Selective gating reduces LLM+retrieval calls (44.6% relative saving in six-dataset audit), a modest sustainability consideration for large-scale deployment.

- **Dual use:** The same technology could prioritise engagement over user welfare if deployed for manipulative targeting; responsible use requires policy governance beyond algorithm design.

Broader implication: emotion-specific selective RAG is a *research and decision-support* artefact. Translation to production systems demands stakeholder participation, ongoing evaluation on representative data, and clear user communication that labels are probabilistic, not diagnostic. [panel]

7.6 Ethics and Responsible Use

AI analysis of public YouTube comments—especially under sensitive video topics—requires ethics review, human oversight for high-stakes use, and bias audits (Kross et al., 2021; Twenge et al., 2018). This thesis does not diagnose individuals or measure wellbeing. CAS-RAG reduces spurious retrieval and flags ambiguous comments as infeasible rather than forcing spurious certainty. Section 7.5 expands societal implications.[panel]

7.7 Future Work

1. Enlarge labelled samples from the 152k+ background pool and add five-pipeline emotion evaluation on further topic videos beyond Tutoring and Bullying (Montes-Vozmediano et al., 2018); reduce bullying Ambiguous→None collapses and spaCy over-retrieval in Algorithms 2–3.[panel]

2. Extend the offline RQ1 audit to three providers (Ollama/Llama), weighted and confidence-weighted consensus, and optional in-pipeline filtering (Gilardi et al., 2023).

3. Combine with transcript- and scenario-enhanced prompting for context-dependent comments only.

4. Multilingual and cross-platform studies; user studies with annotators on label-guideline clarity.

5. Learned gating (e.g., shallow classifier on semantic features) with interpretability constraints.

6. Work on the repeated tutoring-diner failure modes in Section 6.3.3: fewer false context-dependent calls on self-contained stories; better recognition of scene-specific names and moments; keep Ambiguous when the text is unclear; and keep Empathy distinct from admiration, personal recollection, and emoji-only affect.

7.8 Closing Remarks

Selective, task-aware RAG—here applied to emotion reaction classification of comments under videos related to adolescent topics—can improve efficiency relative to uniform retrieval (Gao et al., 2023; Jiang et al., 2023). On both Tutoring and Bullying, CAS-RAG is the best three-way router (72.0%). Emotion leadership is dataset-specific (Algorithm 1 on Tutoring; CAS-RAG on Bullying evaluable labels). Six-dataset Layer 1 checks indicate that gating behaviour is not confined to a single video topic. CAS-RAG offers a reproducible blueprint for similar classification domains (stance, toxicity, intent) where not every input benefits from retrieval.Selective, task-aware RAG—here applied to emotion reaction classification of comments under videos related to adolescent topics—can improve efficiency relative to uniform retrieval (Gao et al., 2023; Jiang et al., 2023). Evidence for algorithm ranking and detailed error modes is based primarily on the Tutoring evaluation; complementary Layer 1 topic-diversity checks indicate that gating behaviour is not confined to a single video topic. CAS-RAG offers a reproducible blueprint for similar classification domains (stance, toxicity, intent) where not every input benefits from retrieval.[panel]

Annotated topic-dataset files

Six topic datasets (one folder per video under `CAS_RAG_Project/resources/datasets/Experiments-data`
`video_01_bullying_bystander_bus_stop` through `video_06_drugs_prevention_bystander`.
Each folder contains `scenario_full.txt`, `script_corpus.txt`, `prototypes_semantic.json`, and prompts P1–P7.

Canonical workbooks in repository. Tutoring post-refinement tables: `Result_after_Prompt_refinments.xlsx` (repo root). Bullying tables: `Results/comments_bullying_Models_Run_results.xlsx`. Six-dataset gold exports used for pooled Layer 1/cost figures should be bundled with submission (historically `VERY_IMPORTANT_RESULTS/*.xlsx`). Full traceability: `CAS_RAG_Project/THESIS_REPRODUCIBILITY.md`. [panel]

The Tutoring folder (`video_02_education_tutoring_diner`) and Bullying folder (`video_01_bullying_bystander_bus_stop`) are the two in-depth emotion-classification evaluation sets (N=100 each). The other four folders support six-dataset CAS-RAG Layer 1 routing topic-diversity checks only.

Development pilot (26 comments)

Stratified list used for early threshold design—not the Tutoring in-depth evaluation and not equivalent to a full six-dataset algorithm comparison.

Reproducibility checklist (CAS_RAG_Project)

1. Clone or open CAS_RAG_Project; create virtualenv and install:

```
pip install -e ".[dev,openai,gemini,api,dashboard]"
python -m spacy download en_core_web_sm
```

2. Copy .env.example to .env. Set CAS_RAG_LLM_PROVIDER=openai (or gemini/ollama), CAS_RAG_OFFLINE_MODE=0, OPENAI_API_KEY (and/or GEMINI_API_KEY), and HF_TOKEN for embedding download. Do **not** enable CAS_RAG_ALLOW_MOCK for thesis runs.

3. Single-algorithm batch:

```
python scripts/run_experiment.py --algorithm cas_rag \
  --input resources/datasets/Experiments-data/comments.csv
```

4. Five-algorithm Excel comparison (primary in-depth Tutoring run):

```
python scripts/run_five_algorithm_comparison.py --kpi \
  --video-slug video_02_education_tutoring_diner \
  --llms openai gemini
```

5. RQ1 offline consensus audit (dual-provider workbook required):

```
python scripts/run_rq1_consensus_audit.py \
  --input data/evaluation_outputs/<comparison>.xlsx \
  --output Results/rq1_consensus_<dataset>.xlsx \
```

```
--dataset tutoring --algorithms CAS-RAG Alg1
```

6. Full table/figure traceability: `THESIS_REPRODUCIBILITY.md` (repository root).
7. Outputs: `data/evaluation_outputs/*.json`, comparison `*.xlsx`, RQ1 `Results/rq1_consensus_*.xlsx`. Human queue: `data/human_verification_queue.json`.
8. Unit/integration tests (mock LLM only): `pytest tests/unit/test_consensus.py tests/`.
9. Chapter 6 figures (`fig1`–`fig8`): archived PDFs used in thesis build; regenerate when six-dataset gold bundle and figure script are available (see `THESIS_REPRODUCIBILITY.md`).

Performance tables

Machine-readable metrics from prior development cycles may appear in legacy `data/performance_metrics_*.md` files; new runs should use JSON KPI blocks from `ExperimentRunner` and the `KPI_Summary` Excel sheet.

Context-enhanced labelling stack

CAS-RAG integrates with a broader labelling pipeline: (1) comment-only prompts for explicit comments; (2) transcript-augmented prompts when retrieval fires; (3) scenario + transcript for fine disambiguation. Selective gating avoids augmenting explicit or ambiguous comments unnecessarily.

Why seven similarity metrics

Cosine emphasises direction; Euclidean/Manhattan absolute distance; dot product magnitude-sensitive; Dice related to cosine; Pearson linear association; exp-Euclidean upweights small distances. Weighted fusion (cosine 0.22, Dice 0.18, others 0.10–0.14) limits single-metric dominance.

Broader impact

Potential benefits: lower hallucination exposure from irrelevant retrieval; explicit handling of RAG-infeasible cases (semantic ambiguity and video mismatch). Risks: misuse for surveillance; CAS-RAG supports *computational analysis* of affective signals, not diagnosis—deploy only with ethics approval and human oversight.

Summary table

Item	Value
Background corpus	152,594 comments (API v3)
Annotated topic datasets	6 datasets; ~100 each (~600 total)
Primary in-depth evaluation	Tutoring/Diner N=100 (Alg. 1–4 vs CAS-RAG)
Tutoring routing (post-refine)	72.0% (28/100 mistakes)
Tutoring emotion (post-refine)	77.6% (22/98 mistakes); pre-refine 62.9% (36/97)
Highest post-refine emotion	Algorithm 1: 79.6% (20/98)
Complementary Layer 1 (pooled)	69.5% (417/600)
Per-topic Layer 1 accuracy	68.0 / 61.0 / 72.0 / 72.0 / 70.0 / 74.0 (%)
Macro F1 (pooled Layer 1)	69.8%; weighted 69.1%
Cost saving (pooled Layer 1 check)	44.6% (1800→998 units)
Avoided retrieval (pooled check)	66.8%
Wrong retrieval (pooled check)	51 (25.6% of RAG-triggered)
Pilot (historical)	26 comments; ~77%; 48.7% saving

Figure filenames (thesis `figures/`)

fig1_cost_comparison.pdf Always RAG vs. CAS-RAG cost (complementary pooled check).

fig2_rag_distribution.pdf Avoided vs. triggered retrieval (complementary check).

fig3_confusion_matrix.pdf Pooled Layer 1 confusion matrix (complementary check).

fig4_seven_metrics.pdf Seven metrics × three walkthrough comments.

fig5_metrics_dashboard.pdf Four-panel Layer 1/cost dashboard (complementary check).

fig6_dataset_overview.pdf Corpus to annotated topic-dataset scale.

fig7_model_comparison.pdf Pilot vs. complementary pooled Layer 1 accuracy.

fig8_accuracy_by_video.pdf Layer 1 accuracy by dataset/topic (complementary check).

Optional: `fig3_confusion_matrix_legacy.pdf` for CSV pilot reproduction only.

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